Analytic Combinatorics and Large Language Models

An Expository Introduction

RACA Notes

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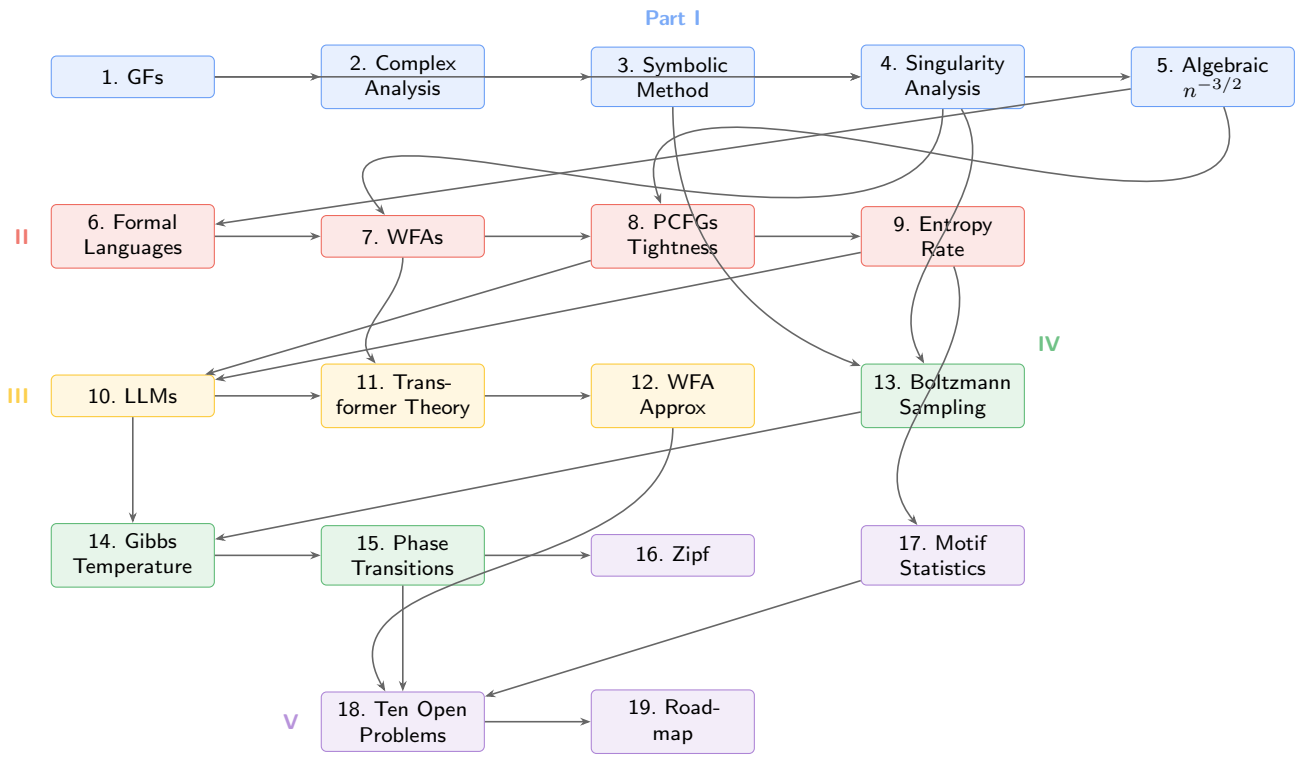
Notation

Symbol	Meaning
Generating functions and asymptotics	
$A(z), B(z), C(z), \ldots$	Ordinary generating functions (OGFs)
$\hat{A}(z)$	Exponential generating function (EGF) of A
$[z^n]A(z)$	Coefficient of z^n in $A(z)$
$[z^n/n!]\hat{A}(z)$	$n! \cdot [z^n]\hat{A}(z)$; the n-th EGF coefficient
R	Radius of convergence (Cauchy–Hadamard)
ρ	Location of dominant singularity; equals R for aperiodic GFs
α	Singularity exponent in $(1 - z/\rho)^{-\alpha}$
$f(n) \sim g(n)$	$\lim_{n\to\infty} f(n)/g(n) = 1$
Combinatorial constructions	
$\mathcal{E}$	Neutral class (one object of size 0; OGF $= 1$)
$\mathcal{Z}$	Atomic class (one object of size 1; OGF $= z$)
$\mathrm{SEQ}(\mathcal{A})$	Sequences of $\mathcal{A}$-objects; OGF $= 1/(1 - A(z))$
$\mathrm{SET}(\mathcal{A})$	Unordered sets; EGF $= \exp \hat{A}(z)$
$\mathrm{CYC}(\mathcal{A})$	Cyclic sequences; EGF $= \log(1/(1 - \hat{A}(z)))$
Formal languages and automata	
Σ	Finite alphabet
Σ^*	Set of all finite strings over Σ
EOS	End-of-sequence token
$L \subseteq \Sigma^*$	A formal language
$(\alpha, \{A_\sigma\}, \beta)$	Weighted finite automaton: initial vector, transition matrices, final vector
$A = \sum_\sigma A_\sigma$	Aggregate transition matrix of a WFA
$\rho(A)$	Spectral radius of matrix A
Probability and information theory	
μ	Probability distribution on Σ^* (LLM output distribution)
$p(\sigma \mid h)$	Conditional next-token probability given history h
h	Shannon entropy rate (bits or nats per symbol)
$H(X_1^n)$	Block entropy of a length-n string
Statistical mechanics / temperature	
$E(w) = -\log \mu(w)$	Energy of string w under model μ
$\beta = 1/T$	Inverse temperature
$Z(\beta) = \sum_w \mu(w)^\beta$	Partition function
$\mu_\beta(w) \propto \mu(w)^\beta$	Gibbs distribution at inverse temperature β
β_c	Critical inverse temperature (phase transition point)
Named constants	
$\varphi = (1 + \sqrt{5})/2$	Golden ratio (≈ 1.618)
$\hat{\varphi} = (1 - \sqrt{5})/2$	Conjugate of the golden ratio (≈ -0.618)
$\Gamma(\alpha)$	Euler gamma function

Throughout, log denotes the natural logarithm unless a base is indicated (as in $\log_2$).

Chapter Dependencies

The following diagram shows the essential prerequisite structure. A solid arrow from Chapter i to Chapter j means that j builds directly on material from i. Chapters within the same part share a common color. The main spine of the book runs 1–2–4–5–8–10; the two branching threads are the Boltzmann/temperature path (13–14–15) and the WFA/Transformer path (7–11–12). Chapters 16–17 can be read after their prerequisites without the intervening material; Chapters 18–19 synthesize everything.



Preface

This book is an extended expository introduction to the research territory where analytic combinatorics meets large language models. It is written for a student who knows calculus, linear algebra, and a little probability, has perhaps seen complex numbers, and wants to understand both halves of the subject from scratch: the symbolic method and singularity analysis of Flajolet and Sedgewick on one side, and the probabilistic, formal-language, and statistical-mechanics views of neural language models on the other.

The organizing claim of the book is that these two traditions, despite rarely citing each other, are talking about the same mathematical objects: partition functions are generating functions, critical temperatures are radii of convergence, phase transitions are singularities, rational-series theorems from automata are dominant-pole theorems from analytic combinatorics. The research frontier consists largely of building bridges that the two communities have, until recently, avoided building. The final chapters make those bridges explicit and give a student a plausible starting point.

Part I develops the classical machinery of analytic combinatorics from zero, up to the $n^{-3/2}\rho^{-n}$ asymptotic for unambiguous context-free languages. Part II turns to weighted formal languages, probabilistic grammars, tightness, and the entropy-rate identity $h = \log(1/R)$. Part III introduces autoregressive language models and the formal-language theory of transformers, then surveys the weighted-finite-automaton approximation pipeline. Part IV is devoted to Boltzmann sampling, Gibbs-measure views of temperature decoding, and phase transitions. Part V covers Zipf's law, motif statistics, and the ten open problems we currently see at the frontier.

Exercises appear in selected chapters but are not the point. The point is to get a reader who starts knowing none of this to the point where they can open an arXiv paper from 2025 on, say, phase transitions in Pythia or WFA extraction from a Mistral model, and see what is going on.

Part I

Analytic Combinatorics from Scratch

This part develops the core toolkit of analytic combinatorics, assuming no prior exposure. We begin with generating functions as formal algebraic objects (Chapter 1), promote them to analytic functions of a complex variable (Chapter 2), and introduce the symbolic method that translates combinatorial structure directly into generating-function equations (Chapter 3). The payoff comes in Chapters 4 and 5: the Flajolet–Odlyzko transfer theorem extracts precise asymptotics from singularities, and the Chomsky–Schützenberger theorem shows that unambiguous context-free languages always produce the universal $n^{-3/2}$ exponent. By the end of Part I, the reader will have a self-contained workflow: specification $\to$ functional equation $\to$ singularity $\to$ asymptotic formula.

Chapter 1

What is a Generating Function?

A sequence of numbers $(a_n)_{n\geq 0}$ is the natural output of a counting problem: how many binary strings of length n, how many trees with n nodes, how many ways to parse a sentence of length n according to a given grammar. Studying the sequence directly forces us to deal with each index separately, and recurrences — while precise — often resist closed-form extraction. The central idea of generating function theory is to encode the entire sequence as a single algebraic object,

$$A(z) \;=\; \sum_{n\geq 0} a_n z^n,$$

and then let the algebra of formal power series do the bookkeeping. This is not mere notational convenience. Operations on sequences that would otherwise require index gymnastics correspond to clean algebraic operations on $A(z)$:

$$a_{n-1} \longleftrightarrow zA(z), \qquad \sum_{k=0}^{n} a_k b_{n-k} \longleftrightarrow A(z)B(z), \qquad a_n - a_{n-1} \longleftrightarrow (1-z)A(z)$$

after the appropriate initial-term corrections. Recurrences, which are linear or polynomial relations among the a_n, become *functional equations* for $A(z)$, and those equations can often be solved in closed form. The solution then exposes structure invisible in the raw sequence.

Definition 1.1. The set of *formal power series* over $\mathbb{Q}$, denoted $\mathbb{Q}[[z]]$, consists of all expressions $A(z) = \sum_{n\geq 0} a_n z^n$ with $a_n \in \mathbb{Q}$. Two series are added term by term: $(A+B)(z) = \sum_n (a_n + b_n) z^n$. Their product is defined by

$$(A \cdot B)(z) \;=\; \sum_{n\geq 0} c_n z^n, \qquad c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

This product is well-defined because the coefficient of z^n depends on only finitely many pairs (a_k, b_{n-k}), namely $k = 0, 1, \ldots, n$.

The product formula follows from collecting all pairs of terms $a_k z^k$ and $b_m z^m$ whose exponents add to n: $z^k \cdot z^m = z^n$ requires $m = n - k$, so the coefficient of z^n in the product is $a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$. At this stage z is a *formal indeterminate* — it carries no numerical value, and questions of convergence are entirely irrelevant. In $\mathbb{Q}[[z]]$ one can always add, subtract, and multiply series, and the proposition below will show exactly when division is possible. (Algebraists call $\mathbb{Q}[[z]]$ a *commutative ring*; this terminology is not needed for what follows.)

Definition 1.2. The *coefficient extraction operator* $[z^n]$ picks out the n-th coefficient of a power series:

$$[z^n]A(z) \;:=\; a_n.$$

It is linear because addition in $\mathbb{Q}[[z]]$ is term by term:

$$[z^n](A+B) = [z^n]\sum_{m\geq 0}(a_m+b_m)z^m = a_n+b_n = [z^n]A + [z^n]B.$$

For products, one simply reads off the coefficient of z^n in the Cauchy product:

$$[z^n](A\cdot B) = \sum_{k=0}^{n} a_k b_{n-k}.$$

Proposition 1.3. *Let $A(z) = \sum_{n\geq 0} a_n z^n \in \mathbb{Q}[[z]]$. Then $A(z)$ has a multiplicative inverse in $\mathbb{Q}[[z]]$ if and only if $a_0 \neq 0$.*

Proof. Suppose first that $a_0 \neq 0$. We seek a series $D(z) = \sum_{n\geq 0} d_n z^n$ such that

$$A(z)D(z) = 1.$$

Comparing coefficients of equal powers of z determines the d_n recursively. The constant term gives

$$a_0 d_0 = 1,$$

so $d_0 = 1/a_0$. For $n \geq 1$, the coefficient of z^n in $A(z)D(z)$ must be 0, so

$$a_0 d_n + a_1 d_{n-1} + \cdots + a_n d_0 = 0.$$

Therefore

$$d_n = -\frac{a_1 d_{n-1} + a_2 d_{n-2} + \cdots + a_n d_0}{a_0}.$$

Each d_n is uniquely determined by the earlier coefficients $d_0, \ldots, d_{n-1}$, so this recursion produces one and only one series $D(z)$ with $A(z)D(z) = 1$.

Conversely, if $a_0 = 0$, then the constant term of $A(z)D(z)$ is always $a_0 d_0 = 0$ for every series $D(z)$. Such a product can never equal the constant series 1. So no inverse exists in that case. □

Lemma 1.4. *In the ring of formal power series,*

$$\frac{1}{1-x} = \sum_{n\geq 0} x^n.$$

Proof. Let $G(x) = \sum_{n\geq 0} x^n$. Then

$$(1-x)G(x) = \sum_{n\geq 0} x^n - \sum_{n\geq 0} x^{n+1} = 1 + \sum_{n\geq 1}(1-1)x^n = 1.$$

So $G(x)$ is the multiplicative inverse of $1-x$. □

Two operations on power series appear repeatedly. The *shift* by z sends

$$A(z) \longmapsto zA(z) = \sum_{n\geq 1} a_{n-1} z^n,$$

so $[z^n](zA(z)) = a_{n-1}$. Likewise,

$$(1-z)A(z) = a_0 + \sum_{n\geq 1}(a_n - a_{n-1})z^n,$$

so multiplying by $1 - z$ encodes first differences after the constant term. The *formal derivative* is

$$A'(z) = \sum_{n \geq 1} n a_n z^{n-1},$$

defined term by term with no analytic content. It still satisfies the usual product rule: if $f(z) = \sum a_n z^n$ and $g(z) = \sum b_n z^n$, then

$$[z^{n-1}](fg)' = n \sum_{k=0}^{n} a_k b_{n-k},$$

while

$$[z^{n-1}](f'g + fg') = \sum_{k=1}^{n} k a_k b_{n-k} + \sum_{k=0}^{n-1} (n-k) a_k b_{n-k} = n \sum_{k=0}^{n} a_k b_{n-k},$$

so $(fg)' = f'g + fg'$ coefficient by coefficient.

The method: from recurrences to functional equations

The central technique of this chapter is a three-step recipe for converting a recurrence into a closed-form generating function.

1. **Define** the generating function $A(z) = \sum_{n \geq 0} a_n z^n$.
2. **Multiply** every term of the recurrence $a_n =$ (expression in $a_{n-1}, a_{n-2}, \ldots$) by z^n and **sum** over all valid n. Recognize each resulting sum as an expression involving $A(z)$.
3. **Solve** the resulting equation for $A(z)$.

Step 2 is the key move. The slogan is that a_{n-k} turns into $z^k A(z)$, but one must watch the lower limit of summation. The clean identity is

$$\sum_{n \geq k} a_{n-k} z^n = z^k A(z).$$

More generally, if the sum begins at $n = N$, then after the substitution $m = n - k$ one gets

$$\sum_{n \geq N} a_{n-k} z^n = z^k \sum_{m \geq N-k} a_m z^m = z^k \big(A(z) - a_0 - a_1 z - \cdots - a_{N-k-1} z^{N-k-1}\big).$$

So the only nuisance is bookkeeping for the missing initial terms. Once those are handled, the recurrence — an infinite family of equations, one for each n — collapses into a single equation for the single unknown $A(z)$.

We now demonstrate this on three progressively harder examples.

Binary strings

Let b_n denote the number of binary strings of length n. A direct count already gives $b_n = 2^n$, since each of the n positions can be filled by either 0 or 1. We now re-derive the same answer by generating functions in order to illustrate the method. Any binary string of length $n \geq 1$ is obtained by appending one of two symbols to a string of length $n - 1$, giving

$$b_n = 2b_{n-1} \qquad (n \geq 1), \qquad b_0 = 1.$$

Define $B(z) = \sum_{n\geq 0} b_n z^n$, multiply the recurrence by z^n, and sum over $n \geq 1$:

$$\sum_{n\geq 1} b_n z^n \;=\; 2z \sum_{n\geq 1} b_{n-1} z^{n-1}.$$

The left side is $B(z) - b_0 = B(z) - 1$. For the right side, let $m = n - 1$. Then $m \geq 0$, so

$$\sum_{n\geq 1} b_{n-1} z^{n-1} = \sum_{m\geq 0} b_m z^m = B(z).$$

Thus

$$B(z) - 1 = 2zB(z), \qquad \text{so} \qquad B(z) = \frac{1}{1-2z}.$$

By the formal geometric-series lemma,

$$\frac{1}{1-2z} = \sum_{n\geq 0} (2z)^n = \sum_{n\geq 0} 2^n z^n,$$

so $[z^n]B(z) = 2^n$ as expected. The functional-equation route is more instructive than the direct count because it still works when no closed form is obvious beforehand.

Fibonacci numbers

The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2} \qquad (n \geq 2).$$

Set $F(z) = \sum_{n\geq 0} F_n z^n$. Multiply the recurrence by z^n and sum over $n \geq 2$:

$$\sum_{n\geq 2} F_n z^n \;=\; z \sum_{n\geq 2} F_{n-1} z^{n-1} \;+\; z^2 \sum_{n\geq 2} F_{n-2} z^{n-2}.$$

The left side is

$$\sum_{n\geq 2} F_n z^n = F(z) - F_0 - F_1 z = F(z) - z.$$

For the first sum on the right, let $m = n - 1$. Then $m \geq 1$, so

$$\sum_{n\geq 2} F_{n-1} z^{n-1} = \sum_{m\geq 1} F_m z^m = F(z) - F_0 = F(z).$$

For the second, let $m = n - 2$. Then $m \geq 0$, so

$$\sum_{n\geq 2} F_{n-2} z^{n-2} = \sum_{m\geq 0} F_m z^m = F(z).$$

Therefore

$$F(z) - z = zF(z) + z^2 F(z),$$

and hence

$$F(z) = \frac{z}{1 - z - z^2}.$$

To extract F_n explicitly, we factor the denominator and use partial fractions. The roots of $x^2 - x - 1 = 0$ are

$$x = \frac{1 \pm \sqrt{5}}{2},$$

so we define

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \hat{\varphi} = \frac{1-\sqrt{5}}{2}.$$

Then $\varphi + \hat{\varphi} = 1$ and $\varphi\hat{\varphi} = -1$, so

$$1 - z - z^2 = (1-\varphi z)(1-\hat{\varphi} z).$$

Because the denominator has two distinct linear factors, elementary algebra tells us that a proper rational function with this denominator can be written in the form

$$\frac{A}{1-\varphi z} + \frac{B}{1-\hat{\varphi} z}.$$

So we seek constants A and B such that

$$F(z) = \frac{z}{(1-\varphi z)(1-\hat{\varphi} z)} = \frac{A}{1-\varphi z} + \frac{B}{1-\hat{\varphi} z}.$$

Multiply both sides by $(1-\varphi z)(1-\hat{\varphi} z)$:

$$z = A(1-\hat{\varphi} z) + B(1-\varphi z).$$

Comparing constant terms gives $A + B = 0$, so $B = -A$. Comparing coefficients of z gives

$$1 = -A\hat{\varphi} - B\varphi = A(\varphi - \hat{\varphi}) = A\sqrt{5},$$

so $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$. Therefore

$$F(z) = \frac{1}{\sqrt{5}} \cdot \frac{1}{1-\varphi z} - \frac{1}{\sqrt{5}} \cdot \frac{1}{1-\hat{\varphi} z}.$$

Applying the geometric-series lemma to each factor,

$$\frac{1}{1-\varphi z} = \sum_{n\geq 0} \varphi^n z^n, \qquad \frac{1}{1-\hat{\varphi} z} = \sum_{n\geq 0} \hat{\varphi}^n z^n.$$

Reading off the coefficient of z^n gives Binet's formula:

$$F_n = \frac{\varphi^n - \hat{\varphi}^n}{\sqrt{5}}.$$

Since $|\hat{\varphi}| < 1$, the second term becomes negligible as $n \to \infty$, so

$$F_n \sim \frac{\varphi^n}{\sqrt{5}}.$$

This asymptotic now follows directly from the explicit formula. A deeper preview is that Chapter 4 will recover the same conclusion from the pole of $F(z)$ at $z = 1/\varphi$, showing how singularities encode growth rates [19].

Catalan numbers

The Catalan numbers C_n count full binary trees (every internal node has exactly two children) with n internal nodes; equivalently, they count triangulations of a convex $(n+2)$-gon, Dyck paths of length $2n$, and well-formed parenthesizations of length n, among many other combinatorial families. The sequence begins

$$1,\ 1,\ 2,\ 5,\ 14,\ 42, \ldots$$

and satisfies the recurrence

$$C_n = \sum_{k=0}^{n-1} C_k\, C_{n-1-k}, \qquad C_0 = 1.$$

Here is the structural reason. A full binary tree with $n \geq 1$ internal nodes has a root. If the left subtree has k internal nodes, then the right subtree must have the remaining $n-1-k$. For $n = 3$, the three possibilities are $(k, n-1-k) = (0,2), (1,1), (2,0)$.

Define $C(z) = \sum_{n\geq 0} C_n z^n$. Multiply the recurrence by z^n and sum over $n \geq 1$. The left side is

$$\sum_{n\geq 1} C_n z^n = C(z) - 1.$$

For the right side, first expand the square of the series:

$$C(z)^2 = \Big(\sum_{i\geq 0} C_i z^i\Big)\Big(\sum_{j\geq 0} C_j z^j\Big) = \sum_{m\geq 0}\Big(\sum_{k=0}^{m} C_k C_{m-k}\Big) z^m.$$

So the coefficient of z^{n-1} in $C(z)^2$ is exactly

$$[z^{n-1}]\, C(z)^2 = \sum_{k=0}^{n-1} C_k C_{n-1-k}.$$

Therefore

$$\sum_{n\geq 1}\Big(\sum_{k=0}^{n-1} C_k C_{n-1-k}\Big) z^n = z \sum_{n\geq 1} [z^{n-1}]\, C(z)^2\, z^{n-1} = z\, C(z)^2.$$

Equating left and right sides gives

$$C(z) - 1 = z\, C(z)^2,$$

or equivalently

$$C(z) = 1 + z\, C(z)^2.$$

This is our first *quadratic* functional equation. We are solving a quadratic equation in the unknown $C(z)$, with coefficients in the field $\mathbb{Q}(z)$. Rearranging,

$$z\, C(z)^2 - C(z) + 1 = 0.$$

Applying the quadratic formula with $a = z$, $b = -1$, $c = 1$ yields

$$C(z) = \frac{1 \pm \sqrt{1-4z}}{2z}.$$

Lemma 1.5. *There is a unique formal power series*

$$S(z) = 1 + s_1 z + s_2 z^2 + \cdots$$

such that $S(z)^2 = 1 - 4z$. We denote this series by $\sqrt{1-4z}$.

Proof. Write

$$S(z)^2 = (1 + s_1 z + s_2 z^2 + \cdots)^2.$$

Comparing constant terms forces $1^2 = 1$, so the constant term must indeed be 1. For $n \geq 1$, the coefficient of z^n in $S(z)^2$ has the form

$$2s_n + (\text{expression involving only } s_1, \ldots, s_{n-1}).$$

Since the coefficient of z in $1 - 4z$ is -4 and all higher coefficients are 0, this determines s_n uniquely from the earlier coefficients. So there is exactly one such formal power series. $\square$

Now choose the sign. Since $\sqrt{1-4z}$ has constant term 1, the numerator

$$1+\sqrt{1-4z}$$

has constant term 2, and dividing by $2z$ would produce a z^{-1} term, so it cannot define a formal power series. By contrast,

$$1-\sqrt{1-4z}$$

has constant term 0, so division by $2z$ can still produce an honest power series. Therefore

$$C(z)=\frac{1-\sqrt{1-4z}}{2z}.$$

Theorem 1.6 (Generalized binomial theorem)**.** *For any rational number α,*

$$(1+w)^{\alpha}=\sum_{n\geq 0}\binom{\alpha}{n}w^n$$

as a formal power series, where

$$\binom{\alpha}{n}=\frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-n+1)}{n!},\qquad \binom{\alpha}{0}=1.$$

This is a formal-power-series statement; no analytic convergence is being used here. We will use the theorem as a black box in this chapter and discuss analytic convergence only in Chapter 2.

Applying it with $\alpha=\frac{1}{2}$ and $w=-4z$ gives

$$\sqrt{1-4z}=(1-4z)^{1/2}=\sum_{n\geq 0}\binom{1/2}{n}(-4z)^n.$$

We next simplify the coefficients. For $n\geq 1$,

$$\binom{1/2}{n}=\frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)\cdots(\frac{1}{2}-n+1)}{n!}=\frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})\cdots(\frac{3-2n}{2})}{n!}.$$

There are n factors of 2 in the denominator, so a factor of 2^{-n} appears automatically. The remaining odd factors contribute a sign $(-1)^{n-1}$ and the product

$$1\cdot 1\cdot 3\cdot 5\cdots(2n-3)=(2n-3)!!.$$

To convert the double factorial into an ordinary factorial, note that

$$(2m)!=(1\cdot 3\cdot 5\cdots(2m-1))(2\cdot 4\cdot 6\cdots 2m)=(2m-1)!!\cdot 2^m m!.$$

Hence

$$(2m-1)!!=\frac{(2m)!}{2^m m!}.$$

Setting $m=n-1$ gives

$$(2n-3)!!=\frac{(2n-2)!}{2^{n-1}(n-1)!}.$$

Substituting this back into the formula for $\binom{1/2}{n}$, we obtain

$$\binom{1/2}{n}=\frac{(-1)^{n-1}}{2^n n!}(2n-3)!!=\frac{(-1)^{n-1}}{2^n n!}\cdot\frac{(2n-2)!}{2^{n-1}(n-1)!}=\frac{(-1)^{n-1}(2n-2)!}{2^n n!\,2^{n-1}(n-1)!}=\frac{(-1)^{n-1}}{2^{2n-1}n}\binom{2n-2}{n-1}\qquad (n\geq 1).$$

Now substitute the binomial expansion into $C(z)$:

$$C(z) = \frac{1}{2z}\Big(1 - \sum_{n\geq 0}\binom{1/2}{n}(-4z)^n\Big) = -\frac{1}{2z}\sum_{n\geq 1}\binom{1/2}{n}(-4z)^n.$$

The coefficient of z^n is therefore

$$[z^n]C(z) = -\frac{1}{2}\binom{1/2}{n+1}(-4)^{n+1}.$$

Using the formula above with $n+1$ in place of n,

$$[z^n]C(z) = -\frac{1}{2}\cdot\frac{(-1)^n}{2^{2n+1}(n+1)}\binom{2n}{n}\cdot(-1)^{n+1}4^{n+1}.$$

Now

$$(-1)^n(-1)^{n+1} = -1,$$

so the two minus signs cancel:

$$[z^n]C(z) = \frac{4^{n+1}}{2^{2n+2}(n+1)}\binom{2n}{n} = \frac{1}{n+1}\binom{2n}{n}.$$

Thus

$$C_n = \frac{1}{n+1}\binom{2n}{n},$$

the standard closed form for the n-th Catalan number.

Remark 1.7. The formula for $C(z)$ shows that this generating function is *algebraic*: it satisfies a polynomial equation over $\mathbb{Q}(z)$. What we have *not* proved yet is the full analytic story about its singularities and the asymptotic behavior of C_n. Chapter 5 will show that the square-root singularity at $z = 1/4$ leads to

$$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}},$$

and will state carefully the hypotheses under which an $n^{-3/2}$ factor really appears. Unambiguous context-free grammars form an important later example, but that connection is only a preview at this stage.

Ordinary versus exponential generating functions

The generating functions above are *ordinary generating functions* (OGFs): the sequence element a_n is literally the coefficient of z^n. For *unlabeled* combinatorial structures, where we care only about size and shape, OGFs are the natural choice.

But many problems in combinatorics — and later, in probabilistic models of strings — involve *labeled* structures, where the n atoms in a structure carry distinct labels from $\{1, \ldots, n\}$ and the assignment of labels matters. The canonical example is permutations: the number of permutations of n distinct elements is $n!$. Combining two labeled structures requires choosing which labels go where. If an A-structure uses k labels and a B-structure uses the remaining $n-k$, then one must first choose the k labels for the A-part. For example, when $n = 4$ and $k = 2$, there are $\binom{4}{2} = 6$ ways to decide which two labels go to the A-structure. In general the total count is

$$\sum_{k=0}^{n}\binom{n}{k}a_k b_{n-k}.$$

This is called *binomial convolution*. The extra binomial coefficient is exactly the cost of distributing labels.

Definition 1.8. The *exponential generating function* (EGF) of a sequence $(a_n)_{n\geq 0}$ is

$$\hat{A}(z) \;=\; \sum_{n\geq 0} a_n \frac{z^n}{n!}.$$

The key algebraic fact is that binomial convolution corresponds to ordinary multiplication of EGFs. Multiply the series and collect all pairs of terms whose total degree is n:

$$[z^n](\hat{A}\cdot\hat{B}) = \sum_{k=0}^{n} \frac{a_k}{k!}\frac{b_{n-k}}{(n-k)!} = \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k} a_k b_{n-k},$$

since $\binom{n}{k} = n!/(k!(n-k)!)$. So

$$n!\,[z^n](\hat{A}\cdot\hat{B})$$

is exactly the binomial convolution, which is why labeled structures prefer EGFs.

We write $[z^n/n!]\hat{A}(z)$ as shorthand for $n!\,[z^n]\hat{A}(z) = a_n$, that is, the coefficient of $z^n/n!$ in the EGF.

Example 1.9. Consider the sequence $a_n = n!$, counting permutations. Its OGF is

$$\sum_{n\geq 0} n!\, z^n.$$

For any fixed $z \neq 0$, the ratio of successive terms has absolute value

$$\frac{(n+1)!|z|^{n+1}}{n!|z|^n} = (n+1)|z| \to \infty,$$

so this series does not converge analytically anywhere except at $z = 0$. Its EGF, on the other hand, is

$$\hat{S}(z) = \sum_{n\geq 0} n!\frac{z^n}{n!} = \sum_{n\geq 0} z^n = \frac{1}{1-z},$$

by the geometric-series lemma. The coefficient $[z^n/n!]\hat{S}(z) = 1$ means that there is one underlying *linear-order shape* of size n; the $n!$ different labelings of that shape are precisely the $n!$ permutations.

The distinction between OGF and EGF is not merely aesthetic: using an OGF for a labeled structure or an EGF for an unlabeled one leads to incorrect functional equations. The *symbolic method*, developed in Chapter 3, makes this assignment systematic. There we will define the constructions SEQ, SET, CYC, and MSET and translate them into generating-function operations.

Road map

This chapter has introduced generating functions as formal algebraic objects. The exact results proved here were algebraic: recurrences became functional equations, and those equations led to closed forms. The asymptotic remarks were only previews. The next chapter treats z as a complex variable and develops the analytic tools needed to justify those previews. One reason complex numbers matter is that singularities away from the real axis can still control the coefficients of a power series. Chapter 3 then introduces the symbolic method, turning combinatorial grammar rules directly into generating-function equations. Chapters 4 and 5 return to the Fibonacci and Catalan examples and prove carefully how poles and square-root singularities control coefficient growth. Later, when we connect these ideas to language models, we will also need to distinguish carefully between counting generating functions and probability-weighted generating functions rather than treating them as the same object.

Exercise 1.10. Find the ordinary generating function for the sequence $a_n = 3^n$ (ternary strings of length n) by both methods: (a) summing the geometric series directly, and (b) applying the recurrence-to-GF recipe to the recurrence $a_n = 3a_{n-1}$, $a_0 = 1$.

Exercise 1.11. Use the generating function method to solve the recurrence $a_n = 2a_{n-1} + 1$ for $n \geq 1$, with $a_0 = 0$. (Hint: multiply by z^n and sum. You will need to handle $\sum_{n\geq 1} z^n = z/(1-z)$ for the constant term on the right side. Use partial fractions on the resulting $A(z)$ to find a closed form for a_n.)

Exercise 1.12. Verify the generalized binomial coefficient formula $\binom{1/2}{n} = (-1)^{n-1}/(2^{2n-1}n) \cdot \binom{2n-2}{n-1}$ for $n = 1, 2, 3, 4$ by computing both sides directly.

Further Reading

- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19] — The definitive reference for analytic combinatorics. Part A (Chapters I–II) covers formal power series, OGFs, EGFs, and the translation dictionary between combinatorial constructions and generating-function operations with full proofs and extensive examples.

- **A. Nijenhuis and H. S. Wilf**, *Combinatorial Algorithms* (Academic Press, 2nd ed., 1978) [42] — Focuses on algorithms for random generation and enumeration of combinatorial structures; includes generating-function derivations as part of the algorithmic analysis, with a hands-on computational style that complements the more theoretical approach of the present chapter.

- **H. S. Wilf**, *generatingfunctionology* (2nd ed., Academic Press, 1994; freely available online) — A short, readable introduction to generating functions aimed at undergraduates. Covers OGFs and EGFs through worked problems; an excellent companion for anyone finding the present chapter's pace too quick.

- **R. P. Stanley**, *Enumerative Combinatorics*, Vol. 1 (Cambridge, 1997) — Chapter 1 gives a thorough algebraic treatment of formal power series and the ring $\mathbb{Q}[[z]]$; Chapter 4 covers rational and algebraic generating functions in depth appropriate for graduate study.

- **D. E. Knuth**, *The Art of Computer Programming*, Vol. 1, §1.2.9 — A classic account of generating functions as a tool for solving recurrences, with worked examples drawn from algorithm analysis; especially useful for readers interested in the connection to computer science.

Chapter 2

Just Enough Complex Analysis

Chapter 1 treated z as a formal symbol, manipulated according to algebraic rules with no thought of convergence or magnitude. The real power of analytic combinatorics comes from promoting z to a genuine complex variable and studying how the coefficients $[z^n]A(z)$ are controlled by the analytic behavior of the resulting function $A(z)$. This chapter develops the minimum complex analysis needed for that move.

Some of the material below will be proved carefully: the root test, the Cauchy–Hadamard formula for the radius of convergence, the coefficient-extraction formula on a circle, and the basic exponential bound that follows from it. Other facts are genuinely deeper theorems of complex analysis. When that happens, I will say so explicitly and use them as black boxes rather than pretending they are obvious. The standard treatment of all topics below is [19], Part A.

2.1 Complex Numbers and Convergence

Recall that

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}, \qquad i^2 = -1.$$

The *modulus* of $z = x + iy$ is

$$|z| = \sqrt{x^2 + y^2}.$$

If $z \neq 0$, we can also describe it by an angle. Angles are only defined modulo 2π, and here we choose the *principal argument* $\arg z \in (-\pi, \pi]$.

Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

is a standard identity, derivable from the power-series definitions of e^x, $\sin x$, and $\cos x$. It allows every nonzero complex number to be written in *polar form*

$$z = re^{i\theta},$$

where $r = |z| > 0$ and $\theta = \arg z$. Multiplication becomes especially transparent in this form:

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)},$$

because exponentials add exponents. So moduli multiply and arguments add.

Convergence in $\mathbb{C}$ is measured by the modulus: a sequence (z_k) converges to L if $|z_k - L| \to 0$. A series $\sum_k w_k$ *converges absolutely* if $\sum_k |w_k| < \infty$.

Proposition 2.1. *If $\sum_k |w_k|$ converges, then $\sum_k w_k$ converges in $\mathbb{C}$.*

Proof. Write $w_k = u_k + iv_k$ with $u_k, v_k \in \mathbb{R}$. Then

$$|u_k| \le |w_k|, \qquad |v_k| \le |w_k|.$$

So $\sum_k u_k$ and $\sum_k v_k$ both converge absolutely as real series. Hence

$$\sum_k w_k = \sum_k u_k + i \sum_k v_k$$

converges in $\mathbb{C}$. □

The geometric series is the first analytic example. For every integer $N \ge 0$,

$$\sum_{k=0}^{N} w^k = \frac{1 - w^{N+1}}{1 - w}, \qquad w \ne 1.$$

If $|w| < 1$, then $|w|^{N+1} \to 0$, so the finite sums converge to

$$\sum_{k \ge 0} w^k = \frac{1}{1 - w}.$$

This is the same identity as in ordinary real algebra; the only extra ingredient is that the modulus tells us exactly when $w^{N+1} \to 0$.

2.2 Power Series and Radius of Convergence

Let

$$A(z) = \sum_{n \ge 0} a_n z^n$$

be a power series with coefficients in $\mathbb{C}$. Treating z as a complex variable, the set of z for which the series converges absolutely is always a disk centered at the origin. What happens on the boundary of that disk can be subtle: some series converge at certain boundary points and diverge at others.

Definition 2.2. For a real sequence (x_n), the *limit superior* $\limsup_{n\to\infty} x_n$ is the largest subsequential limit of the sequence. Equivalently, it is the eventual upper envelope of the tail values $x_N, x_{N+1}, x_{N+2}, \dots$ as $N \to \infty$.

Theorem 2.3 (Root test)**.** *Let $\sum_{n\ge 0} u_n$ be a series of complex numbers, and let*

$$\lambda = \limsup_{n\to\infty} |u_n|^{1/n}.$$

If $\lambda < 1$, then the series converges absolutely. If $\lambda > 1$, then $u_n \not\to 0$, so the series diverges. If $\lambda = 1$, the root test by itself does not decide what happens.

Proof. Suppose $\lambda < 1$. Choose a number r with

$$\lambda < r < 1.$$

By the definition of lim sup, all sufficiently large terms satisfy

$$|u_n|^{1/n} \le r.$$

So for all large n,

$$|u_n| \le r^n.$$

Since $\sum r^n$ is a convergent geometric series, comparison shows that $\sum |u_n|$ converges, so $\sum u_n$ converges absolutely.

Now suppose $\lambda > 1$. Choose r with

$$1 < r < \lambda.$$

Then infinitely many n satisfy $|u_n|^{1/n} \geq r$; otherwise all sufficiently large terms would have $|u_n|^{1/n} < r$, forcing $\limsup |u_n|^{1/n} \leq r$, a contradiction. For those infinitely many n,

$$|u_n| \geq r^n \geq 1.$$

So u_n cannot tend to 0, and the series diverges. □

Theorem 2.4 (Cauchy–Hadamard)**.** *Let*

$$A(z) = \sum_{n \geq 0} a_n z^n, \qquad L = \limsup_{n \to \infty} |a_n|^{1/n}.$$

Define the radius of convergence R *by*

$$\frac{1}{R} = L,$$

with the conventions $1/\infty = 0$ *and* $1/0 = \infty$*. Then the series converges absolutely for every* z *with* $|z| < R$ *and diverges for every* z *with* $|z| > R$*.*

Proof. Apply the root test to the series

$$\sum_{n \geq 0} a_n z^n.$$

Its nth term is $u_n = a_n z^n$, so

$$|u_n|^{1/n} = |a_n|^{1/n} |z|.$$

Therefore

$$\limsup_{n \to \infty} |u_n|^{1/n} = |z| \limsup_{n \to \infty} |a_n|^{1/n} = |z| L = \frac{|z|}{R}.$$

If $|z| < R$, then $|z|/R < 1$, so the root test gives absolute convergence. If $|z| > R$, then $|z|/R > 1$, so the terms do not tend to 0 and the series diverges. □

The extreme cases fit naturally into the same statement. If $R = \infty$, the series converges for every complex z; such a function is called *entire*. If $R = 0$, the series diverges for every nonzero z.

The combinatorial meaning of the radius is that it captures exponential growth. When $0 < R < \infty$, define

$$\theta(n) = a_n R^n.$$

Then

$$|\theta(n)|^{1/n} = |a_n|^{1/n} R,$$

so

$$\limsup_{n \to \infty} |\theta(n)|^{1/n} = R \limsup_{n \to \infty} |a_n|^{1/n} = 1.$$

Thus one may write

$$a_n = R^{-n} \theta(n),$$

where the factor $\theta(n)$ is *subexponential* in the sense above. Later chapters will show how the singularity type controls this finer factor.

Example 2.5. For

$$B(z) = \frac{1}{1-2z} = \sum_{n\geq 0} 2^n z^n,$$

we have $|2^n|^{1/n} = 2$, so $R = 1/2$. The growth rate $2^n = (1/2)^{-n}$ matches the rule R^{-n} exactly.

For the Fibonacci generating function

$$F(z) = \frac{z}{1-z-z^2},$$

the denominator vanishes at

$$z = \frac{-1 \pm \sqrt{5}}{2}.$$

These are approximately 0.618 and -1.618, so the root of smaller modulus is $1/\varphi \approx 0.618$, where $\varphi = (1+\sqrt{5})/2$. Thus the nearest singularity lies at distance $1/\varphi$ from the origin, so $R = 1/\varphi$. The later asymptotic statement

$$F_n \sim \frac{\varphi^n}{\sqrt{5}}$$

is a preview of the transfer machinery proved in Chapter 4, not something earned from the radius alone.

2.3 Analytic Functions

Definition 2.6. An *open set* $U \subseteq \mathbb{C}$ is a region such that every point of U is surrounded by a small disk lying completely inside U.

Definition 2.7. A function $f : U \to \mathbb{C}$ is *holomorphic* on U (also called *analytic*) if it is complex-differentiable at every point of U.

Theorem 2.8 (Basic equivalence, used without proof)**.** *For a function on an open set, the following conditions are equivalent:*

1. *it is complex-differentiable;*
2. *it satisfies the Cauchy–Riemann equations and is infinitely differentiable;*
3. *it is locally equal to a convergent power series.*

This equivalence is one of the deepest beginner-level theorems in complex analysis. We will use it as a black box. The surprising part is that complex differentiability is much more rigid than real differentiability: a real differentiable function need not equal any power series, but a holomorphic function automatically does.

Theorem 2.9 (Power-series calculus, used without proof)**.** *Inside its disk of convergence, a power series may be differentiated and integrated term by term, and the resulting power series has the same radius of convergence.*

As a consequence, if

$$A(z) = \sum_{n\geq 0} a_n z^n$$

has radius of convergence $R > 0$, then it defines a holomorphic function on the disk $|z| < R$.

Definition 2.10. Let f be holomorphic on a punctured disk

$$0 < |z - z_0| < \varepsilon.$$

If f cannot be extended holomorphically to the center point z_0, then z_0 is called a *singularity* of f. Any singularity of minimal modulus is called *dominant*; there may be several.

Theorem 2.11 (Nearest-singularity principle, used without proof)**.** *The radius of convergence of a power series centered at* 0 *equals the distance from* 0 *to the nearest singularity of the holomorphic function represented by the series.*

This theorem is one of the main bridges from complex analysis to asymptotics. It explains why complex singularities matter even when the coefficients are real and positive.

2.4 The Geometric Series as Prototype

The function

$$f(z) = \frac{1}{1-z}$$

is the model example. For $|z| < 1$ the geometric series converges absolutely, and its sum is

$$\frac{1}{1-z} = \sum_{n \geq 0} z^n.$$

The equality is valid only inside the disk $|z| < 1$ when interpreted analytically.

Definition 2.12. A function has a *simple pole* at $z = \rho$ if $(z - \rho)f(z)$ stays finite and nonzero as $z \to \rho$.

The function $1/(1-z)$ is holomorphic on $\mathbb{C} \setminus \{1\}$ and has a simple pole at $z = 1$. The same formula still makes sense at points such as $z = 2$ or $z = -3$, even though the power series $\sum z^n$ diverges there. Extending a holomorphic function beyond its original disk of convergence in this way is called *analytic continuation*. The uniqueness of analytic continuation is another major theorem that we will not prove here.

This example already illustrates the nearest-singularity principle: the singularity at $z = 1$ sits exactly one unit from the origin, so the radius of convergence is 1.

2.5 Recovering Coefficients from a Circle

To extract coefficients analytically, we integrate around circles. If g is continuous on $|z| = r$, we define

$$\oint_{|z|=r} g(z)\, dz$$

to mean the ordinary real integral obtained by parameterizing the circle as

$$z = re^{it}, \qquad 0 \leq t \leq 2\pi,$$

so that

$$dz = ire^{it}\, dt.$$

Proposition 2.13 (Coefficient formula on a circle)**.** *Let*

$$f(z) = \sum_{m \geq 0} a_m z^m$$

have radius of convergence R, and choose $0 < r < R$. Then for every $n \geq 0$,

$$a_n = [z^n]f(z) = \frac{1}{2\pi i} \oint_{|z|=r} \frac{f(z)}{z^{n+1}}\, dz.$$

Proof. Because $r < R$, the series for $f(re^{it})$ converges absolutely and uniformly in t, so we may integrate term by term. Parameterizing the circle gives

$$\frac{1}{2\pi i}\oint_{|z|=r}\frac{f(z)}{z^{n+1}}\,dz = \frac{1}{2\pi}\int_0^{2\pi} f(re^{it})\,r^{-n}e^{-int}\,dt.$$

Now expand

$$f(re^{it}) = \sum_{m\geq 0} a_m r^m e^{imt}.$$

Substituting,

$$\frac{1}{2\pi}\int_0^{2\pi}\sum_{m\geq 0} a_m r^{m-n}e^{i(m-n)t}\,dt = \sum_{m\geq 0} a_m r^{m-n}\cdot\frac{1}{2\pi}\int_0^{2\pi} e^{i(m-n)t}\,dt.$$

The integral on the right is 1 when $m = n$ and 0 otherwise, so only one term survives:

$$\frac{1}{2\pi i}\oint_{|z|=r}\frac{f(z)}{z^{n+1}}\,dz = a_n.$$

□

This is the most concrete version of coefficient extraction. A deeper theorem, Cauchy's integral formula, extends the same idea to arbitrary holomorphic functions and derivatives; we will use that broader result later as a black box.

For estimates we also need the size of f on the circle. Because f is continuous and the circle $|z| = r$ is compact, the maximum

$$M(r) = \max_{|z|=r}|f(z)|$$

exists.

Proposition 2.14 (Circle bound)**.** *With the same hypotheses as above,*

$$|a_n| \leq \frac{M(r)}{r^n}.$$

Proof. Starting from the coefficient formula and parameterizing the circle,

$$|a_n| \leq \frac{1}{2\pi}\int_0^{2\pi}|f(re^{it})|\,r^{-n}\,dt \leq \frac{1}{2\pi}\int_0^{2\pi} M(r)\,r^{-n}\,dt = \frac{M(r)}{r^n}.$$

□

This gives an exponential upper bound. If we can choose a larger circle, the factor r^{-n} improves. A deeper theorem of complex analysis says that, as long as we stay in a region where f is holomorphic, such contour integrals may be deformed without changing their value. We will use that theorem as a black box. The important caution is this: the bound above explains why nearby singularities control the *best exponential rate*, but it does *not* by itself determine the exact asymptotic formula. For that, one needs local analysis near the singularity.

2.6 Prototype Singularities

This section is a catalogue of model cases, not the full transfer theorem. We compute the coefficients for a few prototype singular behaviors and use them as guides. Chapter 4 will state the general theorem that turns these prototypes into systematic asymptotics.

For generating functions with nonnegative coefficients, a classical theorem of Pringsheim says that the point $z = R$ on the positive real axis is always a singularity. We will use that theorem as a black box when needed. There may still be additional dominant singularities of the same modulus elsewhere in the complex plane.

Before discussing square roots and logarithms, one warning is needed. In complex analysis, $\sqrt{\cdot}$ and log are not globally single-valued functions. To make sense of them analytically, one chooses a *branch*. Near the origin we use the principal branch, which is holomorphic on a slit domain obtained by removing a ray from the singular point. Since our power-series expansions are centered at 0, this local branch is the one that matters.

Poles. For a positive integer k,

$$f(z) = \frac{1}{(1 - z/\rho)^k}$$

has a pole of order k at $z = \rho$. Using the generalized binomial theorem from Chapter 1 with $\alpha = -k$ and $w = -z/\rho$, one obtains

$$\frac{1}{(1 - z/\rho)^k} = \sum_{n \geq 0} \binom{n + k - 1}{k - 1} \rho^{-n} z^n.$$

Thus

$$[z^n] f(z) = \binom{n + k - 1}{k - 1} \rho^{-n}.$$

The binomial coefficient is a polynomial in n of degree $k - 1$. Indeed,

$$\binom{n + k - 1}{k - 1} = \frac{(n + k - 1)(n + k - 2) \cdots (n + 1)}{(k - 1)!},$$

so dividing by n^{k-1} gives

$$\frac{\binom{n+k-1}{k-1}}{n^{k-1}} = \frac{(1 + \frac{k-1}{n})(1 + \frac{k-2}{n}) \cdots (1 + \frac{1}{n})}{(k - 1)!} \to \frac{1}{(k - 1)!}.$$

Hence

$$\binom{n + k - 1}{k - 1} \sim \frac{n^{k-1}}{(k - 1)!}.$$

So a simple pole contributes a pure exponential ρ^{-n}, a double pole contributes about $n\rho^{-n}$, a triple pole contributes about $\frac{n^2}{2}\rho^{-n}$, and so on.

Square-root branch points. Consider the prototype

$$f(z) = \sqrt{1 - z/\rho} = (1 - z/\rho)^{1/2}.$$

By the generalized binomial theorem,

$$f(z) = \sum_{n \geq 0} \binom{1/2}{n} (-z/\rho)^n.$$

Chapter 1 proved that

$$\binom{1/2}{n} = \frac{(-1)^{n-1}}{2^{2n-1} n} \binom{2n - 2}{n - 1} \qquad (n \geq 1),$$

so for $n \geq 1$,

$$[z^n]f(z) = -\frac{1}{2^{2n-1}n}\binom{2n-2}{n-1}\rho^{-n}.$$

Now apply Stirling's theorem, which we use here as a standard asymptotic fact:

$$m! \sim \sqrt{2\pi m}\left(\frac{m}{e}\right)^m.$$

With $m = n - 1$,

$$\binom{2m}{m} = \frac{(2m)!}{(m!)^2} \sim \frac{\sqrt{4\pi m}(2m/e)^{2m}}{2\pi m(m/e)^{2m}} = \frac{4^m}{\sqrt{\pi m}}.$$

Therefore

$$\binom{2n-2}{n-1} \sim \frac{4^{n-1}}{\sqrt{\pi n}},$$

and hence

$$[z^n]\sqrt{1 - z/\rho} \sim -\frac{1}{2\sqrt{\pi}}\, n^{-3/2}\rho^{-n}.$$

This is a *prototype* calculation: it suggests the $n^{-3/2}$ behavior that later chapters will prove for a broad class of genuine square-root singularities.

Logarithmic singularities. Inside $|w| < 1$, the geometric series may be integrated term by term:

$$\frac{1}{1-w} = \sum_{n\geq 0} w^n.$$

Integrating from 0 to w gives

$$-\log(1-w) = \sum_{n\geq 1} \frac{w^n}{n},$$

where we use the branch of the logarithm that is real at $w = 0$. Substituting $w = z/\rho$ yields

$$\log \frac{1}{1 - z/\rho} = \sum_{n\geq 1} \frac{\rho^{-n}}{n} z^n,$$

so

$$[z^n] \log \frac{1}{1 - z/\rho} = \frac{\rho^{-n}}{n}.$$

The correction factor is now $1/n$. Logarithms arise naturally in labelled structures involving cycles; Chapter 3 will define the symbolic operators SET and CYC that lead to such terms.

Essential singularities. Functions such as

$$e^{1/(1-z)}$$

have an essential singularity at $z = 1$. We mention the term only for orientation: its precise definition uses Laurent series, which we do not develop here. The point is that its behavior near the singularity is much more irregular than the clean pole and branch-point cases above, so the elementary catalogue is no longer enough.

2.7 The Big Picture

The moral of this chapter is simple, but it must be stated carefully.

1. The radius of convergence tells us the exponential scale of the coefficients.
2. Nearby singularities explain why that exponential scale is what it is.

3. Model singularities suggest what the subexponential correction should look like.

What we do *not* yet have is the general theorem turning the third step into a rigorous automatic procedure. That theorem is the transfer theorem of Flajolet and Odlyzko [18], developed in Chapter 4 and explained in detail in [19].

So the practical program is:

1. derive a usable generating function by algebra or combinatorial structure;
2. locate the dominant singularities on the boundary of the disk of convergence;
3. identify the local singular behavior near those points;
4. only then apply an appropriate transfer theorem.

Even this summary comes with caveats: there may be several dominant singularities, branch choices may matter, and upper bounds alone do not yield exact asymptotics. Still, the basic philosophy is already visible: singularities are not annoyances to be avoided but the source of coefficient information.

Exercise 2.15. Find the radius of convergence of

$$\sum_{n\geq 1} \frac{z^n}{n^2}.$$

The series also converges at $z = 1$. Explain why convergence at a boundary point does not settle what happens to analytic continuation beyond the disk $|z| < 1$.

Exercise 2.16. Show that

$$\frac{1}{(1-z)^3} = \sum_{n\geq 0} \binom{n+2}{2} z^n$$

by applying the generalized binomial theorem from Chapter 1 with $\alpha = -3$ and $w = -z$. Then verify that

$$\binom{n+2}{2} \sim \frac{n^2}{2},$$

consistent with the pole catalogue above.

Exercise 2.17. The series

$$\sum_{n\geq 0} n!\, z^n$$

has radius of convergence $R = 0$; verify this using Cauchy–Hadamard. Why is this an example where the formal power series from Chapter 1 is still a perfectly valid algebraic object, but does not define an analytic function on any nontrivial disk around 0?

Further Reading

- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19], Part B — Chapters IV–VI develop the complex-analytic foundations used throughout this book: analytic continuation, the transfer theorem, and singularity analysis in full generality. Reading Part B alongside the present chapter is strongly recommended.
- **P. Flajolet and A. M. Odlyzko**, "Singularity analysis of generating functions," *SIAM J. Discrete Math.* (1990) [18] — The original paper introducing the transfer theorem and Δ-domains; more concise than the textbook treatment and useful for seeing the theorem in its original form.

- **L. V. Ahlfors**, *Complex Analysis* (3rd ed., McGraw-Hill, 1979) — The standard graduate-level reference for classical complex analysis. Chapters 1–4 cover all the material in the present chapter (convergence, analyticity, Cauchy's theorem, singularity classification) with full proofs; Chapter 5 introduces residues and contour integration.
- **E. C. Titchmarsh**, *The Theory of Functions* (2nd ed., Oxford, 1939) — A more classical treatment, particularly strong on power series (Chapter 1) and contour integration and residues (Chapter 3); useful as a second reference for the Cauchy–Hadamard theorem and the classification of isolated singularities.
- **P. Henrici**, *Applied and Computational Complex Analysis*, Vol. 1 (Wiley, 1974) — Covers power series and analytic functions with an emphasis on effective computation and coefficient extraction; bridges the gap between pure complex analysis and the algorithmic side of generating functions.

Chapter 3

The Symbolic Method

The central ambition of analytic combinatorics is to pass from a structural description of a class of objects to an equation satisfied by its generating function, and then — via the analytic machinery of Chapter 2 — to asymptotic counting formulas. The *symbolic method* is the first half of that program. It is a dictionary translating set-theoretic constructions on combinatorial classes into algebraic operations on generating functions.

There is one important warning at the outset. A symbolic specification is not just a vague resemblance between formulas and objects. It must describe each object *exactly once*. If the same object can be built in two different ways, the generating function will overcount. So the symbolic method works best when a combinatorial decomposition is really a bijection between objects and structured tuples, sequences, or sets of simpler objects. The standard reference for the material in this chapter is [19], Part A.

3.1 Combinatorial Classes

Definition 3.1. A *combinatorial class* is a pair $(\mathcal{C}, |\cdot|)$ where $\mathcal{C}$ is a set and

$$|\cdot| : \mathcal{C} \to \mathbb{Z}_{\geq 0}$$

is a *size function* such that, for every $n \geq 0$, the level set

$$\mathcal{C}_n = \{\gamma \in \mathcal{C} : |\gamma| = n\}$$

is finite.

Write $c_n = |\mathcal{C}_n|$. The *ordinary generating function* (OGF) of $\mathcal{C}$ is

$$C(z) = \sum_{\gamma \in \mathcal{C}} z^{|\gamma|} = \sum_{n \geq 0} c_n z^n.$$

The second equality comes from grouping together all objects of the same size n: there are exactly c_n of them. The finiteness of each level set is what makes this a valid formal power series, because the coefficient of z^n is the finite number c_n.

Three archetypal classes will recur throughout the book.

- Binary words over $\{a, b\}$, with size equal to word length.
- Plane trees, where the children of each node are ordered from left to right.
- Permutations, which are naturally *labeled* objects and therefore call for exponential generating functions rather than ordinary ones.

For example, the plane trees of sizes 1 and 2 look like a single root, and a root with one child; and a size-3 permutation is something like $(2, 3, 1)$, where the labels matter.

Two tiny classes serve as universal building blocks.

Definition 3.2. The *neutral class* $\mathcal{E} = \{\epsilon\}$ contains one object of size 0, so its OGF is $E(z) = 1$. The *atomic class* $\mathcal{Z} = \{\bullet\}$ contains one object of size 1, so its OGF is $Z(z) = z$.

These are the combinatorial analogues of the constants 1 and z. For instance, a word of length 3 built from three atoms has the symbolic form

$$\mathcal{Z} \times \mathcal{Z} \times \mathcal{Z}.$$

The rest of the symbolic method consists of combining $\mathcal{E}$ and $\mathcal{Z}$ through a small family of constructors.

3.2 The Translation Dictionary for OGFs

Proposition 3.3 (Symbolic transfer, unlabeled case)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be combinatorial classes with OGFs $A(z)$ and $B(z)$. Then the following constructions translate into the following OGF operations.*

1. ***Disjoint union.*** *If every object is tagged as coming from the $\mathcal{A}$ side or the $\mathcal{B}$ side, then*

$$\mathcal{A} + \mathcal{B} \longleftrightarrow A(z) + B(z).$$

2. ***Cartesian product.*** *If objects are ordered pairs (a, b) with size $|(a, b)| = |a| + |b|$, then*

$$\mathcal{A} \times \mathcal{B} \longleftrightarrow A(z)B(z).$$

3. ***Sequence.*** *If $\mathcal{A}$ contains no object of size 0, then finite sequences of $\mathcal{A}$-objects satisfy*

$$\mathrm{SEQ}(\mathcal{A}) \longleftrightarrow \frac{1}{1 - A(z)}.$$

The word *disjoint* matters. If the same underlying object could belong to both classes, it would be counted twice. When two classes are not literally disjoint, one uses a *tagged disjoint union*: an object is paired with a label saying which side it came from.

Proof. For disjoint union, every object belongs to exactly one summand, so the OGF simply splits:

$$\sum_{\gamma \in \mathcal{A}+\mathcal{B}} z^{|\gamma|} = \sum_{\alpha \in \mathcal{A}} z^{|\alpha|} + \sum_{\beta \in \mathcal{B}} z^{|\beta|} = A(z) + B(z).$$

For Cartesian products, we use the additive size rule:

$$\sum_{(a,b) \in \mathcal{A} \times \mathcal{B}} z^{|(a,b)|} = \sum_{(a,b)} z^{|a|+|b|} = \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} z^{|a|} z^{|b|} = A(z)B(z).$$

At the coefficient level,

$$[z^n]A(z)B(z) = \sum_{k=0}^{n} a_k b_{n-k},$$

which has the expected combinatorial meaning: to build a size-n pair, choose a size-k object from $\mathcal{A}$ and a size-$(n-k)$ object from $\mathcal{B}$. Only finitely many splits $k + (n - k) = n$ are possible.

For sequences, a finite list of $\mathcal{A}$-objects can have length $0, 1, 2, \ldots$, so as a class,

$$\mathrm{SEQ}(\mathcal{A}) = \mathcal{E} + \mathcal{A} + \mathcal{A}^2 + \mathcal{A}^3 + \cdots,$$

where $\mathcal{A}^k$ means a k-fold Cartesian product and therefore represents ordered sequences of length k. The union is disjoint because the sequence length is part of the data: a 2-term sequence is not the same kind of object as a 3-term sequence.

Why do we exclude size-0 objects from $\mathcal{A}$? Because otherwise local finiteness can fail. If $\mathcal{A}$ had even one size-0 object, then a size-1 sequence could be padded with arbitrarily many zero-size entries, producing infinitely many distinct sequences of total size 1. By contrast, if every object of $\mathcal{A}$ has size at least 1, then a sequence of total size n can have length at most n. So only finitely many sequence lengths contribute to the coefficient of z^n.

Now apply the union and product rules:

$$\mathrm{SEQ}(\mathcal{A}) \longleftrightarrow 1 + A(z) + A(z)^2 + A(z)^3 + \cdots.$$

Because $\mathcal{A}$ has no size-0 object, the constant term of $A(z)$ is 0. So Chapter 1's formal geometric-series identity applies and gives

$$1 + A(z) + A(z)^2 + A(z)^3 + \cdots = \frac{1}{1 - A(z)}.$$

Equivalently, the coefficient of z^n on the left counts all sequences of total size n, and only finitely many powers $A(z)^k$ can contribute because $k \leq n$. □

Remark 3.4. The constructor

$$\mathrm{SEQ}_{\geq k}(\mathcal{A})$$

means sequences of length at least k. Structurally, that is

$$\mathrm{SEQ}_{\geq k}(\mathcal{A}) = \mathcal{A}^k \times \mathrm{SEQ}(\mathcal{A}),$$

because one first chooses the required first k entries and then any additional tail. Therefore its OGF is

$$\frac{A(z)^k}{1 - A(z)}.$$

In particular,

$$\mathrm{SEQ}_{\geq 1}(\mathcal{A}) \longleftrightarrow \frac{A(z)}{1 - A(z)}.$$

3.3 Marking Variables and Bivariate Generating Functions

A second variable can track a statistic at the same time as size. Let $\chi : \mathcal{C} \to \mathbb{Z}_{\geq 0}$ be a parameter, and define the bivariate OGF

$$C(z, u) = \sum_{\gamma \in \mathcal{C}} z^{|\gamma|} u^{\chi(\gamma)} = \sum_{n,k \geq 0} c_{n,k} z^n u^k,$$

where $c_{n,k}$ counts objects of size n with statistic value k. The ordinary OGF is recovered by setting $u = 1$.

To speak about an *expected value* at size n, we must put a probability distribution on the finite set $\mathcal{C}_n$. In this chapter we use the *uniform distribution* on $\mathcal{C}_n$. Then

$$\mathbf{E}[\chi \mid |\gamma| = n] = \frac{[z^n]\partial_u C(z, u)\big|_{u=1}}{[z^n]C(z, 1)}.$$

Indeed,

$$\partial_u u^{\chi(\gamma)} = \chi(\gamma) u^{\chi(\gamma)-1},$$

so after setting $u = 1$,

$$[z^n]\partial_u C(z,u)\big|_{u=1} = \sum_{|\gamma|=n} \chi(\gamma).$$

Dividing by $|\mathcal{C}_n| = [z^n]C(z,1)$ gives the uniform mean.

A related construction is *pointing*. Suppose each size-n object is built from exactly n atomic positions counted by the variable z. Then the pointed class $\mathcal{C}^\bullet$ is obtained by choosing one distinguished atom. Every size-n object therefore gives rise to exactly n pointed objects, so

$$C^\bullet(z) = zC'(z).$$

For example, a 4-letter word has exactly 4 possible pointings, one for each letter position.

3.4 Labeled Classes and EGFs

For labeled structures, the n atoms carry distinct labels from $\{1, \ldots, n\}$. A labeled object is therefore an underlying shape together with a bijection from its n atomic positions to the label set $[n]$. The natural generating function is the *exponential generating function* (EGF)

$$\hat{C}(z) = \sum_{n\geq 0} c_n \frac{z^n}{n!}.$$

The factor $1/n!$ removes the arbitrary naming of the labels and leaves only the structural count. For example, a labeled object of size 3 uses the labels $\{1, 2, 3\}$ exactly once each, assigned to its three atomic positions.

Disjoint union works exactly as before. The crucial new rule is the labeled product. Suppose $\hat{\mathcal{A}}$ and $\hat{\mathcal{B}}$ are labeled classes with EGFs $\hat{A}(z)$ and $\hat{B}(z)$. To build a size-n product object, choose k of the n labels for the $\hat{\mathcal{A}}$ part and give the remaining $n-k$ labels to the $\hat{\mathcal{B}}$ part. Therefore

$$n![z^n]\,\hat{A}(z)\hat{B}(z) = \sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k},$$

which is exactly the number of labeled product objects of size n. This is the same binomial-convolution calculation from Chapter 1, now interpreted combinatorially.

If a labeled class has no size-0 object, then the sequence rule again has the same formal shape:

$$\mathrm{SEQ}(\hat{\mathcal{A}}) \longleftrightarrow \frac{1}{1-\hat{A}(z)}.$$

The underlying reason is that an ordered list of labeled components still has no symmetry: once the list order is fixed, there is nothing to divide out.

The next labeled constructor is *set*. Here the order of the components is forgotten.

Proposition 3.5 (Set constructor in the labeled setting)**.** *If $\hat{\mathcal{A}}$ has no size-0 object, then*

$$\widehat{\mathrm{SET}(\mathcal{A})}(z) = \exp(\hat{A}(z)).$$

Proof. Let $\mathrm{SET}_k(\mathcal{A})$ denote sets of exactly k labeled $\mathcal{A}$-objects. First pretend the k components are ordered. Then the EGF is $\hat{A}(z)^k$, because we are taking a k-fold labeled product. But a set forgets the order of its k components, and there are exactly $k!$ ways to order them. So

$$\widehat{\mathrm{SET}_k(\mathcal{A})}(z) = \frac{\hat{A}(z)^k}{k!}.$$

Now sum over all $k \geq 0$:

$$\widehat{\mathrm{SET}(\mathcal{A})}(z) = \sum_{k \geq 0} \frac{\hat{A}(z)^k}{k!} = \exp(\hat{A}(z)).$$

The exclusion of size-0 objects is again needed to prevent infinitely many zero-size components from appearing. □

As a tiny example, the class of finite sets of labeled atoms is $\mathrm{SET}(\mathcal{Z})$, so its EGF is

$$\exp(z).$$

The coefficient of $z^n/n!$ is 1, which matches the fact that on a given label set $[n]$ there is exactly one n-element set.

Cyclic structures are subtler. The standard labeled rule is

$$\widehat{\mathrm{CYC}(\mathcal{A})}(z) = \log \frac{1}{1 - \hat{A}(z)}.$$

We record this formula here because it is important and useful, but we will treat it as a black-box rule in this chapter. A full proof belongs to the theory of combinatorial species; see [6] or [19].

3.5 Worked Examples

Example 3.6 (Binary words)**.** Let $\mathcal{A} = \{a\}$ and $\mathcal{B} = \{b\}$, each a copy of $\mathcal{Z}$. A binary word is a finite sequence of one-letter choices, so

$$\mathcal{W} = \mathrm{SEQ}(\mathcal{A} + \mathcal{B}).$$

Since each one-letter choice contributes z, the class $\mathcal{A} + \mathcal{B}$ has OGF

$$z + z = 2z.$$

Therefore

$$W(z) = \frac{1}{1 - 2z} = \sum_{n \geq 0} 2^n z^n.$$

So there are 2^n binary words of length n.

Example 3.7 (Plane trees and Catalan numbers)**.** A plane tree consists of a root together with an ordered list of plane trees attached to that root as children. The children are ordered, so a sequence is exactly the right constructor. If size means number of nodes, then the root contributes one atom and the child list contributes a sequence of smaller trees:

$$\mathcal{T} = \mathcal{Z} \times \mathrm{SEQ}(\mathcal{T}).$$

Proposition 3.3 therefore gives

$$T(z) = \frac{z}{1 - T(z)}.$$

Rearranging,

$$T(z)^2 - T(z) + z = 0.$$

Solving the quadratic yields

$$T(z) = \frac{1 \pm \sqrt{1 - 4z}}{2}.$$

Chapter 1 showed that there is a unique formal square root with constant term 1, and the condition $T(0) = 0$ forces the minus sign. So

$$T(z) = \frac{1 - \sqrt{1 - 4z}}{2}.$$

Chapter 1 also computed the coefficients of $\sqrt{1 - 4z}$, which gives

$$[z^n]T(z) = \frac{1}{n}\binom{2n-2}{n-1} \qquad (n \geq 1).$$

This is the Catalan number C_{n-1}. The index shift comes from our size convention: a plane tree with n nodes corresponds to the $(n-1)$st Catalan number because Catalan numbers are often indexed by edges or by internal nodes rather than total nodes.

This example is our first encounter with a square-root expression coming from a symbolic specification. Later chapters will explain when such equations really do force square-root singularities and $n^{-3/2}$ asymptotics; that conclusion is only a preview here.

Example 3.8 (Integer compositions)**.** A positive integer of value k can be modeled as a block of exactly k atoms, because the block size records the integer's value. So the class of positive integers is

$$\mathcal{I} = \mathrm{SEQ}_{\geq 1}(\mathcal{Z}), \qquad I(z) = \frac{z}{1-z}.$$

A composition is then a nonempty sequence of such blocks:

$$\mathcal{K} = \mathrm{SEQ}_{\geq 1}(\mathcal{I}).$$

Therefore

$$K(z) = \frac{I(z)}{1 - I(z)} = \frac{z/(1-z)}{1 - z/(1-z)} = \frac{z}{1-2z}.$$

So

$$[z^n]K(z) = 2^{n-1} \qquad (n \geq 1).$$

This matches the direct gaps argument: between n ones there are $n-1$ gaps, and each gap is either cut or not cut.

Example 3.9 (Bivariate marking: number of a's in a binary word)**.** Let $\chi(w)$ count the number of letters a in a binary word w. Each letter contributes one factor of z for its size. In addition, the letter a contributes a factor of u while the letter b contributes no extra factor. So a single letter contributes either zu or z, and an arbitrary binary word has bivariate OGF

$$W(z,u) = \frac{1}{1 - (zu + z)} = \frac{1}{1 - z(u+1)}.$$

Now expand the geometric series:

$$W(z,u) = \sum_{n \geq 0} z^n (u+1)^n.$$

Therefore

$$[z^n u^k]W(z,u) = [u^k](u+1)^n = \binom{n}{k}.$$

So among the 2^n binary words of length n, exactly $\binom{n}{k}$ contain k letters a.

Using the uniform distribution on binary words of length n, the mean number of a's is

$$\frac{[z^n]\partial_u W(z,u)\big|_{u=1}}{[z^n]W(z,1)} = \frac{[z^n]\frac{z}{(1-z(u+1))^2}\big|_{u=1}}{2^n} = \frac{[z^n]\frac{z}{(1-2z)^2}}{2^n} = \frac{n2^{n-1}}{2^n} = \frac{n}{2},$$

as symmetry suggests.

3.6 Why the Symbolic Method Matters

The examples above share a common pattern: write down what an object *is*, check that the decomposition is unique, apply the translation dictionary, and obtain a functional equation for the generating function. That equation may be rational, algebraic, or — in the labeled setting — involve familiar transcendental functions such as exp and log.

At this stage, however, one must be careful not to overstate what has been proved. This chapter establishes the *combinatorial-to-algebraic translation*. It does *not* yet prove the later analytic claims about singularities or asymptotics. For example:

- Using only $+$, $\times$, and SEQ often leads to rational expressions in generating functions, and clearing denominators can produce polynomial equations.
- Many context-free grammars later translate into polynomial systems in several generating functions.
- Generic square-root singularities often lead to $n^{-3/2}$ factors.

But each of these statements requires additional hypotheses and later theorems. In particular, it is *not* true that every algebraic generating function automatically has $n^{-3/2}$ asymptotics: rational functions are algebraic too, and poles lead to different behavior.

Chapter 5 will return to context-free grammars and explain carefully when grammar equations produce algebraic generating functions and when generic square-root asymptotics really appear. The symbolic method is the combinatorial front end of that theory, not the whole theory by itself.

The connection to language models must also be kept precise. In this chapter we are studying *counting* generating functions attached to combinatorial classes. Later chapters will introduce weighted and probabilistic generating functions as separate objects. The symbolic method supplies the counting side of that dictionary, but one should not identify counting OGFs, probability generating functions, and partition functions without saying exactly what has been weighted and how.

Exercise 3.10. Let $\mathcal{P}$ be the class of balanced parenthesis strings (Dyck words). Write a symbolic specification for $\mathcal{P}$ using a decomposition at the first return to height zero, derive the OGF, and verify that

$$[z^{2n}]P(z) = C_n.$$

Exercise 3.11. A *unary-binary tree* is either a leaf (one node) or an internal node with exactly one child or exactly two children. Write a symbolic equation for the class $\mathcal{U}$ of unary-binary trees (size = number of nodes), derive the OGF as a root of a quadratic, and identify its radius of convergence.

Exercise 3.12 (Challenge)**.** Using the labeled setting and the standard cycle-construction rule recorded above, let $\mathcal{S}$ be the class of permutations and let $\mathcal{D}$ be the class of derangements. Starting from the specification

$$\mathcal{D} = \mathrm{SET}(\mathrm{CYC}_{\geq 2}(\mathcal{Z})),$$

derive the EGF of derangements and confirm that the probability a random permutation of $[n]$ is a derangement tends to e^{-1} as $n \to \infty$.

Further Reading

- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19], Chapters I–II — The definitive treatment of the symbolic method, covering

both the unlabeled (OGF) and labeled (EGF) translation dictionaries, with proofs, an extensive worked-example catalogue, and the full derivation of the SET, CYC, and MSET rules omitted from the present chapter.

- **P. Flajolet and M. Soria**, "General combinatorial schemas: Gaussian limit distributions and exponential tails," *Discrete Mathematics* 114 (1993), 159–180 — Shows how bivariate generating functions and the symbolic method together yield limit laws (Gaussian distributions of parameters) for broad families of combinatorial structures; a readable illustration of how the bivariate marking machinery of this chapter extends to probability.

- **R. P. Stanley**, *Enumerative Combinatorics*, Vol. 2 (Cambridge, 1999) — Chapter 5 covers trees and the Lagrange inversion formula; Chapter 6 develops algebraic, D-finite, and noncommutative generating functions. Together they extend the symbolic-method viewpoint to more advanced territory.

- **A. Nijenhuis and H. S. Wilf**, *Combinatorial Algorithms* (Academic Press, 2nd ed., 1978) [42] — Develops algorithms for random generation and enumeration of combinatorial structures (set partitions, permutations, labeled trees); the algorithmic perspective complements the symbolic-method viewpoint of this chapter.

- **F. Bergeron, G. Labelle, and P. Leroux**, *Combinatorial Species and Tree-Like Structures* (Cambridge, 1998) — The standard reference for the theory of species, the categorical underpinning of the symbolic method; essential for readers who want to understand why the translation dictionary works at the level of bijections rather than just coefficient identities.

Chapter 4

Singularity Analysis

Chapter 2 showed how the radius of convergence captures the exponential scale of a coefficient sequence. This chapter adds the next level of information. Under suitable analyticity hypotheses, the *local form* of a generating function near its dominant singularity determines the polynomial correction to that exponential growth. That principle is the content of the Flajolet–Odlyzko transfer theorem [18].

This chapter has two different goals. First, it explains the main theorem and the shape of its hypotheses. Second, it works through the most important model cases so the theorem does not feel like a black box dropped from the sky. What it does *not* do is give a beginner-friendly full proof of the transfer theorem; the complex-analytic proof uses more contour machinery than we want to develop here. Whenever we are only giving the roadmap, I will say so explicitly.

4.1 Exponential Growth Rate Revisited

Let

$$A(z) = \sum_{n \geq 0} a_n z^n$$

have radius of convergence ρ. By Cauchy–Hadamard,

$$\limsup_{n \to \infty} |a_n|^{1/n} = \rho^{-1}.$$

So for every $\varepsilon > 0$, eventually

$$|a_n|^{1/n} \leq \rho^{-1} + \varepsilon,$$

and therefore

$$|a_n| \leq (\rho^{-1} + \varepsilon)^n$$

for all sufficiently large n. In this sense, ρ^{-n} is the correct exponential scale.

What the radius does *not* determine is the subexponential correction. For example,

$$n^{1/2}\rho^{-n} \qquad \text{and} \qquad n^{-3/2}\rho^{-n}$$

have the same exponential part because

$$\lim_{n \to \infty} (n^{1/2})^{1/n} = \lim_{n \to \infty} (n^{-3/2})^{1/n} = 1.$$

The difference between those two sequences lies entirely in the polynomial prefactor. Singularity analysis is what recovers that finer information.

4.2 Dominant Singularities and Periodicity

Definition 4.1. Assume $A(z)$ has finite radius of convergence $\rho < \infty$. A *dominant singularity* of A is a point ζ with $|\zeta| = \rho$ such that the holomorphic function represented by the power series on $|z| < \rho$ cannot be analytically continued through ζ.

The phrase "cannot be continued through ζ" means that no holomorphic extension exists on any neighborhood of ζ that agrees with the original power series on the part of that neighborhood lying inside the disk of convergence.

There may be more than one dominant singularity. The simplest example is

$$\frac{1}{1-z^2} = \frac{1}{2}\left(\frac{1}{1-z} + \frac{1}{1+z}\right).$$

Its radius of convergence is 1, and the dominant singularities are $z = 1$ and $z = -1$. The coefficient of z^n is

$$\frac{1}{2}(1 + (-1)^n),$$

so the second singularity contributes the alternating factor $(-1)^n = (-1)^{-n}$. This is why multiple dominant singularities can produce oscillation instead of a clean single-term asymptotic.

To keep the main theorem readable, it is common to restrict first to the *aperiodic* case.

Definition 4.2. A coefficient sequence (a_n) is *aperiodic* if

$$\gcd\{n \geq 0 : a_n \neq 0\} = 1.$$

If the gcd is $d > 1$, then all nonzero coefficients lie on a lattice of step d, and one typically expects d different dominant singularities related by roots of unity.

In many positive combinatorial classes, aperiodicity is the condition that rules out forced periodic oscillation. For the rest of the chapter, we focus on the cleanest situation: a unique dominant singularity at a positive real point $z = \rho$.

4.3 Rational Generating Functions as a Prototype

The rational case can be understood exactly, and it provides the first model for the general theorem.

Proposition 4.3. *For every integer $m \geq 1$,*

$$\frac{1}{(1-z/\rho)^m} = \sum_{n\geq 0} \binom{n+m-1}{m-1} \rho^{-n} z^n.$$

Consequently,

$$[z^n]\frac{1}{(1-z/\rho)^m} = \binom{n+m-1}{m-1}\rho^{-n} \sim \frac{n^{m-1}}{(m-1)!}\rho^{-n}.$$

Proof. Apply the generalized binomial theorem from Chapter 1 with $\alpha = -m$ and $w = -z/\rho$:

$$(1-w)^{-m} = \sum_{n\geq 0} \binom{-m}{n}(-w)^n.$$

The standard identity

$$\binom{-m}{n}(-1)^n = \binom{n+m-1}{m-1}$$

gives the displayed series.

For the asymptotic, write

$$\binom{n+m-1}{m-1} = \frac{(n+m-1)(n+m-2)\cdots(n+1)}{(m-1)!}.$$

Dividing by n^{m-1} yields

$$\frac{\binom{n+m-1}{m-1}}{n^{m-1}} = \frac{(1+\frac{m-1}{n})(1+\frac{m-2}{n})\cdots(1+\frac{1}{n})}{(m-1)!} \to \frac{1}{(m-1)!}.$$

So

$$\binom{n+m-1}{m-1} \sim \frac{n^{m-1}}{(m-1)!}.$$

□

Now take a general proper rational function $A(z) = P(z)/Q(z)$. Partial fractions express it as a finite sum of terms of the form

$$\frac{c_{j,m}}{(1-z/\zeta_j)^m},$$

where the ζ_j are the poles of A. Each such term contributes a polynomial in n times ζ_j^{-n}. If ρ is the unique pole of smallest modulus, then every other term is exponentially smaller because

$$\left|\frac{\rho}{\zeta_j}\right|^n \to 0 \qquad \text{whenever } |\zeta_j| > \rho.$$

So the nearest pole determines the asymptotic behavior.

Example 4.4 (Fibonacci numbers)**.** Chapter 1 showed that

$$F(z) = \frac{z}{1-z-z^2}$$

is the generating function of the Fibonacci sequence. Its poles are at

$$z = \frac{-1 \pm \sqrt{5}}{2} = \frac{1}{\varphi}, -\varphi.$$

The dominant pole is $\rho = 1/\varphi$, and it is simple. The partial fraction formula from Chapter 1 gives

$$F(z) = \frac{1/\sqrt{5}}{1-\varphi z} - \frac{1/\sqrt{5}}{1-\hat{\varphi} z}.$$

Therefore

$$F_n = \frac{\varphi^n - \hat{\varphi}^n}{\sqrt{5}},$$

and in particular

$$F_n \sim \frac{\varphi^n}{\sqrt{5}}.$$

Since $|\hat{\varphi}| = 1/\varphi < 1$, the second term is exponentially smaller, and for every $n \geq 1$ its absolute value is at most

$$\frac{\varphi^{-n}}{\sqrt{5}} \leq \frac{1}{\sqrt{5}\,\varphi} < \frac{1}{2}.$$

4.4 The Transfer Theorem

Rational functions are only the beginning. Catalan numbers, trees, and many recursive combinatorial classes lead to algebraic singularities such as square roots. The general tool is the transfer theorem.

First we need the domain on which the local singular model is valid.

Definition 4.5. For $\rho > 0$, $\phi \in (0, \pi/2)$, and $\eta > \rho$, the Δ-*domain at* ρ is

$$\Delta(\rho, \phi, \eta) = \{z \in \mathbb{C} : |z| < \eta,\ z \neq \rho,\ |\arg(z - \rho)| > \phi\}.$$

A function is Δ-*analytic at* ρ if it is analytic in some such domain.

Geometrically, this is a disk of radius η centered at the origin with a small wedge removed around the ray starting at ρ and extending to the right. The removed wedge is exactly the place where a branch cut for $(1 - z/\rho)^\alpha$ is allowed to sit.

Definition 4.6. For $\alpha > 0$, the Gamma function is

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\, dt.$$

It extends to noninteger complex values by analytic continuation. The special values we need are

$$\Gamma(k) = (k-1)! \quad (k \in \mathbb{Z}_{>0}), \qquad \Gamma(1/2) = \sqrt{\pi}, \qquad \Gamma(-1/2) = -2\sqrt{\pi}.$$

Theorem 4.7 (Flajolet–Odlyzko transfer theorem, used as a major black box)**.** *Let $\rho > 0$, and suppose $A(z)$ is Δ-analytic at ρ. Assume that for some real*

$$\alpha \notin \mathbb{Z}_{\leq 0}$$

and some nonzero constant K,

$$A(z) = K\left(1 - \frac{z}{\rho}\right)^{-\alpha} + o\left(\left(1 - \frac{z}{\rho}\right)^{-\alpha}\right)$$

as $z \to \rho$ within the Δ-domain, where the fractional power is taken on the branch determined by that slit domain. Then

$$[z^n]A(z) \sim \frac{K}{\Gamma(\alpha)} n^{\alpha-1} \rho^{-n}.$$

A few comments are worth making immediately.

- The exclusion $\alpha \in \mathbb{Z}_{\leq 0}$ avoids the poles of the Gamma function and also avoids cases where the model becomes analytic instead of singular.
- The phrase "as $z \to \rho$ within the Δ-domain" means we approach ρ while staying away from the slit, so the chosen branch of the fractional power remains single-valued.
- The notation $o((1 - z/\rho)^{-\alpha})$ means that after dividing the error term by $(1 - z/\rho)^{-\alpha}$, the quotient tends to 0 in that same slit-domain limit.
- In practice one often writes a local expansion as "analytic background + singular term." The theorem is then applied to the first singular term, because that is the term controlling the leading asymptotic.

Remark 4.8 (Proof roadmap only). The rigorous proof uses Cauchy's coefficient formula, contour deformation, a contour that passes near the singularity while avoiding the branch cut, and a rescaling that produces the Gamma integral. We do *not* prove those steps in full here. The roadmap is:

1. write $[z^n]A(z)$ as a contour integral;
2. deform the contour so that only a neighborhood of the dominant singularity matters;
3. replace $A(z)$ by its local singular model near ρ;
4. rescale by $z = \rho(1 - t/n)$, which produces the factors ρ^{-n} and $n^{\alpha-1}$ and leads to the Gamma integral.

Full details are in [19], Chapter VI.

4.5 Catalogue of Common Exponents

The theorem becomes much easier to remember once a few model cases are recorded.

Poles. If

$$A(z) \sim \frac{K}{(1 - z/\rho)^m}$$

with $m \in \mathbb{Z}_{>0}$, then $\alpha = m$ and

$$[z^n]A(z) \sim \frac{K}{(m-1)!} n^{m-1} \rho^{-n}.$$

For $m = 1$ this is pure exponential growth. For $m = 2$ it is $n\rho^{-n}$. This matches the exact rational-function computations above.

Square-root singularity. If the singular part looks like

$$K\left(1 - \frac{z}{\rho}\right)^{1/2},$$

then equivalently it is of the form $(1 - z/\rho)^{-\alpha}$ with $\alpha = -1/2$. The theorem gives

$$[z^n]A(z) \sim \frac{K}{\Gamma(-1/2)} n^{-3/2} \rho^{-n} = -\frac{K}{2\sqrt{\pi}} n^{-3/2} \rho^{-n}.$$

This is the prototype source of the famous $n^{-3/2}$ factor. The sign belongs to the singular coefficient K, not to the whole generating function, which may have a nonzero analytic background.

Inverse square-root singularity. If the singular part is

$$\frac{K}{\sqrt{1 - z/\rho}},$$

then $\alpha = 1/2$, so

$$[z^n]A(z) \sim \frac{K}{\sqrt{\pi}} n^{-1/2} \rho^{-n}.$$

A standard model example is

$$\frac{1}{\sqrt{1 - 4z}} = \sum_{n \geq 0} \binom{2n}{n} z^n,$$

whose coefficients satisfy $\binom{2n}{n} \sim 4^n/(\sqrt{\pi n})$.

Logarithmic singularity. Logarithms are best treated as a separate case rather than by pretending that "$\alpha \to 0$" is a literal theorem. Chapter 2 already showed that

$$\log \frac{1}{1 - z/\rho} = \sum_{n \geq 1} \frac{\rho^{-n}}{n} z^n,$$

so directly

$$[z^n] \log \frac{1}{1 - z/\rho} = \frac{\rho^{-n}}{n}.$$

4.6 Worked Example: Catalan Numbers

Example 4.9 (Catalan numbers)**.** Chapter 1 derived the Catalan generating function

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}.$$

The dominant singularity is at

$$\rho = \frac{1}{4},$$

because that is where the square root becomes singular. The apparent pole at $z = 0$ is removable: the numerator also vanishes at $z = 0$, so the quotient has a finite value there.

Since

$$1 - \frac{z}{\rho} = 1 - 4z,$$

we can rewrite the square root directly in the transfer-theorem variable. Also,

$$\frac{1}{2z} = \frac{1}{2\rho} + O\left(1 - \frac{z}{\rho}\right) = 2 + O\left(1 - \frac{z}{\rho}\right)$$

as $z \to \rho$. Therefore

$$C(z) = \left(2 + O\left(1 - \frac{z}{\rho}\right)\right)\left(1 - \sqrt{1 - \frac{z}{\rho}}\right).$$

Multiplying out and keeping the first nonanalytic term,

$$C(z) = 2 - 2\left(1 - \frac{z}{\rho}\right)^{1/2} + O\left(1 - \frac{z}{\rho}\right).$$

The constant term 2 is part of the analytic background. The leading singular term is

$$-2\left(1 - \frac{z}{\rho}\right)^{1/2}.$$

So in Theorem 4.7 we take

$$K = -2, \qquad \alpha = -\frac{1}{2}, \qquad \rho = \frac{1}{4}.$$

Applying the theorem,

$$C_n \sim \frac{-2}{\Gamma(-1/2)} n^{-3/2} \rho^{-n} = \frac{-2}{-2\sqrt{\pi}} n^{-3/2} 4^n = \frac{4^n}{\sqrt{\pi}\, n^{3/2}}.$$

This is exactly the classical Catalan asymptotic.

4.7 Summary and Preview

The transfer theorem completes the main analytic-combinatorics workflow in principle:

1. derive a generating function from a combinatorial description;
2. locate the dominant singularity and identify the leading singular term;
3. verify the hypotheses carefully enough to apply the transfer theorem;
4. read off the leading asymptotic.

The phrase "verify the hypotheses carefully" matters. The theorem is powerful, but it is not a push-button rule. One may still need real work to solve the functional equation, identify the dominant singularity, prove uniqueness or aperiodicity, and normalize the local expansion correctly.

Chapter 5 will apply this machinery to algebraic generating functions arising from context-free grammars. In many generic nondegenerate situations, those systems lead to square-root singularities and hence to an $n^{-3/2}$ factor. But that conclusion is not automatic from "algebraic" or "context-free" alone; the precise hypotheses will matter.

Later chapters will ask whether weighted and probabilistic objects related to language models show analogous singular behavior. That is a motivating research direction, not something established by the present chapter.

Further Reading

- **P. Flajolet and A. M. Odlyzko**, "Singularity analysis of generating functions," *SIAM J. Discrete Math.* (1990) [18] — The original paper proving the transfer theorem in the form used throughout this chapter; contains the Hankel-contour proof and the full catalogue of singular types, and is more concise than the book-length treatment in [19].
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19], Chapter VI — The textbook development of singularity analysis, including the Δ-domain framework, a detailed treatment of the Hankel contour argument, and worked examples covering random trees, urns, and maps.
- **M. Drmota**, *Random Trees* (Springer, Vienna, 2009) [14] — A monograph on random trees whose first half is an accessible introduction to the singularity analysis of algebraic and D-finite generating functions; Chapter 2 is especially useful for understanding how the characteristic system of the present chapter extends to systems of equations.
- **A. Odlyzko**, "Asymptotic enumeration methods," in *Handbook of Combinatorics*, Vol. 2 (Elsevier, 1995) — A comprehensive survey of coefficient-extraction techniques from generating functions, covering the transfer theorem, saddle-point methods, and the Darboux method; excellent for seeing the transfer theorem in context alongside other asymptotic tools.
- **C. Banderier and M. Drmota**, "Formulae and asymptotics for coefficients of algebraic functions," *Combinatorics, Probability and Computing* (2015) [3] — Proves universal limit laws for coefficients of algebraic generating functions beyond the leading term; relevant for understanding why generic algebraic singularities often produce the $n^{-3/2}$ exponent, but also why hypotheses matter.

Chapter 5

Algebraic Generating Functions and the $n^{-3/2}$ Law

Chapter 4 explained how a square-root singularity leads to an $n^{-3/2}$ coefficient law. This chapter studies one important source of such singularities: nonlinear algebraic systems coming from unambiguous context-free structure. The conclusion is striking, but it must be stated carefully. The exponent $-3/2$ is *not* forced by being context-free all by itself. Simple unambiguous context-free languages can still have rational generating functions and pole-type asymptotics.

A good warning example is

$$a^* = \{\varepsilon, a, aa, aaa, \dots\},$$

generated by the unambiguous grammar

$$S \to aS \mid \varepsilon.$$

Its generating function is

$$S(z) = 1 + zS(z) = \frac{1}{1-z},$$

so the coefficients are $1, 1, 1, \dots$, not of the form $Cn^{-3/2}\rho^{-n}$. Thus the square-root law belongs to a *nonlinear generic regime*, not to every context-free language without exception.

The goal of the chapter is to explain the pipeline

$$\text{unambiguous grammar} \Longrightarrow \text{algebraic generating function} \Longrightarrow \text{generic square-root singularity} \Longrightarrow n^{-3/2}\rho^{-n},$$

while also making clear where the hidden hypotheses live.

5.1 Unambiguous Context-Free Grammars

A *context-free grammar* is a tuple

$$\mathcal{G} = (V, \Sigma, S, \mathcal{P}),$$

where V is a finite set of nonterminals, Σ is a finite terminal alphabet, $S \in V$ is the start symbol, and $\mathcal{P}$ is a finite set of productions of the form

$$A \to \alpha, \qquad A \in V, \quad \alpha \in (V \cup \Sigma)^*.$$

The language $L(\mathcal{G})$ consists of all terminal strings derivable from S.

A grammar is *unambiguous* if every word in the language has a unique parse tree. Equivalently — a standard fact from formal-language theory — every word has a unique leftmost derivation. This matters because our generating functions count *distinct words*; if one word had several parse trees, a naive grammar-to-generating-function translation would overcount it.

5.2 From Productions to Polynomial Systems

The translation rule is straightforward once written symbol by symbol. Suppose a production has the form

$$A \to X_1 X_2 \cdots X_m,$$

where each X_i is either a terminal, a nonterminal, or the empty word ε.

- A terminal contributes a factor of z because it adds one unit of word length.
- A nonterminal B contributes a factor of $B(z)$, the generating function for words derivable from B.
- The empty word contributes 1.

Multiplying those contributions gives the monomial associated with the production. Summing over all productions with left-hand side A gives an equation for $A(z)$.

This is exactly the symbolic-method logic from Chapter 3: concatenation becomes multiplication, alternative productions become addition, and unambiguity is what ensures the resulting count is by words rather than by derivations.

Example 5.1 ($a^n b^n$)**.** The grammar

$$S \to aSb \mid \varepsilon$$

generates $\{a^n b^n : n \geq 0\}$. The production aSb contributes

$$z \cdot S(z) \cdot z = z^2 S(z),$$

and the production ε contributes 1. So

$$S(z) = 1 + z^2 S(z), \qquad S(z) = \frac{1}{1 - z^2}.$$

This is rational, not square-root. Its coefficients are exactly 1 for even n and 0 for odd n.

Example 5.2 (Plane trees revisited)**.** The symbolic specification

$$\mathcal{T} = \mathcal{Z} \times \mathrm{SEQ}(\mathcal{T})$$

from Chapter 3 is not literally a grammar production, but it obeys the same translation logic: a root contributes z and the ordered list of children contributes $1/(1 - T(z))$. Thus

$$T(z) = \frac{z}{1 - T(z)},$$

which rearranges to the polynomial equation

$$T(z)^2 - T(z) + z = 0.$$

This is our first nonlinear algebraic example and will serve as the model case below.

After standard grammar normalization, one often obtains a finite polynomial system

$$Y_i(z) = \Phi_i(z, Y_1(z), \ldots, Y_k(z)), \qquad i = 1, \ldots, k,$$

with each Φ_i a polynomial having nonnegative coefficients. Chomsky normal form is a convenient way to keep the dependence at most quadratic in the nonterminal variables, though we will not need the full details of that conversion here.

5.3 Why the Generating Function Is Algebraic

The decisive structural theorem is the following.

Theorem 5.3 (Chomsky–Schützenberger, used as a major black box)**.** *If L is generated by an unambiguous context-free grammar and c_n denotes the number of words of length n in L, then the ordinary generating function*

$$L(z) = \sum_{n \geq 0} c_n z^n$$

is algebraic over $\mathbb{Q}(z)$.

One standard proof translates the grammar into a polynomial system for the nonterminal generating functions and then uses elimination theory to extract a single polynomial equation for the start-symbol series. The elimination step is advanced algebra, so we treat it as a black box here. The key point to keep in mind is simpler: a finite unambiguous grammar creates a finite algebraic system, and that finiteness is what forces algebraicity.

There is also a uniqueness issue. Polynomial systems can have many formal solutions in principle, but the grammar singles out one nonnegative power-series solution by coefficient recursion in length: the coefficient of z^n is determined by the number of derivations producing words of length n, and unambiguity prevents a hidden multiplicity problem.

5.4 Puiseux Expansions: What Algebraic Singularities Look Like

Algebraic functions do not have arbitrary singularities.

Theorem 5.4 (Puiseux, used as a major black box)**.** *Let $F(z)$ be an algebraic function satisfying a polynomial equation*

$$P(z, F(z)) = 0.$$

Near any singularity ρ of a chosen branch of F, there is a positive integer d and an expansion of the form

$$F(z) = \sum_{k \geq k_0} a_k \left(1 - \frac{z}{\rho}\right)^{k/d},$$

valid in a slit neighborhood of ρ.

What does this mean in practice?

- The generating function defined by its power series at 0 gives one distinguished branch.
- Near a singularity, that branch may continue into a local expression involving fractional powers.
- Integer exponents belong to the analytic part; the first genuinely fractional exponent is the first genuinely singular term.

Thus algebraic singularities are much more rigid than arbitrary singularities. They are poles or branch points of rational order, never essential singularities.

For quadratic equations, the generic fractional exponent is $1/2$, which is why square roots appear so often. But "generic" is the key word: algebraicity alone does not force a square root in every case, and certainly does not rule out rational degeneracies.

5.5 Why Square Roots Are Generic in the Nonlinear Case

We now explain the mechanism heuristically. This section is a local model, not a full proof.

Suppose a generating function $F(z)$ satisfies

$$P(z, F(z)) = 0,$$

and suppose the relevant branch stays bounded up to its dominant singularity, with finite limit

$$\tau = \lim_{z \to \rho} F(z).$$

If the implicit-function theorem were still active at (ρ, τ), then the branch would continue analytically past ρ. So a singularity must occur where

$$P(\rho, \tau) = 0 \qquad \text{and} \qquad P_y(\rho, \tau) = 0.$$

This pair of equations is called the *characteristic system.*

Now write

$$\Delta z = z - \rho, \qquad \Delta F = F(z) - \tau.$$

Expanding P near (ρ, τ) gives, schematically,

$$0 = P_z \, \Delta z + P_y \, \Delta F + \frac{1}{2} P_{yy} (\Delta F)^2 + P_{zy} \, \Delta z \, \Delta F + \cdots .$$

At the characteristic point, the linear term in ΔF vanishes because $P_y(\rho, \tau) = 0$. If we now guess that the first nontrivial balance is

$$\Delta F = O(\sqrt{\Delta z}),$$

then the mixed term $\Delta z \, \Delta F$ is of order $(\Delta z)^{3/2}$, smaller than both Δz and $(\Delta F)^2$. So the leading balance becomes

$$P_z \, \Delta z + \frac{1}{2} P_{yy} (\Delta F)^2 \approx 0.$$

Provided

$$P_z(\rho, \tau) \neq 0 \qquad \text{and} \qquad P_{yy}(\rho, \tau) \neq 0,$$

one obtains

$$\Delta F \asymp K \sqrt{\Delta z}$$

for some nonzero constant K. Since

$$\Delta z = -\rho \left(1 - \frac{z}{\rho} \right),$$

this is the same as saying that the singular term is some branch-dependent constant times

$$\sqrt{1 - z/\rho}.$$

That is the local origin of the square-root law.

The word *generic* means that higher degeneracies require extra coincidences. If one also had $P_{yy}(\rho, \tau) = 0$, then a cubic or still higher balance would appear instead. Such cases exist, but they are not the default.

5.6 The Generic $n^{-3/2}$ Law

The theorem actually used in analytic combinatorics is not "every context-free language has a square-root singularity." It is a more carefully scoped theorem about nonlinear positive algebraic systems.

Theorem 5.5 (Generic square-root law for nonlinear context-free systems)**.** *Consider an unambiguous context-free grammar whose counting generating functions form a nonlinear polynomial system with nonnegative coefficients. Assume the standard structural hypotheses from the Drmota–Lalley–Woods type theory: the relevant dependency graph is strongly connected / irreducible, the system is proper and genuinely nonlinear, and the resulting coefficient sequence is aperiodic. Then the start-symbol generating function has a unique dominant singularity $\rho > 0$ of square-root type,*

$$L(z) = g(z) - h(z)\sqrt{1 - z/\rho},$$

with g and h analytic near ρ and $h(\rho) > 0$. Consequently,

$$c_n \sim C\, n^{-3/2}\rho^{-n}$$

for some constant $C > 0$.

Informally, these hypotheses mean the following. Build a dependency graph on the nonterminals, with an edge $A \to B$ whenever B appears on the right-hand side of a production for A. "Strongly connected" or "irreducible" means every nonterminal can influence every other through this graph. "Genuinely nonlinear" means the polynomial system is not merely linear in the unknown generating functions; some right-hand side must contain a product of unknown series, so rational degeneracies like a^* are excluded. "Aperiodic" means

$$\gcd\{n \geq 0 : c_n \neq 0\} = 1.$$

Finally, "proper" is a shorthand for the standard well-foundedness assumptions excluding pathological epsilon-cycles and ensuring that the coefficient recursion is meaningful.

This theorem deliberately excludes the linear and rational cases illustrated earlier. Languages such as a^* or $\{a^n b^n\}$ remain context-free and unambiguous, but their generating functions are rational, so their coefficient asymptotics are controlled by poles rather than square-root branch points.

5.7 Applying the Transfer Theorem

Assume we are in the setting of Theorem 5.5, so that

$$L(z) = g(z) - h(z)\sqrt{1 - z/\rho}.$$

The singular information sits in the second term. By Chapter 4,

$$[z^n]\,(1 - z/\rho)^{1/2} \sim -\frac{1}{2\sqrt{\pi}} n^{-3/2}\rho^{-n}.$$

Therefore

$$[z^n]\left(-h(\rho)\sqrt{1 - z/\rho}\right) \sim \frac{h(\rho)}{2\sqrt{\pi}} n^{-3/2}\rho^{-n}.$$

In the theorem setting, the analytic remainder extends farther than the singular term and therefore contributes exponentially smaller coefficients, so

$$c_n \sim \frac{h(\rho)}{2\sqrt{\pi}} n^{-3/2}\rho^{-n}.$$

This is the promised $n^{-3/2}$ law.

5.8 Worked Example: Plane Trees Revisited

The plane-tree equation from Chapter 3 is

$$T(z)^2 - T(z) + z = 0.$$

Here

$$P(z, F) = F^2 - F + z.$$

The characteristic system is

$$P(\rho, \tau) = 0, \qquad P_F(\rho, \tau) = 2\tau - 1 = 0.$$

So $\tau = 1/2$, and substituting into $P = 0$ gives

$$\frac{1}{4} - \frac{1}{2} + \rho = 0, \qquad \rho = \frac{1}{4}.$$

The local characteristic-system picture predicts a square-root singularity, and the exact quadratic formula confirms it:

$$T(z) = \frac{1 - \sqrt{1 - 4z}}{2} = \frac{1}{2} - \frac{1}{2}\sqrt{1 - z/\rho}.$$

So here

$$h(\rho) = \frac{1}{2}.$$

The generic law gives

$$[z^n]T(z) \sim \frac{1/2}{2\sqrt{\pi}} n^{-3/2} 4^n = \frac{4^n}{4\sqrt{\pi} n^{3/2}}.$$

Since $[z^n]T(z) = C_{n-1}$, this agrees with the classical Catalan asymptotic

$$C_{n-1} \sim \frac{4^{n-1}}{\sqrt{\pi}(n-1)^{3/2}} \sim \frac{4^n}{4\sqrt{\pi} n^{3/2}}.$$

5.9 What This Does and Does Not Let Us Infer

The $n^{-3/2}$ law is mathematically strong, but it should not be used carelessly.

- If one proves that a counting generating function belongs to the nonlinear square-root regime above, then the exponent $-3/2$ is a theorem.
- If one sees a finite-data fit with exponent near $-3/2$, that is only *heuristic evidence* for such a regime, not a proof.
- Other exponents point to other singular models. For instance, a simple pole corresponds to exponent 0, a logarithmic singularity gives exponent -1, and an inverse square root gives exponent $-1/2$.

So empirical exponent fitting can be suggestive, but only once one also remembers the limitations of finite data, periodicity, and model mismatch. Chapter 19 will return to these issues in a more explicit research-roadmap style.

Remark 5.6. Aperiodicity is not the only hypothesis that matters. Linear / rational cases are another important source of exceptions to the square-root law. So "aperiodic" should never be read as a synonym for "therefore $n^{-3/2}$."

Summary of Part I

Chapters 1–5 together provide the basic analytic-combinatorics pipeline.

1. Encode a combinatorial class by a generating function.
2. Use structural equations to identify the relevant analytic object.
3. Locate and classify the dominant singularity.
4. Apply a transfer theorem to recover coefficient asymptotics.

For a broad and important family of *nonlinear* algebraic systems, the outcome is the universal-looking law

$$c_n \sim Cn^{-3/2}\rho^{-n}.$$

But this is a theorem about a specific regime, not about all context-free languages indiscriminately.

The rest of the book moves from pure counting toward weighted and probabilistic models. That shift will require new care, because counting OGFs, probability generating functions, and partition sums are related objects, not identical ones.

Further Reading

- **N. Chomsky and M. P. Schützenberger**, "The algebraic theory of context-free languages," in *Computer Programming and Formal Systems* (North-Holland, 1963) [12] — The original paper proving algebraicity for unambiguous context-free languages.
- **M. Drmota**, *Random Trees* (Springer, Vienna, 2009) [14] — A detailed source for characteristic systems, square-root singularities, and nonlinear algebraic systems.
- **C. Banderier and M. Drmota**, "Formulae and asymptotics for coefficients of algebraic functions," *Combinatorics, Probability and Computing* (2015) [3] — Develops generic asymptotic laws for algebraic systems well beyond the simplest tree examples.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19], Chapter VII — Covers algebraic functions, Newton polygons, and Puiseux expansions from the analytic-combinatorics point of view.

Part II

Formal Languages and Probabilistic Sources

Part I treated combinatorial classes abstractly. This part grounds the theory in formal languages — the subsets of Σ^* generated by grammars and recognized by machines. Chapter 6 surveys the Chomsky hierarchy (regular, context-free, context-sensitive, recursively enumerable) and its connection to the rational/algebraic GF dichotomy. Chapter 7 introduces weighted finite automata, whose generating functions are automatically rational, and the Kleene–Schützenberger theorem. Chapter 8 asks when a probabilistic grammar or an autoregressive model defines a genuine probability distribution (tightness), connecting Booth–Thompson's 1973 criterion with Du et al.'s 2023 measure-theoretic framework. Chapter 9 establishes the entropy-rate identity $h = \log(1/R)$ and carefully distinguishes the counting GF from the length PGF — a distinction that will matter throughout Parts III–V.

Chapter 6

Formal Languages and the Chomsky Hierarchy

Part I developed analytic combinatorics in a setting-agnostic way: one starts with a class of objects, encodes it by a generating function, and studies the singularities of that function. This chapter connects that viewpoint to a classical source of combinatorial classes, namely *formal languages*. The Chomsky hierarchy organizes those languages into four broad levels:

1. regular,
2. context-free,
3. context-sensitive,
4. recursively enumerable.

The goal here is mostly orienting rather than proving every theorem from scratch. Results such as Myhill–Nerode, Kleene's theorem, the pumping lemmas, and the equivalence between context-free grammars and pushdown automata are major theorems of formal-language theory. I will use them as imported facts and focus on the analytic-combinatorial lessons they support.

The main lesson is one-way, not one-to-one: regular languages always lead to rational length generating functions, and unambiguous context-free languages lead to algebraic ones. But moving higher in the hierarchy does *not* create a perfect analytic classification. Higher-level languages can still have simple generating functions when counted by length. So the hierarchy is a guide to tractability, not a complete dictionary from language class to analytic type.

6.1 Alphabets, Strings, and Languages

Definition 6.1. An *alphabet* Σ is a finite nonempty set. A *string* over Σ is a finite sequence

$$w = \sigma_1\sigma_2\cdots\sigma_k, \qquad \sigma_i \in \Sigma.$$

Its *length* is $|w| = k$. The unique string of length 0 is the *empty string* ε.

For example, over $\Sigma = \{a, b\}$, the word 'abba' has length 4.

The set Σ^* of all finite strings over Σ is called the *free monoid* on Σ; this simply means that strings can be concatenated and that ε acts as the identity element. We write

$$\Sigma^n = \{w \in \Sigma^* : |w| = n\}, \qquad \Sigma^+ = \Sigma^* \setminus \{\varepsilon\}.$$

A *language* is any subset $L \subseteq \Sigma^*$.

The counting object relevant for this book is

$$L(z) = \sum_{n \geq 0} |L \cap \Sigma^n| z^n,$$

which records the number of accepted words of each length. For the full language Σ^* over a two-letter alphabet,

$$|\Sigma^n| = 2^n, \qquad \sum_{n \geq 0} 2^n z^n = \frac{1}{1 - 2z}.$$

6.2 Finite Automata and Regular Languages

Definition 6.2. A *deterministic finite automaton* (DFA) is a tuple

$$M = (Q, \Sigma, \delta, q_0, F)$$

where Q is a finite set of states, Σ is the input alphabet, $\delta : Q \times \Sigma \to Q$ is the transition function, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of accepting states.

The transition function extends from single letters to whole strings by the recursion

$$\hat{\delta}(q, \varepsilon) = q, \qquad \hat{\delta}(q, w\sigma) = \delta(\hat{\delta}(q, w), \sigma).$$

So to run a DFA on a word, one simply feeds the letters in one at a time. For instance, in the parity automaton below, the word ‘abaa’ toggles parity three times and ends in the odd state.

Example 6.3 (Even number of a’s)**.** Let $\Sigma = \{a, b\}$. The language

$$L = \{w : w \text{ contains an even number of } a\text{'s}\}$$

is recognized by a DFA with two states: “even” and “odd.” Reading an a switches the state, while reading a b leaves it unchanged. The start state is “even,” and it is also the only accepting state.

A *nondeterministic finite automaton* (NFA) allows several possible next states, and usually also ε-transitions. One may therefore think of its transition map as taking values in subsets of states over letters in $\Sigma \cup \{\varepsilon\}$. The key determinization theorem is this:

Theorem 6.4 (NFA = DFA, used as a black box)**.** *Every NFA accepts the same language as some DFA.*

The idea is the subset construction: a DFA state remembers the whole set of NFA states that are currently possible after reading the input so far.

Definition 6.5. A language is *regular* if it is accepted by some DFA (equivalently, by some NFA).

6.3 Myhill–Nerode and Regularity

For a language L, define an equivalence relation on strings by

$$x \sim_L y \iff \forall z \in \Sigma^* \, (xz \in L \iff yz \in L).$$

So two strings are equivalent if they have exactly the same possible futures leading to acceptance.

Theorem 6.6 (Myhill–Nerode, used as a black box)**.** *A language L is regular if and only if $\sim_L$ has only finitely many equivalence classes. In that case, the number of classes is exactly the number of states in the minimal DFA for L.*

The forward direction matches intuition: if two strings lead a DFA to the same state, then every future continuation is processed identically. The converse constructs a DFA whose states are the equivalence classes themselves.

This theorem is a powerful nonregularity test. For

$$L = \{a^n b^n : n \geq 0\},$$

the strings

$$\varepsilon, a, a^2, a^3, \ldots$$

are pairwise inequivalent, because

$$a^i b^i \in L, \qquad a^j b^i \notin L \quad (j \neq i).$$

So $\sim_L$ has infinitely many classes, and the language is not regular.

6.4 Regular Expressions

Regular expressions give a syntactic description of regular languages. Their semantics are:

- $\emptyset$ denotes the empty language,
- ε denotes the language $\{\varepsilon\}$,
- a symbol σ denotes the one-word language $\{\sigma\}$,
- $R + S$ denotes union,
- RS denotes concatenation:

$$L(RS) = \{xy : x \in L(R),\ y \in L(S)\},$$

- R^* denotes Kleene star:

$$L(R^*) = \bigcup_{k \geq 0} L(R)^k.$$

Theorem 6.7 (Kleene, used as a black box)**.** *A language is regular if and only if it is denoted by some regular expression.*

There is an important warning for this book. Regular-expression operations do *not* automatically translate into symbolic-method OGF operations when we count *distinct words.* Union may fail to be disjoint, concatenation may fail to be uniquely decodable, and star may be ambiguous. So the analogy between regexes and symbolic constructors is only safe under uniqueness / disjointness hypotheses. For counting distinct accepted words by length, automata-based linear algebra is often the safer route.

6.5 The Pumping Lemma for Regular Languages

Theorem 6.8 (Pumping lemma for regular languages, used as a black box)**.** *If L is regular, then there exists a pumping length p such that every word $w \in L$ with $|w| \geq p$ can be written as*

$$w = xyz$$

with $|xy| \leq p$, $|y| \geq 1$, and

$$xy^k z \in L \qquad \text{for all } k \geq 0.$$

The proof uses repeated states along the run of a DFA. It is important to remember that the pumping lemma is only a *necessary* condition for regularity, not a sufficient one.

For $\{a^n b^n : n \geq 0\}$, take $w = a^p b^p$. Since the first p symbols are all a, the pumped block y lies entirely inside the initial block of a's. Pumping y changes the number of a's without changing the number of b's, so the pumped word leaves the language.

6.6 Context-Free Grammars and Pushdown Automata

Definition 6.9. A *context-free grammar* (CFG) is a tuple

$$G = (V, \Sigma, P, S)$$

where V is a finite set of nonterminals, Σ is a finite set of terminals, P is a finite set of productions

$$A \to \alpha, \qquad A \in V, \quad \alpha \in (V \cup \Sigma)^*,$$

and $S \in V$ is the start symbol.

A language is *context-free* if it is generated by some CFG.

Example 6.10 ($a^n b^n$)**.** The grammar

$$S \to aSb \mid \varepsilon$$

generates exactly $\{a^n b^n : n \geq 0\}$.

Example 6.11 (Dyck language)**.** Balanced parenthesis strings are generated unambiguously by the first-return grammar

$$S \to (S)S \mid \varepsilon.$$

If z counts individual parentheses, this gives

$$S(z) = 1 + z^2 S(z)^2.$$

So the generating function is

$$S(z) = \frac{1 - \sqrt{1 - 4z^2}}{2z^2},$$

which is algebraic rather than rational.

A *pushdown automaton* (PDA) is a finite automaton equipped with a stack. The stack records pending nested structure, which is exactly what recursive grammatical structure needs.

Theorem 6.12 (CFG = PDA, used as a black box)**.** *A language is context-free if and only if it is accepted by some pushdown automaton.*

The intuition is that grammar expansion creates obligations that must be completed later, and the stack stores those obligations.

6.7 The Pumping Lemma for Context-Free Languages

Theorem 6.13 (Pumping lemma for context-free languages, used as a black box)**.** *If L is context-free, then there exists a pumping length p such that every sufficiently long word in L can be written as*

$$w = uvxyz$$

with

$$|vxy| \leq p, \qquad |vy| \geq 1, \qquad uv^k xy^k z \in L \text{ for all } k \geq 0.$$

Standard proofs use parse trees in a controlled normal form. As with the regular pumping lemma, this is only a necessary condition for context-freeness, not a sufficient one.

To show that

$$\{a^n b^n c^n : n \geq 0\}$$

is not context-free, one studies a long word $a^p b^p c^p$. The short middle region vxy can touch at most two of the three blocks, so pumping it inevitably changes one or two symbol counts while leaving the third fixed. Equal counts cannot be preserved.

6.8 The Chomsky Hierarchy

The hierarchy is best read as a sequence of increasingly powerful language models:

- **Type 3: regular languages.** Recognized by finite automata. Their memory is finite.
- **Type 2: context-free languages.** Recognized by pushdown automata. They can handle one stack of nested dependencies.
- **Type 1: context-sensitive languages.** Recognized by linear-bounded automata.
- **Type 0: recursively enumerable languages.** Recognized by unrestricted Turing machines.

The inclusions are strict.

For example, regular languages are closed under union, intersection, complement, concatenation, and star. Context-free languages are closed under union, concatenation, and star, but not in general under intersection or complement. A standard witness is

$$\{a^n b^n c^k : n, k \geq 0\} \cap \{a^k b^n c^n : k, n \geq 0\} = \{a^n b^n c^n : n \geq 0\},$$

and the right-hand side is not context-free.

We will not dwell on the upper two levels in detail. For this book, they mainly play a cautionary role: once one leaves the regular and unambiguous context-free worlds, the clean analytic methods of Parts I–II become less automatic.

6.9 The Analytic-Combinatorial Payoff

The cleanest implications for this book are these.

1. If L is regular, then its length generating function is rational.
2. If L is generated by an unambiguous CFG, then its length generating function is algebraic.

The first claim can be seen by linear algebra. Let a DFA have states $q_1, \ldots, q_m$, start-state vector u, and accepting-state vector v. Define a matrix A by letting A_{ij} be the number of alphabet symbols that send state q_i to state q_j. Then the number of accepted words of length n is

$$u^T A^n v,$$

so the OGF is

$$L(z) = \sum_{n \geq 0} u^T A^n v \, z^n = u^T (I - zA)^{-1} v,$$

which is rational. The $n = 0$ term automatically handles the empty word: it contributes $u^T v$, which is 1 exactly when the start state is accepting.

The second claim is the Chomsky–Schützenberger theorem from Chapter 5. Here again, the direction is one-way and structural: unambiguous CFGs give algebraic generating functions, but algebraicity alone does not force a unique asymptotic regime without the additional hypotheses discussed in Chapter 5.

What should one say about higher levels? Not that "context-sensitive" means one specific analytic type. The honest statement is weaker: once we leave the regular and unambiguous context-free settings, we lose the general theorems guaranteeing rational or algebraic generating functions. Some higher-level languages may still have rational or algebraic length generating functions, but there is no comparably clean universal structure theorem waiting for us there.

One bound remains universal. For every language $L \subseteq \Sigma^*$ over a finite alphabet,

$$|L \cap \Sigma^n| \leq |\Sigma|^n.$$

So word counts by length are always geometrically bounded. The "exotic" behavior that can arise at higher levels is about complicated singularity structure or the failure of easy continuation theorems, not about supergeometric growth of the counting sequence.

This is why regular and unambiguous context-free languages are analytically privileged: they come with robust structural theorems that turn formal-language descriptions into generating functions we can actually study.

Preview of Chapter 7

Chapter 7 studies *weighted finite automata*, a probabilistic refinement of regular-language automata. There the rational generating-function viewpoint becomes computational: matrices, resolvents, and spectral radii will let us count, weight, and eventually approximate sequence models.

Further Reading

- **J. E. Hopcroft and J. D. Ullman**, *Introduction to Automata Theory, Languages, and Computation* (1979) [24] — the classical reference for automata, grammars, Myhill–Nerode, pumping lemmas, and the Chomsky hierarchy.
- **M. Sipser**, *Introduction to the Theory of Computation* (2013) [52] — a modern, readable introduction to the same material.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for the symbolic and generating-function side of the regular / context-free story.
- **J. Berstel and C. Reutenauer**, *Noncommutative Rational Series with Applications* (2011) [7] — for the algebraic background behind rational series and weighted automata.

Chapter 7

Weighted Finite Automata and Rational Power Series

Chapter 6 studied finite automata that answer a yes/no question about each string. A *weighted finite automaton* (WFA) replaces that binary verdict by a numerical weight. The main exact theorem of the chapter is simple and powerful:

> For a WFA over $\mathbb{R}$ or $\mathbb{C}$, the generating function obtained by summing weights by string length is always rational.

That places WFAs squarely inside the pole-based world of Chapters 1–4.

Two distinctions are important from the start.

1. The definition of a WFA makes sense over an arbitrary semiring, but the resolvent formula $(I - zA)^{-1}$ belongs to the real/complex matrix setting.
2. A *locally stochastic* WFA is not automatically a proper probability distribution on finite strings. Properness is a separate almost-sure-halting condition.

These two distinctions will prevent several common confusions later in the book.

7.1 Semirings

A *semiring* is a set where one can add and multiply, multiplication distributes over addition, and subtraction need not exist. The examples to keep in mind are:

- the Boolean semiring, for ordinary accept/reject automata;
- the nonnegative reals, for probabilistic or weighted counting models;
- the tropical semiring, for shortest-path style problems.

General semiring language is useful because it unifies several automata theories at once. For the analytic results of this chapter, however, we will soon specialize to matrices over $\mathbb{R}$ or $\mathbb{C}$.

7.2 Weighted Finite Automata

Definition 7.1. A *weighted finite automaton* over alphabet Σ with n states is a triple

$$\mathcal{A} = (\alpha, \{A_\sigma\}_{\sigma\in\Sigma}, \beta),$$

where

- α is a $1 \times n$ row vector of initial weights,
- β is an $n \times 1$ column vector of final weights,
- each A_σ is an $n \times n$ transition matrix.

For a word $w = \sigma_1 \cdots \sigma_k$, its weight is

$$f_{\mathcal{A}}(w) = \alpha A_{\sigma_1} \cdots A_{\sigma_k} \beta.$$

The empty word has weight

$$f_{\mathcal{A}}(\varepsilon) = \alpha I \beta = \alpha\beta.$$

The matrix dimensions are worth noticing:

$$(1 \times n)(n \times n) \cdots (n \times n)(n \times 1) = (1 \times 1),$$

so the result is a scalar.

The formula is really a sum over hidden state paths. For a two-letter word $\sigma\tau$,

$$\alpha A_\sigma A_\tau \beta = \sum_{i,j,k} \alpha_i (A_\sigma)_{ij} (A_\tau)_{jk} \beta_k.$$

So one sums over all choices of start state i, intermediate state j, and final state k. This is exactly the path-sum interpretation of matrix multiplication.

In the Boolean semiring, the same formula says whether there exists an accepting path labeled by the word, so WFAs genuinely extend ordinary finite automata.

Example 7.2 (A Fibonacci WFA)**.** Let $\Sigma = \{a\}$ and define

$$\alpha = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad A_a = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Then

$$f_{\mathcal{A}}(a^k) = \alpha A_a^k \beta.$$

A standard induction shows that

$$A_a^k = \begin{pmatrix} F_{k-1} & F_k \\ F_k & F_{k+1} \end{pmatrix} \qquad (k \geq 1),$$

so

$$f_{\mathcal{A}}(a^k) = F_k.$$

Thus a two-state WFA already realizes the Fibonacci sequence.

7.3 Local Stochasticity Versus Properness

Now specialize to nonnegative weights. A WFA is *locally stochastic* if

- the initial vector α is a probability distribution on states, and
- for each state q,

$$\sum_{\sigma \in \Sigma} \sum_{q'} (A_\sigma)_{q,q'} + \beta_q = 1.$$

The intended interpretation is that from state q, the automaton either emits a symbol and changes state, or halts via the final weight β_q.

But this local condition does *not* by itself guarantee a probability distribution on finite strings.

Example 7.3 (A non-halting locally stochastic WFA)**.** Take one state, one letter a, and

$$\alpha = (1), \qquad A_a = (1), \qquad \beta = (0).$$

The local stochasticity condition holds, but the machine loops forever with probability 1 and halts with probability 0. So every finite string receives probability 0, and the total mass on Σ^* is also 0.

This is why we need a second notion.

Definition 7.4. A nonnegative WFA is *proper* if the total weight of all finite strings is exactly 1:

$$\sum_{w \in \Sigma^*} f_{\mathcal{A}}(w) = 1.$$

Equivalently, the model halts almost surely.

Chapter 8 will study properness systematically. For the present chapter, the main point is conceptual: local stochasticity is a step toward probability, not the whole story.

7.4 The Length Generating Function is Rational

From now on we work over $\mathbb{R}$ or $\mathbb{C}$. Define the aggregate matrix

$$A = \sum_{\sigma \in \Sigma} A_\sigma.$$

This forgets which symbol was emitted and keeps only the total one-step weight.

The length generating function of the WFA is

$$F(z) = \sum_{w \in \Sigma^*} f_{\mathcal{A}}(w) z^{|w|}.$$

Group words by length:

$$F(z) = \sum_{k \geq 0} z^k \sum_{|w|=k} f_{\mathcal{A}}(w).$$

A word of length k is exactly a k-tuple of symbols, so

$$\sum_{|w|=k} f_{\mathcal{A}}(w) = \sum_{\sigma_1, \ldots, \sigma_k \in \Sigma} \alpha A_{\sigma_1} \cdots A_{\sigma_k} \beta.$$

By distributivity,

$$\sum_{\sigma_1, \ldots, \sigma_k \in \Sigma} A_{\sigma_1} \cdots A_{\sigma_k} = \left(\sum_{\sigma \in \Sigma} A_\sigma \right)^k = A^k.$$

This step uses only distributivity, so it already makes sense over semirings. Hence

$$F(z) = \sum_{k \geq 0} z^k \alpha A^k \beta = \alpha \left(\sum_{k \geq 0} (zA)^k \right) \beta.$$

Now the real/complex specialization matters. As a *formal* matrix identity,

$$(I - zA) \sum_{k \geq 0} (zA)^k = I,$$

so

$$\sum_{k\geq 0}(zA)^k = (I - zA)^{-1}$$

as a matrix power series. Analytically, the same Neumann series converges for

$$|z| < 1/\rho(A),$$

where $\rho(A)$ is the spectral radius of A. So the same algebraic expression serves two roles: formally it is a power-series identity, and inside the disk of convergence it is an honest analytic rational function.

Theorem 7.5 (Rationality of WFA length generating functions)**.** *For a WFA over $\mathbb{R}$ or $\mathbb{C}$,*

$$F(z) = \alpha(I - zA)^{-1}\beta, \qquad A = \sum_{\sigma\in\Sigma} A_\sigma.$$

In particular, $F(z)$ is a rational function of z.

Because

$$(I - zA)^{-1} = \frac{\operatorname{adj}(I - zA)}{\det(I - zA)},$$

each entry is rational. The possible poles occur among the reciprocals of the nonzero eigenvalues of A, though cancellations can remove some of them after multiplication by α and β.

7.5 Rational Asymptotics and What Must Be Assumed

Chapter 4 analyzed rational generating functions in terms of their poles. The correct moral for WFAs is therefore this:

> The asymptotics are controlled by the dominant pole of the scalar rational function $F(z) = \alpha(I - zA)^{-1}\beta$, not merely by the raw eigenvalue multiplicities of A.

A safe statement is the following.

Proposition 7.6. *Suppose $F(z) = \alpha(I - zA)^{-1}\beta$ has a unique dominant pole at $z = 1/\lambda$ of order m. Then*

$$[z^n]F(z) \sim C\,\lambda^n n^{m-1}$$

for some constant $C \neq 0$.

This formulation avoids several pitfalls at once:

- multiple poles on the spectral circle can create oscillations,
- the boundary vectors α and β can cancel some eigen-directions,
- and the relevant order is the actual pole order of the scalar function, not merely the algebraic multiplicity of an eigenvalue of A.

For proper stochastic WFAs, the coefficients

$$p_n = [z^n]F(z) = \sum_{|w|=n} f_{\mathcal{A}}(w)$$

form the length distribution. In the simple aperiodic case with a unique simple dominant pole, this total probability decays geometrically with n.

Example 7.7 (Fibonacci again)**.** For the Fibonacci WFA,

$$A = A_a = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

so

$$F(z) = \alpha(I - zA)^{-1}\beta = \frac{z}{1 - z - z^2}.$$

This is the same generating function as in Chapters 1 and 4, and its dominant pole at $z = 1/\varphi$ gives

$$[z^n]F(z) = F_n \sim \frac{\varphi^n}{\sqrt{5}}.$$

The WFA viewpoint reaches the same asymptotic through matrix structure.

7.6 Recognizable Versus Rational Series

There is also a noncommutative version of the story. A formal series over words,

$$S = \sum_{w \in \Sigma^*} c_w \, w,$$

can be viewed as assigning a coefficient directly to each word rather than first collapsing everything to a length variable. Such a series is called *recognizable* if it comes from a WFA, and *rational* if it is built from finite series using sum, concatenation, and star.

Theorem 7.8 (Kleene–Schützenberger, used as a black box)**.** *A noncommutative formal series is recognizable if and only if it is rational.*

The star operation again requires the analogue of a zero constant term, for the same reason as in Chapter 3: one must avoid infinitely many ways to build the same total length. If we now send each word w to the commutative monomial $z^{|w|}$, concatenation of words becomes multiplication of powers of z because lengths add. That is the bridge from noncommutative rationality to the ordinary rational function $F(z)$ above.

7.7 What WFAs Can and Cannot Express Exactly

The rationality theorem gives an exact limitation. An exact WFA model always has a rational length generating function. Therefore it cannot exactly realize a genuine square-root algebraic length law such as the Catalan/Dyck behavior studied in Chapters 3–5.

This is one way to read the probabilistic hierarchy discussed by Icard [26]: finite-state weighted models sit at a rational level, while context-free probabilistic models can produce algebraic length behavior above that level. The exact details of that hierarchy are not needed here. The main message is simpler: exact WFA models occupy the rational regime.

7.8 Why This Matters for Language Models

A large language model is not literally a WFA. Still, the WFA viewpoint matters in two different ways.

First, it provides a tractable exact model class. When a sequence model really is finite-state weighted, its length generating function is rational and the pole analysis of Part I applies exactly.

Second, it provides a research program for approximation. If some finite-state surrogate approximates a language model well on finite strings or finite contexts, that may still be useful.

But one must be careful: closeness of distributions or coefficients on a finite range does *not* automatically imply closeness of dominant singularities or asymptotic regimes. Chapter 12 returns to this issue and formulates approximation targets more carefully.

A good rule of thumb is:

- exact WFA model ⇒ exact rational length generating function;
- observed square-root algebraic correction ⇒ evidence against an exact WFA description;
- useful finite-length approximation does not automatically imply the same asymptotic singularity type.

We will use that distinction repeatedly later.

Further Reading

- **M. Droste, W. Kuich, and H. Vogler** (eds.), *Handbook of Weighted Automata* (2009) [15] — the standard general reference for weighted automata over semirings.
- **J. Berstel and C. Reutenauer**, *Noncommutative Rational Series with Applications* (2011) [7] — for recognizable versus rational series and the noncommutative Kleene–Schützenberger theorem.
- **T. Icard**, "Calibrating generative models: the probabilistic Chomsky–Schützenberger hierarchy" (2020) [26] — for one formal version of the rational-versus-algebraic probabilistic hierarchy.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for the rational-function asymptotics that make the WFA theorem analytically useful.

Chapter 8

Probabilistic Grammars and Tightness

The chapters so far mostly studied counting generating functions. This chapter changes the question. We now care about objects that are supposed to define genuine probability distributions on finite outputs. In that setting, local normalization is not enough. A recursive model may still place positive probability on never terminating, so that the total mass on finite strings is strictly less than 1.

That is the conceptual core of the chapter. For probabilistic grammars and autoregressive language models alike, we must distinguish:

- local conditional probabilities,
- the global probability of finite outputs,
- and the question of almost-sure termination.

The condition

$$F(1) = 1$$

for a length probability generating function is the right probabilistic sanity check. It says we truly have a probability distribution on finite lengths. It is *not* a blanket prerequisite for analytic combinatorics as a formal method; defective generating functions can still be studied analytically. What fails first when $F(1) < 1$ is the intended probability interpretation.

8.1 Probabilistic Context-Free Grammars

A *probabilistic context-free grammar* (PCFG) is a context-free grammar in which, for each nonterminal A, the productions with left-hand side A are assigned probabilities summing to 1.

If a derivation uses productions $r_1, r_2, \ldots, r_m$, then its probability is

$$\prod_{i=1}^{m} p(r_i).$$

The probability of a terminal string w is the sum of those derivation probabilities over all derivations producing w. So ambiguity matters exactly as it did in the counting setting, except that now we sum probabilities rather than counts.

Example 8.1. Suppose the grammar has one rule of probability q and one rule of probability $1 - q$. Then a derivation using the first rule three times and the second twice has probability $q^3(1 - q)^2$.

8.2 Branching-Process Intuition

A PCFG derivation can be viewed as a population of unfinished nonterminals. Each time a nonterminal is expanded, it disappears and is replaced by whatever nonterminals occur on the right-hand side of the chosen production. That is the branching-process viewpoint.

For example, consider

$$S \to SS \quad (q), \qquad S \to a \quad (1-q).$$

One expansion of S produces either two new copies of S or none. So the expected number of new nonterminals created by one old nonterminal is

$$2q.$$

If this expectation is large, the derivation tends to keep reproducing unfinished work instead of terminating.

8.3 Consistency and the First-Moment Matrix

Definition 8.2. A PCFG is *consistent* if the total probability of finite terminal derivations is 1.

Equivalently, starting from the start symbol, the derivation terminates almost surely.

The standard summary statistic is the *first-moment matrix* M. Its entry $M_{A,B}$ is the expected number of occurrences of nonterminal B created by one random expansion of A.

For the one-nonterminal grammar above,

$$M = [2q].$$

So the spectral radius is just

$$\rho(M) = 2q.$$

Booth and Thompson analyzed consistency through this matrix, but one must quote the sharp theorem carefully. The safe lessons for this book are:

- if $\rho(M) < 1$, then the grammar is subcritical and terminates almost surely;
- if $\rho(M) > 1$, then there is positive probability of infinite derivation;
- the critical case $\rho(M) = 1$ is delicate and can contain both ordinary critical branching behavior and degenerate loops.

The toy grammar

$$S \to S \quad (1)$$

shows why care is needed: here $\rho(M) = 1$, but the derivation never terminates. So any blanket "iff $\rho(M) \leq 1$" formulation needs extra nondegeneracy hypotheses.

8.4 Length PGFs and What 'F(1)=1' Means

Let $q_{A,n}$ be the probability that a derivation rooted at nonterminal A yields a terminal string of length exactly n. Define the length PGF

$$F_A(z) = \sum_{n \geq 0} q_{A,n} z^n.$$

Then

$$F_A(1) = \sum_{n \geq 0} q_{A,n}$$

is exactly the total probability that A eventually produces a finite terminal string.

So for the start symbol S,

$$F_S(1) = 1$$

means consistency, while

$$F_S(1) < 1$$

means that positive probability mass has escaped into infinite derivations.

That distinction is probabilistic, not purely analytic. A defective PGF may still have a radius of convergence and singularities; it simply no longer encodes a genuine probability distribution on finite outputs.

8.5 Chi's Theorem

Chi proved an important result for PCFG estimation from treebanks.

Theorem 8.3 (Chi, used as a black box)**.** *A maximum-likelihood PCFG estimated from a finite treebank is consistent.*

For this book, the right interpretation is modest: maximum-likelihood estimation removes one foundational obstruction, namely defectiveness. It does *not* automatically guarantee all the later hypotheses one might want for singularity analysis, such as algebraicity, aperiodicity, or a specific dominant singularity type.

8.6 Autoregressive Language Models

Now turn to autoregressive models over a vocabulary Σ with an explicit end-of-sequence token EOS. Such a model specifies conditional probabilities

$$p(\cdot \mid h)$$

for every history $h \in \Sigma^*$. The probability of a finite string

$$w = w_1 w_2 \cdots w_n$$

is

$$p(w) = p(\text{EOS} \mid w_1 \cdots w_n) \prod_{t=1}^{n} p(w_t \mid w_1 \cdots w_{t-1}).$$

Again, local normalization at each history does not by itself imply that

$$\sum_{w \in \Sigma^*} p(w) = 1.$$

The model may keep generating forever with positive probability.

8.7 Tightness and the Hazard Criterion

To formalize this, one enlarges the sample space to include infinite strings as well as finite ones. We use that measure-theoretic construction as a black box and focus on the resulting definition.

Definition 8.4. An autoregressive language model is *tight* if it halts almost surely, equivalently if the total probability of finite strings is 1.

The key quantity is the average stopping hazard at time t. Let

$$\widetilde{p}_{\mathrm{EOS}}(t)$$

denote the conditional probability of halting at step t, averaged over all histories of length $t-1$ that have survived that long.

The crucial product formula is

$$\Pr(T \geq t+1) = \prod_{s=1}^{t} \big(1 - \widetilde{p}_{\mathrm{EOS}}(s)\big),$$

where T is the random halting time. This simply says: to survive past time t, the process must fail to halt at each earlier step.

From this one gets the tightness criterion used by Du et al.

Theorem 8.5 (Du et al., used as a black box)**.** *An autoregressive language model is tight if and only if*

$$\sum_{t=1}^{\infty} \widetilde{p}_{\mathrm{EOS}}(t) = \infty.$$

The intuition is that the infinite product above vanishes exactly when the cumulative stopping hazard is large enough. So divergent total hazard means almost-sure halting.

A very useful sufficient condition is the following. If there exists $\varepsilon > 0$ such that

$$p(\mathrm{EOS} \mid h) \geq \varepsilon \qquad \text{for every history } h,$$

then certainly

$$\widetilde{p}_{\mathrm{EOS}}(t) \geq \varepsilon$$

for every t, so the divergent-series criterion holds and the model is tight.

8.8 Examples

Example 8.6 (A non-tight model)**.** Suppose along a deterministic path the EOS probability at time t is

$$\widetilde{p}_{\mathrm{EOS}}(t) = 2^{-t}.$$

Then

$$\sum_{t \geq 1} 2^{-t} < \infty,$$

so the model is not tight: with positive probability it generates forever.

Example 8.7 (Barely tight and heavy-tailed)**.** Suppose instead that

$$\widetilde{p}_{\mathrm{EOS}}(t) = \frac{1}{t+1}.$$

Then

$$\Pr(T \geq n+1) = \prod_{t=1}^{n} \frac{t}{t+1} = \frac{1}{n+1},$$

so

$$\Pr(T = n) = \Pr(T \geq n)\widetilde{p}_{\mathrm{EOS}}(n) = \frac{1}{n(n+1)} \sim \frac{1}{n^2}.$$

This model is tight, but only barely. Its expected length diverges because

$$\sum_{n \geq 1} n \Pr(T = n) \asymp \sum_{n \geq 1} \frac{1}{n}$$

diverges.

The point is not that all borderline models have the same singularity type. They do not. Some critical branching models produce square-root singularities, while some heavy-tailed sequential hazards lead to softer singular behavior. The only safe universal message is that near-critical termination can produce heavy tails and long outputs.

8.9 Transformer Tightness

Du et al. prove a tightness theorem for a broad class of softmax transformer language models. We treat that theorem as a black box here.

Theorem 8.8 (Du et al., black box)**.** *Under the hypotheses of their framework, softmax transformer language models are tight.*

We deliberately do *not* reproduce the proof sketch. The real argument is architectural and measure-theoretic, and a vague appeal to "softmax is always positive" is not enough to justify it. For this chapter, the safe takeaway is simply that there are important modern model classes for which tightness has been proved, not guessed.

8.10 The Common Lesson

The PCFG story and the autoregressive story are parallel.

- In a PCFG, the local normalization is over productions expanding one nonterminal.
- In an autoregressive model, the local normalization is over next-token probabilities given a history.

In both settings, local normalization does not by itself imply a proper distribution on finite outputs. One still needs an almost-sure termination theorem.

Once properness is known, the length PGF satisfies

$$F(1) = 1.$$

That is the correct probability-theoretic checkpoint. It does not, by itself, determine rationality, algebraicity, or singularity type. Those are separate structural questions.

Preview of Chapter 9

The next chapter turns to entropy and typical-set growth. There we will need to keep several objects distinct: counting generating functions, length PGFs, and entropy-related quantities. Tightness is the first step because it tells us that the probabilistic objects we write down really are probability distributions on finite outputs.

Further Reading

- **T. L. Booth and R. A. Thompson**, "Applying probability measures to abstract languages" (1973) [9] — the classical source on consistency for probabilistic grammars.
- **Z. Chi**, "Statistical properties of probabilistic context-free grammars" (1999) [11] — for consistency of maximum-likelihood PCFG estimation from treebanks.
- **L. Du et al.**, "A measure-theoretic characterization of tight language models" (2023) [16] — the main reference for tightness of autoregressive models.
- **R. Cotterell et al.**, "Formal aspects of language modeling" (2023) [13] — for a broader probabilistic perspective on language models as measures on strings.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for the analytic side of probability generating functions once properness has been established.

Chapter 9

Entropy Rate and the Identity $h = \log(1/R)$

This chapter builds a bridge between information theory and generating functions. The bridge is real, but one must keep the objects straight. There are three related quantities that are easy to confuse:

- the Shannon entropy rate of a probabilistic source,
- the growth rate of a typical set,
- and the length distribution of outputs from a halting model.

Only the second of these is directly encoded by the counting-type generating function that leads to the identity

$$h = \log_2(1/R). \tag{9.1}$$

A main goal of the chapter is to make that distinction explicit.

9.1 Entropy Rate

Let $X_1, X_2, \ldots$ be a stationary stochastic process over a finite alphabet Σ. Define the block entropy

$$H_n := H(X_1^n) = -\sum_{w \in \Sigma^n} p(w) \log_2 p(w),$$

where $p(w) = \Pr(X_1 \cdots X_n = w)$.

Proposition 9.1. *For a stationary finite-alphabet source, the limit*

$$h = \lim_{n \to \infty} \frac{H_n}{n}$$

exists. Moreover,

$$h = \lim_{n \to \infty} H(X_n \mid X_1^{n-1}).$$

Proof sketch. The chain rule gives

$$H_{m+n} = H(X_1^{m+n}) \leq H(X_1^m) + H(X_{m+1}^{m+n}).$$

By stationarity,

$$H(X_{m+1}^{m+n}) = H(X_1^n),$$

so (H_n) is subadditive:

$$H_{m+n} \leq H_m + H_n.$$

A standard subadditivity theorem therefore implies that H_n/n has a limit.

Also,

$$H_n = \sum_{t=1}^{n} H(X_t \mid X_1^{t-1})$$

by the chain rule. Stationarity implies that the sequence of conditional entropies

$$H(X_n \mid X_1^{n-1})$$

is nonincreasing in n and bounded below by 0, so it converges. Its Cesàro average is exactly H_n/n, hence both limits agree. □

The quantity h is the average uncertainty per new symbol once arbitrarily long context is known. It always satisfies

$$0 \leq h \leq \log_2 |\Sigma|.$$

The upper bound is attained by the uniform i.i.d. source; in that case every new symbol is maximally unpredictable.

9.2 Shannon–McMillan–Breiman and Typical Sets

From this point through the typical-set construction, assume in addition that the stationary source is ergodic. The key imported theorem is the asymptotic equipartition property.

Theorem 9.2 (Shannon–McMillan–Breiman, used as a black box)**.** *For an ergodic stationary source with entropy rate h,*

$$-\frac{1}{n} \log_2 p(X_1 \cdots X_n) \to h$$

almost surely.

This motivates the usual typical sets.

Definition 9.3. For $\varepsilon > 0$, the ε-typical set at length n is

$$T_n^{(\varepsilon)} = \left\{ w \in \Sigma^n : \left| -\frac{1}{n} \log_2 p(w) - h \right| < \varepsilon \right\}.$$

For large n, most of the probability mass lies in $T_n^{(\varepsilon)}$, and every typical word has probability between

$$2^{-n(h+\varepsilon)} \quad \text{and} \quad 2^{-n(h-\varepsilon)}.$$

From this one obtains the standard cardinality estimate:

$$(1 - \delta_n) 2^{n(h-\varepsilon)} \leq |T_n^{(\varepsilon)}| \leq 2^{n(h+\varepsilon)},$$

where

$$\delta_n = 1 - \Pr(X_1^n \in T_n^{(\varepsilon)}) \to 0.$$

The lower bound comes from the fact that the typical set carries probability close to 1, while each individual typical word has probability at most $2^{-n(h-\varepsilon)}$.

9.3 A Corrected Typical-Set Identity

The displayed bounds for a *fixed* ε only pin down the exponential growth rate up to an ε-window. To obtain the exact exponent h, we let the window shrink with n.

Choose a sequence

$$\varepsilon_n \downarrow 0$$

slowly enough that

$$\Pr(X_1^n \in T_n^{(\varepsilon_n)}) \to 1.$$

(One can do this because the Shannon–McMillan–Breiman theorem says the deviations from h vanish almost surely.) Now define

$$a_n := \left|T_n^{(\varepsilon_n)}\right|.$$

For completeness one may set $a_0 = 1$; the asymptotic argument concerns large n. There is no single canonical choice of the sequence ε_n, but the exponential growth rate below does not depend on that choice.

From the cardinality bounds we get

$$(1-\delta_n)2^{n(h-\varepsilon_n)} \le a_n \le 2^{n(h+\varepsilon_n)},$$

with $\delta_n \to 0$. Taking logarithms and dividing by n gives

$$h - \varepsilon_n + \frac{1}{n}\log_2(1-\delta_n) \le \frac{1}{n}\log_2 a_n \le h + \varepsilon_n.$$

Since both error terms go to 0, we obtain

$$\frac{1}{n}\log_2 a_n \to h.$$

Therefore

$$a_n^{1/n} \to 2^h.$$

Now form the counting-type generating function

$$A(z) = \sum_{n\ge 0} a_n z^n.$$

By Cauchy–Hadamard,

$$\frac{1}{R} = \limsup_{n\to\infty} a_n^{1/n} = 2^h.$$

So indeed

$$h = \log_2(1/R).$$

This is the corrected form of the identity: it applies to an *entropy-typical counting generating function*, built from a shrinking typicality window.

9.4 A Crucial Distinction: Counting Growth vs. Length Distribution

The identity above does *not* apply to the length PGF of a halting model. If

$$P(z) = \sum_{n\ge 0} p_n z^n, \qquad p_n = \Pr(|W| = n),$$

then

$$P(1) = \sum_{n \geq 0} p_n = 1$$

for a proper model, so the radius of convergence satisfies

$$R_P \geq 1.$$

Thus

$$\log_2(1/R_P) \leq 0,$$

which cannot encode a positive entropy rate.

So we must keep two generating functions separate.

1. A **counting-type typical-set GF**

$$A(z) = \sum a_n z^n,$$

whose coefficients grow like 2^{nh} and whose radius encodes entropy rate.

2. A **length probability generating function**

$$P(z) = \sum p_n z^n,$$

whose singularity structure describes tail behavior of output length.

A concrete example helps. For a halting model with geometric length distribution,

$$p_n = q(1-q)^n,$$

one gets

$$P(z) = \sum_{n \geq 0} q(1-q)^n z^n = \frac{q}{1-(1-q)z},$$

whose radius is

$$R_P = \frac{1}{1-q} > 1.$$

This radius says something about how fast long outputs become unlikely. It says nothing by itself about Shannon entropy rate.

9.5 Finite-State Sources: Topological vs. Shannon Entropy

This is the place where one must be especially careful. For a finite-state automaton or subshift of finite type, the number of *allowed paths* of length n often grows like

$$\rho(M)^n,$$

where M is an adjacency matrix. The corresponding logarithm

$$\log_2 \rho(M)$$

is a *support-growth* or *topological* entropy.

That is not the same as Shannon entropy rate of a probabilistic source in general.

Example 9.4 (Biased one-state source)**.** Take one state and two emitted symbols, 0 and 1, with probabilities 0.9 and 0.1. Every binary word of length n is allowed, so the support-counting growth rate is

$$2^n, \qquad \log_2 2 = 1.$$

But the Shannon entropy rate is

$$h = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 < 1.$$

So path counting and Shannon entropy are genuinely different quantities.

The identity $h = \log_2(1/R)$ in this chapter applies to the entropy-typical counting object, not to naive support counting of all admissible paths.

9.6 Entropy Estimation: Classical Results and Modern Reinterpretation

The historical literature estimates Shannon entropy rate directly, not by literally writing down a typical-set generating function.

Shannon (1951). Shannon's guessing experiments [51] produced upper and lower bounds for the entropy rate of printed English. In the language of this chapter, one may *reinterpret* those estimates as probing the exponential growth rate of near-typical sets. That reinterpretation is modern and conceptual; it is not something Shannon himself formulated in generating-function language.

Kontoyiannis et al. (1998). Kontoyiannis, Algoet, Suhov, and Wyner [29] study entropy-rate estimation through longest-match statistics. Their work is rigorous information theory. The connection to generating functions is again interpretive: these estimators measure the same entropy rate h that appears in the exponential growth of entropy-typical sets.

Modern neural estimates. Takahashi and Tanaka-Ishii [58] and Scheibner, Smith, and Bialek [49] provide increasingly refined empirical estimates of the entropy rate of natural language using neural models and long contexts. It is reasonable to read these as sharper measurements of the same quantity h, but the present chapter does not construct a canonical "English generating function" whose radius is being measured directly.

9.7 A Research Hypothesis: Convergence Rate and Singularity Type

The location of the dominant singularity of a typical-set counting GF determines the entropy rate. A more ambitious question is whether the *type* of that singularity leaves a detectable signature in the finite-context convergence of entropy estimators.

This chapter does *not* prove such a theorem. Instead, we record the idea as a research hypothesis.

- In simple finite-memory models, entropy estimates often converge rapidly.
- In more structured or long-range sources, the convergence can be much slower.
- It is plausible that singularity type on the counting side and estimator convergence on the information-theoretic side are related.

But turning that philosophy into a theorem would require a much more precise bridge than we have built here. So the safe conclusion is only this: long-context entropy drift is suggestive, not diagnostic. It may be a clue about deeper singularity structure, but at present that is part of the research program rather than an established consequence of Chapters 1–9.

Summary

This chapter's strongest rigorous conclusion is the corrected identity

$$h = \log_2(1/R),$$

where R is the radius of convergence of an entropy-typical counting generating function built from a shrinking typicality window. Its strongest conceptual lesson is the distinction between that counting object and the length PGF of a halting model.

The next parts of the book must keep that distinction visible. Support growth, entropy rate, and output-length tails are related, but they are not interchangeable.

Further Reading

- **C. E. Shannon**, "Prediction and entropy of printed English" (1951) [51] — the historical starting point for entropy-rate estimation.
- **I. Kontoyiannis, P. H. Algoet, Y. M. Suhov, and A. J. Wyner**, "Nonparametric entropy estimation for stationary processes and random fields" (1998) [29] — a rigorous source on entropy-rate estimation.
- **W. Szpankowski**, *Average Case Analysis of Algorithms on Sequences* (2001) [57] — for analytic tools around strings, tries, and entropy-related sequence problems.
- **S. Takahashi and K. Tanaka-Ishii**, "Cross entropy of neural language models at infinity—a new bound of the entropy rate" (2018) [58] — for modern neural extrapolation of entropy rate.
- **D. Scheibner, C. Smith, and W. Bialek**, "Entropy rate of natural language revisited" (2025) [49] — for long-context entropy estimation using frontier language models.

Part III

Large Language Models and Their Formal Theory

We now turn to the objects that motivate the entire enterprise: large language models. Chapter 10 introduces an LLM as a probability distribution on Σ^* defined by the autoregressive factorization, and proves that every softmax Transformer is tight. Chapter 11 surveys the formal-language theory of Transformers — the TC^0 complexity ceiling, Hahn's impossibility results, and Merrill–Sabharwal's chain-of-thought breakthrough — explaining what these results mean for the analytic structure of LLM output distributions. Chapter 12 asks: if we cannot describe a Transformer exactly as a weighted finite automaton, how well can we approximate it by one? Four approaches (distillation, architectural equivalence, L^* extraction, spectral/AAK methods) are surveyed, each connecting back to the rational-GF machinery of Parts I–II.

Chapter 10

Autoregressive Language Models

Earlier chapters spoke abstractly about distributions on strings. This chapter fixes the concrete convention used for the rest of the book when the strings come from a large language model. The main conceptual move is simple but important: the mathematical object is the induced distribution on outputs, not the neural network parameters themselves.

To make that precise, we must first settle the sample space cleanly.

10.1 Tokens, Vocabulary, and End-of-Sequence

Let Σ denote the vocabulary of ordinary content tokens, and let

$$\mathrm{EOS} \notin \Sigma$$

be a distinguished end-of-sequence symbol. The model emits tokens from the enlarged alphabet

$$\Sigma \cup \{\mathrm{EOS}\},$$

but the actual finite outputs are strings in

$$\Sigma^*.$$

A generated string ends when the first EOS is emitted. Thus:

- histories during generation are strings in Σ^*,
- outputs are strings in Σ^*,
- and EOS is a stopping action, not a symbol that appears inside the returned output string.

This convention is the cleanest one for the later generating functions. It also matches the length PGF of Chapter 8, where length means "number of non-EOS tokens before termination."

Tokenization still matters. A model over characters, one over words, and one over BPE tokens define genuinely different distributions on Σ^*. All later counting, length, and entropy questions depend on that choice of atomic units.

10.2 The Autoregressive Factorization

An autoregressive language model specifies, for every history $h \in \Sigma^*$, a conditional distribution

$$p(\cdot \mid h) \in \Delta(\Sigma \cup \{\mathrm{EOS}\}).$$

If the model outputs

$$w = w_1 w_2 \cdots w_n \in \Sigma^*,$$

then its probability is

$$p(w) = p(\mathrm{EOS} \mid w_1 \cdots w_n) \prod_{t=1}^{n} p(w_t \mid w_1 \cdots w_{t-1}).$$

The empty output has probability

$$p(\varepsilon) = p(\mathrm{EOS} \mid \varepsilon).$$

Definition 10.1 (Autoregressive language model)**.** An *autoregressive language model* over content vocabulary Σ is a family of conditional distributions

$$\mathcal{M} = \{p(\cdot \mid h) : h \in \Sigma^*\},$$

where each conditional is supported on $\Sigma \cup \{\mathrm{EOS}\}$. The induced output distribution on Σ^* is defined by the factorization above.

The conditionals after a history already containing EOS are irrelevant, because such histories are never reached by the sampling process. One may define them arbitrarily or ignore them entirely.

This is the right abstraction for the book. The network parameters are one way to represent the family $\mathcal{M}$; they are not themselves the primary mathematical object.

10.3 Softmax and Temperature

In practice, modern LLMs compute the conditional distribution by applying softmax to a vector of logits:

$$p(\sigma \mid h) = \frac{e^{\ell_\sigma(h)}}{\sum_{\tau \in \Sigma \cup \{\mathrm{EOS}\}} e^{\ell_\tau(h)}}.$$

For every finite temperature parameter $T > 0$, the tempered distribution is

$$p_T(\sigma \mid h) = \frac{e^{\ell_\sigma(h)/T}}{\sum_\tau e^{\ell_\tau(h)/T}}.$$

At $T = 1$ this is the native model. For every fixed finite $T > 0$, every token in $\Sigma \cup \{\mathrm{EOS}\}$ keeps strictly positive probability. As $T \to 0$, mass concentrates on the set of maximizers of the logits; if there are ties, the limiting distribution is supported on the tied argmax set.

Temperature will matter later when we distinguish local tokenwise rescaling from more global Gibbs-style reweighting of full strings.

10.4 Transformers in One Paragraph

A transformer is one specific neural parameterization of the history-to-logit map

$$h \longmapsto \ell(h).$$

For the mathematics of this chapter, the exact architecture matters less than the existence of that map. The only reason to mention the architecture at all is that some later theorems, such as tightness results of Du et al., are proved for particular model classes rather than for arbitrary conditional families.

10.5 Tightness

Chapter 8 introduced the right probability-theoretic notion.

Definition 10.2. An autoregressive language model is *tight* if the total probability of finite outputs is 1:

$$\sum_{w \in \Sigma^*} p(w) = 1.$$

Equivalently, generation halts almost surely.

For a tight model, the length distribution

$$p_n = \Pr(|W| = n)$$

is a genuine probability distribution on $\mathbb{Z}_{\geq 0}$.

We use the Du et al. result as a black box.

Theorem 10.3 (Du et al., black box)**.** *Under the hypotheses of their framework, softmax transformer language models are tight.*

We deliberately do not reproduce a proof sketch here. Positivity of softmax alone is not enough for a self-contained proof, and the real argument is architectural and measure-theoretic. For this chapter, the safe lesson is simply that there are important LLM classes for which properness has been proved, not merely hoped for.

10.6 Two Generating Functions Attached to an LLM

Once the output distribution is fixed, there are at least two different generating functions one might attach to it.

1. The length PGF. For a tight model, define

$$P(z) = \sum_{n \geq 0} p_n z^n, \qquad p_n = \Pr(|W| = n).$$

This is the right object for questions about how long the model tends to generate before stopping.

2. A counting-type typical-set GF. If one later chooses an entropy notion and a typicality criterion as in Chapter 9, one can count how many outputs of length n are "typical" and form a second generating function

$$A(z) = \sum_{n \geq 0} a_n z^n.$$

This is a counting object, not a length-probability object.

The point of this section is not to define every such counting GF in full detail again. It is to prevent confusion: 'P(z)' and 'A(z)' answer different questions and generally have different radii and different singularity interpretations.

10.7 Prefix Dependence and the Markov Viewpoint

There is one useful formal viewpoint. The sampling process can be seen as a Markov chain on the countable state space of finite prefixes Σ^*. From any history there are only finitely many next-token choices, so the branching is large but still finite at each step.

However, Chapter 9's entropy-rate theory for stationary infinite processes does *not* transfer automatically to this terminated setting. A tight autoregressive model produces a finite output and then stops. To speak about an entropy rate in the stationary sense, one would need an additional construction, such as an associated infinite process or a conditioning scheme. We therefore avoid claiming in this chapter that a generic tight LLM comes with a Chapter 9 entropy rate for free.

What we can say safely is simpler:

- finite-context models behave more like finite-order Markov sources,
- full-prefix models can depend on the entire history,
- and the generating functions of their output distributions need not be rational unless further finite-state structure is present.

10.8 A Unigram Example

Consider the simplest autoregressive model: at every step, independently of history, emit a non-EOS token with total probability $1 - q$ and emit EOS with probability q.

Then

$$\Pr(|W| = n) = q(1-q)^n \qquad (n \geq 0),$$

so the length PGF is

$$P(z) = \sum_{n \geq 0} q(1-q)^n z^n = \frac{q}{1-(1-q)z}.$$

This is a rational function with a simple pole at

$$z = \frac{1}{1-q} > 1.$$

So even the most trivial autoregressive model already fits the rational pole picture of Part I.

Preview

The next two chapters ask how far that rational picture extends.

- Chapter 11 studies what kinds of formal-language behavior transformer architectures can and cannot support.
- Chapter 12 asks under what approximation targets a finite-state weighted model is a useful surrogate for a neural language model.

Those are the chapters where "close to rational" must be made precise. This chapter only sets up the underlying probabilistic object and the two generating functions one might study.

Further Reading

- **A. Vaswani et al.**, "Attention is all you need" (2017) [59] — the original transformer architecture paper.
- **L. Du et al.**, "A measure-theoretic characterization of tight language models" (2023) [16] — the main reference for tightness of autoregressive language models.

- **R. Cotterell et al.**, "Formal aspects of language modeling" (2023) [13] — for a broader formal perspective on language models as probability measures on strings.
- **T. Icard**, "Calibrating generative models: the probabilistic Chomsky–Schützenberger hierarchy" (2020) [26] — for one formal hierarchy comparing rational and algebraic probabilistic models.

Chapter 11

Formal Language Theory of Transformers

This chapter has one guiding question: what do transformer expressivity results suggest about the kinds of generating functions one should expect from transformer-based language models?

The word *suggest* matters. Results in formal-language theory and circuit complexity are primarily about *recognition* and *simulation*: what classes of languages a model can recognize, or what computational devices it can simulate. Generating-function claims are about *counting sequences* or *output distributions*. The bridge between these two worlds is often illuminating, but it is not automatic. So this chapter is best read as a survey plus a modeling stance, not as a chapter that fully proves the analytic conclusions it points toward.

11.1 Circuit Classes as a Measuring Stick

The complexity classes that appear most often in transformer expressivity papers are the following.

Definition 11.1.

- AC^0: constant-depth, polynomial-size Boolean circuits with unbounded-fan-in AND, OR, and NOT gates.
- TC^0: the same, but with threshold gates added.
- NC^1: logarithmic-depth, polynomial-size circuits with bounded fan-in.

Intuitively, AC^0 can aggregate many bits quickly but is weak at global counting. A classical example is parity: deciding whether a binary string has an even number of ones is not in AC^0, but it is in TC^0 by a constant-depth threshold-circuit construction. It is *not* computed by a single threshold gate; the point is only that threshold gates make parity easy in constant depth.

The class NC^1 is stronger because logarithmic depth allows recursive composition. Many expression-evaluation and parsing tasks naturally live there. In this chapter we suppress uniformity details; the cited papers make those conventions precise.

11.2 Single-Pass Transformers and the TC^0 Heuristic

A strong line of work, surveyed in [54], shows that several fixed-depth transformer variants sit inside or near TC^0-like expressivity classes when one studies single-pass recognition.

Theorem 11.2 (Merrill, Sabharwal, Smith 2022, black box)**.** *Saturated transformers of constant depth can be simulated by polynomial-size constant-depth threshold circuits. In particular, the languages they recognize lie in* TC^0*.*

The point for us is not the proof technology. It is the resulting modeling picture: fixed-depth single-pass transformers appear much closer to finite-state or threshold-circuit behavior than to full context-free parsing power.

But this is only a starting point for analytic combinatorics. A recognition upper bound does *not* by itself prove that an output distribution has a rational generating function. What it does is motivate a working hypothesis:

> For single-pass transformers, finite-state / WFA-like approximations are plausible enough to study seriously.

That is a modeling stance, not a theorem yet.

11.3 Impossibility Results as Warnings

Hahn's theorem gives a complementary message.

Theorem 11.3 (Hahn 2020, black box)**.** *Fixed-depth soft-attention transformers cannot uniformly solve certain length-unbounded tasks such as parity and multi-type Dyck recognition.*

The safest interpretation is architectural, not analytic. These theorems say that some globally sensitive or deeply nested behaviors are hard for bounded-depth soft-attention models to represent uniformly across all lengths. They do *not* directly tell us that a particular pole or branch point must or must not appear in a generating function.

So the correct lesson is modest: lower bounds on recognition complexity warn us against treating single-pass transformers as arbitrary pushdown-like devices. They do not by themselves determine the analytic type of an output distribution.

11.4 Exact Rational Regimes: RNNs and Probabilistic Automata

There are, however, exact positive results on the neural side.

Theorem 11.4 (Svete and Cotterell 2023, black box)**.** *A certain class of recurrent language models is exactly equivalent to a corresponding class of probabilistic finite-state automata.*

This matters because Chapter 7 already told us what finite-state weighted models imply analytically: rational length generating functions, pole-based asymptotics, and a finite-state matrix description. So the theorem shows that at least some neural sequence models do sit exactly inside the rational regime.

That does *not* mean all recurrent models do, and it certainly does not mean all transformer models do. The safe conclusion is narrower: there is an exact neural-to-rational corner of the landscape, so rational approximations are not an empty fantasy.

11.5 Transformers Simulating Weighted Automata

Another important direction is constructive simulation.

Theorem 11.5 (Rizvi et al. 2024, black box)**.** *Transformers with suitable attention mechanisms can exactly or approximately simulate weighted finite automata on inputs up to a prescribed length bound.*

Again, the correct analytic takeaway is limited. Exact simulation of a WFA means rational behavior for that simulated system. Approximate simulation on strings up to length T means that finite-state behavior can be reproduced well on a finite range. It does *not* by itself imply

that the transformer's global generating function has approximately the same singularities or asymptotics as the WFA for all lengths.

So the right lesson is:

- exact WFA behavior is available to transformers,
- finite-length WFA approximation is a meaningful program,
- but asymptotic analytic type is a stronger question than finite-length simulation.

That stronger question will be taken up in Chapter 12 rather than assumed here.

11.6 Chain of Thought Changes the Picture

All of the previous discussion concerns models that map an input to an output in a bounded-depth forward pass. Chain-of-thought changes that regime by allowing the model to emit intermediate tokens and then condition on them.

Theorem 11.6 (Merrill and Sabharwal 2024, black box)**.** *Under the hypotheses of their framework, a transformer with polynomially many chain-of-thought steps can realize computations all the way up to polynomial time.*

For our purposes, the crucial implication is qualitative. Once intermediate computation is allowed, the finite-state / rational heuristic becomes much less compelling. The model may now simulate much richer mechanisms than a single-pass bounded-depth device.

What should one conclude analytically? Not that chain-of-thought automatically implies algebraic generating functions or an $n^{-3/2}$ law. That would be far too strong. The safe statement is only this:

> Chain-of-thought makes context-free-like, pushdown-like, and even richer behavior much more plausible than in the single-pass regime.

Whether that plausibility shows up in the actual output-support or output-probability generating functions of a trained model is a separate question.

11.7 A Three-Level Synthesis

The chapter's message can be compressed into three levels of confidence.

1. **Theorems.** We have cited exact results about recognition bounds, exact finite-state equivalences, and exact or finite-length simulation theorems.
2. **Modeling stance.** For single-pass transformers, it is reasonable to investigate rational / WFA-style approximations first.
3. **Open analytic question.** How those expressivity facts constrain the actual generating functions of output distributions is not settled by this chapter alone.

This is the stance the rest of the book will adopt. Single-pass models are treated as candidates for rational approximation. Chain-of-thought models are treated as candidates for richer algebraic or beyond-algebraic behavior. Both are hypotheses to be tested, not conclusions already extracted from the cited complexity theorems.

Preview

Chapter 12 turns this survey perspective into an operational question: if a transformer is WFA-like in a useful sense, what exactly should be approximated, and what analytic conclusions survive that approximation? That is where the approximation target must finally be made precise.

Further Reading

- **L. Strobl et al.**, "What formal languages can transformers express? A survey" (2024) [54] — the best overview of transformer expressivity results.
- **W. Merrill, A. Sabharwal, and N. A. Smith**, "Saturated transformers are constant-depth threshold circuits" (2022) [36] — the foundational TC^0 upper-bound paper.
- **M. Hahn**, "Theoretical limitations of self-attention in neural sequence models" (2020) [23] — for parity and Dyck-style lower bounds on soft-attention models.
- **A. Svete and R. Cotterell**, "Recurrent neural networks and weighted finite-state transducers" (2023) [56] — for an exact finite-state corner of neural sequence modeling.
- **M. Rizvi-Martel et al.**, "Simulating weighted automata over sequences and trees with transformers" (2024) [47] — for constructive transformer simulation of weighted automata.
- **W. Merrill and A. Sabharwal**, "The expressive power of transformers with chain of thought" (2024) [35] — for the chain-of-thought expressivity jump.

Chapter 12

Approximating Language Models by Weighted Finite Automata

Chapter 11 did *not* prove that transformers are weighted finite automata. It suggested something more modest: in the single-pass regime, finite-state approximations are plausible enough to study seriously. This chapter turns that suggestion into a program.

The most important question is not "Which extraction algorithm is best?" It is:

> What exactly do we want the WFA to approximate?

Without a precise answer, one can slide too quickly between four different things:

- exact structure of the original LLM,
- approximation on finite data,
- exact rationality of the surrogate,
- asymptotic conclusions about the original model.

This chapter tries to keep those levels separate.

12.1 Approximation Targets

A WFA surrogate might be asked to preserve any of the following.

1. **String probabilities on a bounded set.** Approximate $\mu(w)$ well for strings w up to some maximum length.
2. **Next-token conditionals on bounded contexts.** Approximate $p(\sigma \mid h)$ well for histories h in some finite family.
3. **Length statistics.** Approximate the length PGF
$$P(z) = \sum_{n \geq 0} p_n z^n.$$
4. **Support structure.** Approximate which strings are effectively possible or impossible.
5. **Low-rank Hankel structure.** Approximate the string function by one whose Hankel matrix has small rank.

These targets are related, but not equivalent. In particular, good approximation on finite lengths or on a validation corpus does *not* automatically imply that the surrogate and the teacher share the same dominant singularities or the same asymptotic regime.

12.2 Line 1: Black-Box Distillation

The simplest idea is to treat the language model as a teacher and fit a small WFA student.

There are two distinct versions of this idea.

Score fitting. One may fit a WFA-style string scorer to match values such as

$$s(w) = \log \mu(w)$$

on a finite corpus. This is useful if the downstream task only needs relative ranking or rescoring. But a fitted score is not automatically a normalized probability distribution.

Distribution fitting. If the goal is to minimize KL divergence or total variation against a genuine probability model, then the surrogate must itself be a proper probabilistic object. In the WFA setting that means using a nonnegative, properly normalized automaton rather than an arbitrary real-valued string function.

The advantage of black-box distillation is practical flexibility: it can be applied even when the internal architecture of the teacher is unavailable. The limitation is equally clear: success on a dataset is only evidence about the chosen target and the sampled region of string space.

12.3 Line 2: Structural Equivalences

Some neural architectures are exactly finite-state or WFA-like in a much stronger sense.

Theorem 12.1 (Structural equivalence, black-box family of results)**.** *Certain recurrent architectures can be written in a form where the update on input symbol x_t is literally of the form*

$$h_t = A_{x_t} \odot h_{t-1}$$

for an appropriate semiring operation $\odot$, making the model a rational recurrence / weighted-automaton-style computation.

This is the cleanest regime analytically because the equivalence is exact, not approximate. In that case Chapter 7 applies directly.

The reason this line matters for the present book is not that transformers reduce to these models. It is that exact finite-state structure really does exist inside the broader neural-model landscape, so the rational viewpoint is not merely decorative.

12.4 Line 3: Query Learning

A different strategy is to ask the model targeted questions rather than fit it passively on a fixed corpus.

Angluin's classical L^* algorithm learns a minimal DFA by using membership and counterexample queries. For language models, that picture is only partially transferable:

- it is natural for accept/reject or thresholded support questions,
- it is much less straightforward for real-valued probabilities,
- and exact equivalence queries are essentially unavailable in practice.

Still, the idea of *adaptive querying* is valuable. It forces the learner to seek out structurally informative strings instead of hoping that a passive corpus already contains them.

12.5 Line 4: Spectral and Hankel Methods

This is the mathematically richest line.

Definition 12.2 (Hankel matrix)**.** Given a string function $f : \Sigma^* \to \mathbb{R}$, its Hankel matrix is the bi-infinite matrix

$$H_f(u, v) = f(uv),$$

indexed by prefixes u and suffixes v.

The importance of the Hankel matrix is captured by a fundamental theorem.

Theorem 12.3 (Finite Hankel rank, black box)**.** *A string function is realizable by a finite-state weighted automaton if and only if its Hankel matrix has finite rank. The minimal number of WFA states equals that rank.*

This is the weighted analogue of the Myhill–Nerode theorem. In the Boolean case, finite-state structure means finitely many future behaviors. In the weighted case, it means the future-behavior rows span a finite-dimensional linear space.

Spectral learning takes a large finite block of the Hankel matrix, computes a low-rank factorization, and reads off a small WFA from that factorization. The exact formulas depend on the chosen finite prefix/suffix bases and on pseudoinverses, so we do not reproduce them in detail here. The core idea is enough: small Hankel rank means small WFA.

In one-letter settings, operator-theoretic results such as the Adamyan–Arov–Krein theorem say that the best low-rank approximation can be taken *within* the Hankel class, not just among arbitrary low-rank matrices. This is mathematically beautiful, but also far beyond the level of a first pass through the subject. For our purposes, it is enough to know that one-letter Hankel approximation is the cleanest controlled setting.

12.6 One-Letter Projections: Useful but Limited

Collapsing a model to a one-letter alphabet means forgetting token identities and keeping only length. Concretely, if

$$p_n = \Pr(|W| = n),$$

then the one-letter projection is exactly the length PGF

$$P(z) = \sum_{n \geq 0} p_n z^n.$$

This is extremely useful for studying length tails, stopping behavior, and the singularity structure of the length distribution.

But it is *not* the same as the counting-type typical-set generating function of Chapter 9. In particular, the one-letter projection is the right object for length statistics, not the object whose radius directly encodes Shannon entropy via

$$h = \log_2(1/R).$$

So the one-letter projection is a feature, not a bug — but it answers a specific question, not every question.

12.7 Scale Realities

All four approximation lines become much harder at modern LLM scale.

- Distillation is limited by how much of string space the training sample reveals.
- Query learning is limited by the impossibility of exact equivalence checks.
- Spectral methods are limited by very large effective Hankel rank and by sampling noise.
- Exact structural equivalences usually apply only to restricted model classes.

This is why small surrogates and controlled synthetic settings matter. For example, one can begin with models trained on data generated from a known PCFG or from a known finite-state source. In those cases the underlying formal structure is known in advance, which makes approximation error interpretable.

12.8 What a WFA Surrogate Actually Buys

Suppose we construct a WFA surrogate

$$\widehat{M} = (\hat{\alpha}, \{\hat{A}_\sigma\}, \hat{\beta})$$

with scalar length generating function

$$\widehat{F}(z) = \hat{\alpha}(I - z\hat{A})^{-1}\hat{\beta}, \qquad \hat{A} = \sum_\sigma \hat{A}_\sigma.$$

Then two things are true.

1. The surrogate has an exact rational length generating function, so Chapters 4 and 7 determine its pole-based asymptotics.
2. Those conclusions apply *exactly* to the surrogate and only *indirectly* to the original LLM.

So the rational toolkit becomes available, but only for the object we have actually extracted. Whether the surrogate's poles are informative about the teacher depends entirely on the chosen approximation target and on how well that target was matched.

A safe formulation is this: WFA approximation is a way of *proposing* a rational model for some aspect of an LLM. It is not, by itself, a proof that the original LLM has rational asymptotics.

Preview

The next chapters use this chapter's distinction aggressively.

- Chapter 13 and beyond will study exact analytic objects such as partition functions and length PGFs.
- Later empirical chapters will ask whether simple rational or algebraic surrogates actually fit observed data.

The approximation target chosen here will determine what those later claims can legitimately mean.

Further Reading

- **G. Lecorvé and P. Motlicek**, "Conversion of recurrent neural network language models to weighted finite-state transducers" (2012) [32] — an early example of black-box distillation.
- **A. T. Suresh et al.**, "Approximating probabilistic models as weighted finite automata" (2021) [55] — a modern distillation reference.
- **G. Weiss, Y. Goldberg, and E. Yahav**, "Extracting automata from recurrent neural networks using queries and counterexamples" (2018) [60] — for the query-learning perspective.
- **G. Rabusseau, T. Li, and D. Precup**, "Connecting weighted automata and recurrent neural networks through spectral learning" (2019) [46] — for spectral reconstruction ideas.
- **C. Lacroce, P. Panangaden, and G. Rabusseau**, "Extracting weighted finite automata from recurrent neural networks using constrained AAK theory" (2021) [31] — for the one-letter AAK perspective.
- **J. Berstel and C. Reutenauer**, *Noncommutative Rational Series with Applications* (2011) [7] — the standard algebraic reference on rational series and Hankel rank.

Exercise 12.4. For the Fibonacci WFA from Chapter 7, compute

$$F(z) = \alpha(I - zA_a)^{-1}\beta$$

explicitly and identify its poles.

Exercise 12.5. Consider the scalar string function over the one-letter alphabet defined by

$$f(1^n) = n + 1.$$

Show that its Hankel matrix has rank 2 and exhibit a 2-state WFA realizing it.

Part IV

Statistical Mechanics Meets Generating Functions

This part builds the bridge between the analytic-combinatorics side of the book and the statistical-physics side. Chapter 13 introduces Boltzmann sampling and makes the central identification: a generating function evaluated at $x = e^{-\beta}$ *is* a partition function, and the radius of convergence corresponds to a critical temperature. Chapter 14 transfers this framework to LLMs, defining the energy $E(w) = -\log \mu(w)$ and showing that temperature decoding is (approximately) sampling from a Gibbs measure — with a precise characterization of when the local and global versions agree. Chapter 15 reviews the growing empirical evidence that LLM output distributions undergo genuine phase transitions as temperature varies, and proposes that these transitions can be explained as singularities of the partition function.

Chapter 13

Boltzmann Sampling

This chapter has one beautiful exact insight at its center:

> For a combinatorial class counted by size, Boltzmann sampling is exactly a Gibbs distribution with energy equal to size, and the ordinary generating function is exactly the partition function.

That statement is precise, not metaphorical. Everything else in the chapter should be read as built on top of that fact.

13.1 Motivation

Suppose we want a uniform random object of size exactly n from a combinatorial class $\mathcal{A}$. A classical approach uses a recursive sampler together with precomputed counting tables. Boltzmann sampling takes a different route: instead of forcing the size to be exactly n, it samples from all sizes at once with a tunable parameter x and then, if necessary, conditions or rejects to recover exact size.

The main point for this book is not the last word on complexity bounds. It is the formula for the distribution itself and how naturally it interacts with generating functions.

13.2 Boltzmann Samplers

Let

$$A(x) = \sum_{n \geq 0} a_n x^n, \qquad a_n = |\mathcal{A}_n|,$$

be the ordinary generating function of a combinatorial class $\mathcal{A}$, and let ρ be its radius of convergence.

Definition 13.1. For a parameter $x \in (0, \rho)$, the *Boltzmann distribution* on $\mathcal{A}$ is

$$\Pr_x(\gamma) = \frac{x^{|\gamma|}}{A(x)}.$$

A random generator with this law is called a *Boltzmann sampler*.

The normalization is immediate:

$$\sum_{\gamma \in \mathcal{A}} \Pr_x(\gamma) = \frac{1}{A(x)} \sum_{n \geq 0} a_n x^n = 1.$$

Every object of the same size receives the same probability. Therefore, conditioning on the event $|\gamma| = n$ produces a uniform random element of $\mathcal{A}_n$.

The size itself is random, with distribution

$$\Pr_x(|\gamma| = n) = \frac{a_n x^n}{A(x)}.$$

Its mean is

$$\mathbb{E}_x[N] = \frac{xA'(x)}{A(x)}.$$

The mean is monotone in x because

$$\frac{d}{dx}\left(\frac{xA'(x)}{A(x)}\right) = \frac{\mathrm{Var}_x(N)}{x} \geq 0.$$

So increasing x pushes mass toward larger objects.

13.3 Recursive Construction from Symbolic Specifications

The symbolic method translates directly into Boltzmann samplers.

Disjoint union. If

$$\mathcal{A} = \mathcal{B} + \mathcal{C}, \qquad A(x) = B(x) + C(x),$$

then choose the $\mathcal{B}$ branch with probability $B(x)/A(x)$ and the $\mathcal{C}$ branch with probability $C(x)/A(x)$.

Cartesian product. If

$$\mathcal{A} = \mathcal{B} \times \mathcal{C}, \qquad A(x) = B(x)C(x),$$

then sample independently from the Boltzmann laws on $\mathcal{B}$ and $\mathcal{C}$ and return the pair.

Sequence. If

$$\mathcal{A} = \mathrm{SEQ}(\mathcal{B}), \qquad A(x) = \frac{1}{1 - B(x)},$$

with $B(x) < 1$, then the number of components has geometric law

$$\Pr(K = k) = (1 - B(x))B(x)^k, \qquad k \geq 0.$$

After drawing K, sample K independent Boltzmann objects from $\mathcal{B}$ and concatenate them.

These are the constructors we need most often. More delicate labeled and cycle constructions exist as well, but they are better treated in a specialized reference than in a one-page summary. The important point here is that the symbolic specification and the sampling rule use the same algebra.

13.4 Tuning and Exact-Size Rejection

If the target is exact size n, a natural choice of parameter x is the one that makes the expected size equal to n:

$$\frac{xA'(x)}{A(x)} = n.$$

This same equation also appears by maximizing the exact-size probability

$$\Pr_x(N = n) = \frac{a_n x^n}{A(x)}$$

with respect to x: differentiate

$$n \log x - \log A(x)$$

and set the derivative to zero.

This tuning equation need not have a solution for every class and every large n; it depends on the expected size actually diverging as $x \uparrow \rho$. After tuning x, one may simply rerun the Boltzmann sampler until the drawn object has the desired size. In many tree-like and admissible classes this is extremely efficient, but the precise complexity depends on local limit behavior and class-specific regularity assumptions. We deliberately avoid stating a one-line universal theorem here.

13.5 The Gibbs Interpretation

Now make the substitution

$$x = e^{-\beta},$$

with β interpreted as inverse temperature. Then

$$\Pr_\beta(\gamma) = \frac{e^{-\beta|\gamma|}}{A(e^{-\beta})}.$$

If we define the energy of a structure to be its size,

$$E(\gamma) = |\gamma|,$$

then this becomes

$$\Pr_\beta(\gamma) = \frac{e^{-\beta E(\gamma)}}{Z(\beta)}, \qquad Z(\beta) = A(e^{-\beta}).$$

So in this exact combinatorial setting:

- the generating function is the partition function,
- Boltzmann weights are Gibbs weights,
- and the size variable is the energy.

This exactness depends on the particular choice of energy $E(\gamma) = |\gamma|$. If one changes the energy, one changes the partition function accordingly.

13.6 Critical Temperature

The radius of convergence determines the critical inverse temperature:

$$\beta_c = -\log \rho.$$

For

$$\beta > \beta_c \qquad (x = e^{-\beta} < \rho),$$

the partition function is finite and the Gibbs distribution is well defined. At the boundary

$$\beta = \beta_c \qquad (x = \rho),$$

convergence depends on the coefficient tail.

If

$$a_n \sim C\rho^{-n} n^{-\alpha},$$

then at criticality

$$a_n \rho^n \sim C n^{-\alpha}.$$

So:

- if $\alpha > 1$, the critical partition function still converges;
- if $\alpha \leq 1$, it diverges.

Even when the critical partition function converges, the expected size may still diverge. For example, if $\alpha = 3/2$, then

$$\sum_n a_n \rho^n \sim \sum_n n^{-3/2}$$

converges, but

$$\sum_n n a_n \rho^n \sim \sum_n n^{-1/2}$$

diverges. This is why critical Boltzmann samplers often produce heavy-tailed sizes.

The physically relevant "high-temperature" limit inside the normalizable regime is therefore not "x close to 1" in general. It is the approach to criticality

$$x \uparrow \rho \qquad \text{or equivalently} \qquad \beta \downarrow \beta_c.$$

13.7 A Free-Energy Viewpoint

It is sometimes useful to isolate the size-n contribution

$$Z_n(\beta) = a_n e^{-\beta n}.$$

This is the size-restricted partition weight. Its exponential rate is governed by

$$\frac{1}{n} \log Z_n(\beta) = -\beta + \frac{1}{n} \log a_n.$$

If

$$a_n \sim C \rho^{-n} n^{-\alpha},$$

then

$$\lim_{n \to \infty} \frac{1}{n} \log a_n = \log(1/\rho),$$

so the leading exponential growth/decay rate is

$$-\beta + \log(1/\rho).$$

This is a combinatorial growth exponent. One may call the negative of this quantity a free energy per unit size if desired, but the important point for this chapter is simpler: the same radius of convergence that controls coefficients also controls the thermodynamic threshold.

We deliberately avoid calling this quantity an entropy rate here. Chapter 9 showed that entropy-rate identities require a carefully specified probabilistic source and a carefully specified counting object.

13.8 Worked Examples

Example 13.2 (Binary strings)**.** For binary strings,

$$A(z) = \frac{1}{1 - 2z}, \qquad \rho = \frac{1}{2}.$$

The Boltzmann sampler draws a geometric length N with

$$\Pr_x(N = n) = (1 - 2x)(2x)^n,$$

and then chooses a uniformly random binary string of that length. The expected size is

$$\mathbb{E}_x[N] = \frac{2x}{1-2x}.$$

The partition function is

$$Z(\beta) = A(e^{-\beta}) = \frac{1}{1-2e^{-\beta}},$$

so

$$\beta_c = \log 2.$$

Example 13.3 (Catalan structures)**.** For Catalan objects,

$$C(z) = \frac{1-\sqrt{1-4z}}{2z}, \qquad \rho = \frac{1}{4},$$

and

$$[z^n]C(z) \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}.$$

At criticality,

$$\Pr_{\beta_c}(N = n) \propto [z^n]C(z)\rho^n \sim \frac{1}{\sqrt{\pi} n^{3/2}}.$$

So the partition function is finite, but the expected size diverges. This is the same $n^{-3/2}$ exponent that earlier chapters encountered through square-root singularities.

Preview of Chapter 14

There is a tempting analogy between Boltzmann sampling and temperature in autoregressive language models. For a global string distribution $\mu(w)$, the Gibbs-style reweighting

$$\mu_\beta(w) \propto \mu(w)^\beta$$

really is a partition-function construction. Practical LLM temperature decoding, however, is usually applied *locally* at each next-token distribution. Those two operations are related, but they are not identical in general. Chapter 14 will focus on that distinction rather than blur it.

Further Reading

- **P. Duchon, P. Flajolet, G. Louchard, and G. Schaeffer**, "Boltzmann samplers for the random generation of combinatorial structures" (2004) [17] — the foundational Boltzmann-sampling paper.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for the full sampler dictionary and analytic background.
- **A. Nijenhuis and H. S. Wilf**, *Combinatorial Algorithms* (1978) [42] — for classical recursive sampling.
- **M. Blondel et al.**, "Autoregressive language models are secretly energy-based models" (2025) [8] — for a modern global energy-based perspective on autoregressive models.
- **T. Kempton and C. Burrell**, "Temperature and local normalization in autoregressive models" (2025) [28] — for the gap between local temperature scaling and global Gibbs reweighting.

13.9 Exercises

Exercise 13.4. For a combinatorial class with generating function $A(x)$, prove the variance identity

$$\mathrm{Var}_x(N) = x\frac{d}{dx}\left(\frac{xA'(x)}{A(x)}\right).$$

Deduce that the expected size is monotone nondecreasing in x.

Exercise 13.5. For binary strings, solve

$$\frac{2x}{1-2x} = n$$

for the tuned parameter $x = x^*(n)$.

Exercise 13.6. Show that for Catalan structures at criticality,

$$\sum_{n\geq 0}[z^n]C(z)\rho^n < \infty$$

but

$$\sum_{n\geq 0} n[z^n]C(z)\rho^n = \infty.$$

Interpret this probabilistically.

Chapter 14

Gibbs Measures and Temperature Decoding

Chapter 13 showed that Boltzmann sampling on combinatorial classes is exactly a Gibbs distribution with energy equal to size. For language models there is a similarly natural energy,

$$E(w) = -\log \mu(w),$$

but now two different temperature operations appear:

1. a *global* Gibbs reweighting of complete strings,
2. the *local* tokenwise temperature scaling used in practice.

The central lesson of the chapter is that these are generally *not the same distribution.*

14.1 Sample Space and Energy

We use the same convention as Chapter 10. Let Σ be the content-token vocabulary and let

$$\mathrm{EOS} \notin \Sigma$$

be a stopping symbol. The model samples tokens from $\Sigma \cup \{\mathrm{EOS}\}$, but the actual finite outputs are strings in Σ^*.

If

$$w = w_1 \cdots w_n \in \Sigma^*,$$

then its probability under the autoregressive model is

$$\mu(w) = p(\mathrm{EOS} \mid w_1 \cdots w_n) \prod_{t=1}^{n} p(w_t \mid w_1 \cdots w_{t-1}).$$

This is the probability that the model emits exactly the content string w and then stops.

Definition 14.1. The *energy* of a complete output string is

$$E(w) = -\log \mu(w).$$

The energy is additive along the sampled path because logarithms turn the autoregressive product into a sum of surprisals.

14.2 The Global Gibbs Distribution

For inverse temperature $\beta > 0$, define the global Gibbs reweighting by

$$\mu_\beta(w) = \frac{\mu(w)^\beta}{Z(\beta)}, \qquad Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$

This is a genuine probability distribution whenever the partition function $Z(\beta)$ is finite.

At $\beta = 1$, if the original model is tight, then

$$Z(1) = \sum_{w \in \Sigma^*} \mu(w) = 1,$$

so $\mu_1 = \mu$.

A useful exact observation is:

- for $\beta \geq 1$, one always has $Z(\beta) \leq 1$ because $0 \leq \mu(w) \leq 1$ and raising to a power at least one can only decrease each term;
- for $0 < \beta < 1$, finiteness is no longer automatic and must be checked.

So the interesting convergence questions begin on the high-temperature side.

14.3 Local Temperature Sampling

In practical decoding, temperature is applied not to complete-string probabilities but to each next-token distribution separately. For a history $h \in \Sigma^*$, define the local partition factor

$$\widetilde{Z}_\beta(h) = \sum_{a \in \Sigma \cup \{\mathrm{EOS}\}} p(a \mid h)^\beta.$$

The locally tempered next-token distribution is

$$\widetilde{p}_\beta(a \mid h) = \frac{p(a \mid h)^\beta}{\widetilde{Z}_\beta(h)}.$$

Iterating these conditionals produces a new output distribution on Σ^*, which we denote by $\widetilde{\mu}_\beta$.

If

$$w = w_1 \cdots w_n,$$

then the local-temperature probability of the complete output w is

$$\widetilde{\mu}_\beta(w) = \left(\prod_{t=1}^{n} \frac{p(w_t \mid w_1 \cdots w_{t-1})^\beta}{\widetilde{Z}_\beta(w_1 \cdots w_{t-1})} \right) \frac{p(\mathrm{EOS} \mid w)^\beta}{\widetilde{Z}_\beta(w)}.$$

14.4 The Exact Comparison Formula

Let

$$\mathrm{Pref}(w) = \{\varepsilon, w_1, w_1 w_2, \ldots, w_1 \cdots w_n\}$$

be the set of prefixes encountered while generating w, including the full content string just before emitting EOS.

Proposition 14.2. *For every complete output string $w \in \Sigma^*$,*

$$\widetilde{\mu}_\beta(w) = \frac{\mu(w)^\beta}{\prod_{h \in \mathrm{Pref}(w)} \widetilde{Z}_\beta(h)}.$$

Hence, whenever $Z(\beta) < \infty$,

$$\frac{\widetilde{\mu}_\beta(w)}{\mu_\beta(w)} = \frac{Z(\beta)}{\prod_{h \in \mathrm{Pref}(w)} \widetilde{Z}_\beta(h)}.$$

Therefore

$$\widetilde{\mu}_\beta = \mu_\beta$$

if and only if the prefix-product

$$\prod_{h \in \mathrm{Pref}(w)} \widetilde{Z}_\beta(h)$$

is the same for every output string w with positive probability.

This is the sharp equality condition. It is stronger than requiring each individual local partition function to be "prefix-independent" in isolation, because the full product also depends on how many prefixes a string traverses.

14.5 Why the Difference Occurs

The global Gibbs distribution reweights *complete strings*. The local rule reweights one-step choices and renormalizes immediately after each prefix. That renormalization forgets how much total future mass lies below each branch.

One useful way to say this is to define the future partition factor

$$G_\beta(h) = \sum_{u \in \Sigma^*} \mu_h(u)^\beta,$$

where μ_h is the continuation distribution from prefix h. Then the global Gibbs conditional at prefix h has the form

$$\mu_\beta(a \mid h) \propto p(a \mid h)^\beta G_\beta(ha),$$

with the special case of halting obtained by taking $a = \mathrm{EOS}$ and $G_\beta(h\,\mathrm{EOS}) = 1$.

The local rule keeps the factor $p(a \mid h)^\beta$ but drops the continuation weight $G_\beta(ha)$. That missing subtree mass is exactly where the distortion comes from.

14.6 A Fully Specified Example

Consider the following autoregressive model over content vocabulary $\Sigma = \{a, b, c, d\}$.

At the root:

$$p(a \mid \varepsilon) = 0.8, \qquad p(b \mid \varepsilon) = 0.2, \qquad p(\mathrm{EOS} \mid \varepsilon) = 0.$$

After prefix a:

$$p(\mathrm{EOS} \mid a) = 0.9, \qquad p(c \mid a) = 0.1.$$

After prefix b:

$$p(\mathrm{EOS} \mid b) = 0.5, \qquad p(d \mid b) = 0.5.$$

After prefixes ac and bd, the model halts with probability 1.

So the four complete outputs are

$$a, \qquad ac, \qquad b, \qquad bd,$$

with probabilities

$$\mu(a) = 0.72, \quad \mu(ac) = 0.08, \quad \mu(b) = 0.10, \quad \mu(bd) = 0.10.$$

Take $\beta = 1/2$.

Global Gibbs. The weights are

$$\sqrt{0.72},\ \sqrt{0.08},\ \sqrt{0.10},\ \sqrt{0.10},$$

whose normalized values are approximately

$$\mu_{1/2}(a) \approx 0.481, \quad \mu_{1/2}(ac) \approx 0.160, \quad \mu_{1/2}(b) \approx 0.179, \quad \mu_{1/2}(bd) \approx 0.179.$$

Local temperature. At the root, the tempered distribution on $\{a, b\}$ is proportional to

$$\sqrt{0.8},\ \sqrt{0.2},$$

so

$$\widetilde{p}_{1/2}(a \mid \varepsilon) \approx 0.667, \qquad \widetilde{p}_{1/2}(b \mid \varepsilon) \approx 0.333.$$

At prefix a, the local tempered split between halting and continuing is proportional to

$$\sqrt{0.9},\ \sqrt{0.1},$$

so

$$\widetilde{p}_{1/2}(\mathrm{EOS} \mid a) = 0.75, \qquad \widetilde{p}_{1/2}(c \mid a) = 0.25.$$

At prefix b, the local partition factors are equal, so

$$\widetilde{p}_{1/2}(\mathrm{EOS} \mid b) = \widetilde{p}_{1/2}(d \mid b) = 0.5.$$

Thus

$$\widetilde{\mu}_{1/2}(a) = 0.500, \quad \widetilde{\mu}_{1/2}(ac) = 0.167, \quad \widetilde{\mu}_{1/2}(b) = 0.167, \quad \widetilde{\mu}_{1/2}(bd) = 0.167.$$

These do not match the global Gibbs probabilities. The low-probability b-subtree is over-promoted because the local procedure does not know that its future mass is smaller.

14.7 Greedy, High Temperature, and Truncation Heuristics

Two limits should be distinguished. Recall that local temperature is usually written as T, with $\beta = 1/T$.

Global Gibbs limit. As $\beta \to \infty$, the global Gibbs measure concentrates on complete strings that maximize $\mu(w)$ globally.

Local decoding limit. As $T \to 0$ in local tokenwise temperature sampling, each next-token distribution concentrates on its local argmax set. This is greedy decoding. Greedy decoding need not coincide with the globally most probable complete string.

At the other extreme, as $\beta \to 0$, the global Gibbs weights flatten. But on a countably infinite set of strings there is no honest uniform distribution to converge to, so "uniform" should only be understood heuristically.

Practical heuristics such as top-k and nucleus sampling remain autoregressive procedures: they still define prefix-conditioned next-token distributions. What they break is not the prefix-Markov structure, but the clean interpretation as either the original model or a global Gibbs reweighting of it.

Preview of Chapter 15

Chapter 15 studies what happens when the global partition function

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta$$

changes analytic character as β varies. The point of the present chapter is that one must first decide which temperature operation is being studied. Local tokenwise temperature and global Gibbs reweighting are different objects, and any talk of "phase transitions in temperature decoding" must say which one it means.

Further Reading

- **M. Blondel et al.**, "Autoregressive language models are secretly energy-based models" (2025) [8] — for global energy-based reformulations of autoregressive models.
- **T. Kempton and C. Burrell**, "Temperature and local normalization in autoregressive models" (2025) [28] — for quantitative bounds on the gap between local and global temperature manipulations.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for the Gibbs/Boltzmann side on the combinatorial end.

14.8 Exercises

Exercise 14.3. Verify the ratio formula for $\widetilde{\mu}_\beta(w)/\mu_\beta(w)$ directly from the definitions.

Exercise 14.4. Construct a memoryless model with two content tokens and one EOS token, compute both μ_β and $\widetilde{\mu}_\beta$, and check that they are generally different when output lengths vary.

Chapter 15

Phase Transitions in Language Model Output

This chapter asks whether temperature variation in language models can reveal genuine large-scale structural changes in the output distribution. The answer has two layers.

1. There is a clean mathematical object: the *global* partition function
$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$
2. There is a practical experimental procedure: vary the *local* decoding temperature used by an LLM and measure changes in its outputs.

Chapter 14 showed that those are not the same thing in general. So the purpose of this chapter is not to pretend the bridge is already proved. It is to explain the global object, summarize what singularity ideas would matter if the right surrogate were available, and then interpret the empirical evidence as suggestive rather than decisive.

15.1 The Global Partition Function

Let μ be the output distribution of a tight autoregressive model on content strings $w \in \Sigma^*$. For inverse temperature $\beta > 0$, define
$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$
This is a Renyi-type partition sum over complete strings.

Basic facts are immediate.

- If the model is tight, then
$$Z(1) = \sum_{w} \mu(w) = 1.$$
- For every $\beta \geq 1$,
$$Z(\beta) \leq 1,$$
because each term satisfies $0 \leq \mu(w)^\beta \leq \mu(w)$.
- For $0 < \beta < 1$, finiteness is not automatic. Small probabilities are enlarged by the exponent, so the sum can in principle diverge.

The corresponding global Gibbs distribution is

$$\mu_\beta(w) = \frac{\mu(w)^\beta}{Z(\beta)}$$

whenever $Z(\beta)$ is finite.

15.2 Why This Is Not an Ordinary OGF

A major conceptual caution is needed here. The function $Z(\beta)$ is not automatically an ordinary generating function of the Chapter 4 form

$$\sum_{n\geq 0} a_n z^n$$

with integer exponents. If we write

$$E(w) = -\log \mu(w),$$

then formally

$$Z(\beta) = \sum_w e^{-\beta E(w)}.$$

Setting $z = e^{-\beta}$ gives an expression of the form

$$\sum_w z^{E(w)},$$

where the exponents are generally arbitrary nonnegative real numbers, not integers. So Chapter 4's transfer theorems do *not* apply directly just by the substitution $z = e^{-\beta}$.

What survives from earlier chapters is more limited but still useful:

- if one can replace the true LLM partition sum by a finite-state or algebraic *surrogate*, then the singularity toolkit of Part I becomes relevant to the surrogate;
- if not, then one should regard the Chapter 4 language as analogy, not theorem.

15.3 What We Mean by a Phase Transition Here

For a mathematically defined global Gibbs family, the cleanest notion is:

> A phase transition is a point where the partition function or a thermodynamic quantity built from it ceases to behave analytically as a function of β.

In finite-size empirical work, one never sees a literal nonanalyticity. Instead one sees sharp crossovers, peak susceptibilities, and finite-size scaling patterns that are interpreted as evidence for an underlying infinite-limit singularity.

For this reason we will be cautious with first-order / second-order terminology. The relevant derivative tests depend on which free-energy normalization one chooses and on what limiting family of systems is being studied. This chapter will therefore emphasize *qualitative singularity types* and empirical signatures rather than trying to force a fragile derivative classification onto every example.

15.4 A Prototype Algebraic Critical Point

The Catalan family remains the cleanest model example. Its counting generating function is

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z}, \qquad [z^n]C(z) \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}.$$

If we build the size-weighted partition sum

$$Z_C(\beta) = \sum_{n \geq 0} C_n e^{-\beta n},$$

then the critical inverse temperature is

$$\beta_c = \log 4.$$

At criticality,

$$C_n e^{-\beta_c n} \sim \frac{1}{\sqrt{\pi} n^{3/2}}.$$

So the critical Boltzmann weights are heavy-tailed with the characteristic exponent $-3/2$. This is a clean combinatorial model of algebraic criticality.

Two cautions are essential.

1. For $\beta < \beta_c$, the physical sum $Z_C(\beta)$ diverges. Analytic continuation of the function beyond that boundary is a different mathematical object, not the original partition sum.
2. This example is about a counting-by-size partition sum, not directly about the LLM partition sum $\sum_w \mu(w)^\beta$.

So Catalan provides a *prototype* of square-root criticality, not a proof that LLM temperature decoding sits in the same universality class.

15.5 Composition Schemes as a Catalogue of Possibilities

Results such as those of Banderier, Kuba, Wagner, and Wallner [4] show that in combinatorial models, critical composition schemes can produce a small catalogue of recurring singular behaviors: poles, square-root branch points, and more degenerate Airy-type phenomena.

For this book, the right takeaway is not that those theorems already apply to LLM partition sums. It is that they supply a menu of plausible universality patterns one might later try to fit or motivate if an appropriate surrogate model is found.

15.6 What the Empirical Papers Actually Vary

The recent empirical papers on LLM "phase transitions" do *not* directly sample from the global Gibbs distribution

$$\mu_\beta(w) \propto \mu(w)^\beta.$$

They vary the practical decoding temperature used locally at each token. Chapter 14 showed that this usually produces a different distribution.

So the empirical story should be read carefully:

- the experiments are about *local temperature decoding*,
- the global partition function provides a conceptual reference object,
- and the relation between the two is part of the open theory, not a settled equivalence.

This does not make the experiments irrelevant. It means they are evidence about a nearby but not identical object.

15.7 Empirical Evidence from LLMs

Several recent studies report sharp temperature-dependent changes in output statistics for models such as GPT-2, Pythia, Mistral, and Llama.

Nakaishi, Nishikawa, and Hukushima (2024). They study GPT-2 across a range of local decoding temperatures and observe a sharp crossover in susceptibility-like observables near a critical window around $T \approx 1$ [40].

Arnold et al. (2024). They extend the picture across model families and scales, again finding robust temperature windows and critical-looking scaling behavior [1].

Mikhaylovskiy (2025). This work goes further and proposes a specific universality-class interpretation for those fitted exponents [38]. That proposal is interesting, but still empirical and model-dependent.

The honest summary is this: the papers present suggestive finite-size evidence for phase-transition-like behavior under local temperature decoding. They do not yet prove that the global partition function $Z(\beta)$ has a specific singularity.

15.8 The Bridging Proposal

Here is the research proposal that makes the chapter worth having.

> Perhaps the sharp empirical crossovers seen under local temperature decoding are shadows of a deeper global singularity structure, which in turn might be captured by a rational or algebraic surrogate for the LLM output distribution.

That proposal has several possible forms.

- In a rational surrogate, the dominant analytic structure would be pole-like.
- In an algebraic surrogate, square-root or related branch-point behavior could appear.
- More complicated surrogates could produce richer singularity landscapes.

What the proposal does *not* say is that one may apply Chomsky–Schützenberger directly to

$$Z(\beta) = \sum_w \mu(w)^\beta.$$

The counting theorems of Chapters 5–6 concern different objects. Any bridge from PCFG-style algebraicity to LLM partition sums would require a new weighted theorem or a new surrogate argument, not a direct reuse of the old counting theorem.

So the bridging proposal should be read as conjectural. It is a disciplined conjecture, but still a conjecture.

Summary

This chapter's safe conclusions are:

1. the global partition function of an LLM is

$$Z(\beta) = \sum_w \mu(w)^\beta;$$

2. this object is not automatically an ordinary generating function in an integer index;
3. local decoding temperature and global Gibbs temperature are different objects;
4. empirical temperature studies therefore provide suggestive but indirect evidence about the global partition function;
5. combinatorial models such as Catalan structures show what algebraic criticality could look like if the right surrogate were found.

That is enough to support a serious theory-meets-evidence program without pretending the bridge is already finished.

Further Reading

- **C. Banderier, M. Kuba, S. Wagner, and M. Wallner**, "Phase transitions of composition schemes" (2024) [4] — for a catalogue of singularity-driven universality patterns in combinatorial models.
- **Y. Nakaishi, Y. Nishikawa, and K. Hukushima**, "Phase transitions in GPT-2" (2024) [40] — for empirical finite-size scaling under local temperature decoding.
- **N. Arnold et al.**, "Phase transitions in large language models and the Ising model" (2024) [1] — for cross-model empirical evidence.
- **N. Mikhaylovskiy**, "Universality of phase transitions in large language models" (2025) [38] — for a stronger universality-class proposal.
- **T. Kempton and C. Burrell**, "Temperature and local normalization in autoregressive models" (2025) [28] — for the gap between local and global temperature operations.

15.9 Exercises

Exercise 15.1. Show directly that if μ is a probability distribution on a countable set, then for every $\beta \geq 1$,

$$\sum_w \mu(w)^\beta \leq 1.$$

Why does the same argument fail for $0 < \beta < 1$?

Exercise 15.2. For the Catalan partition sum

$$Z_C(\beta) = \sum_{n \geq 0} C_n e^{-\beta n},$$

identify the critical inverse temperature and show that the critical weights behave like $n^{-3/2}$.

Part V

Applications, Open Problems, and the Road Ahead

The final part applies the toolkit to specific phenomena and maps the research frontier. Chapter 16 examines Zipf's law through the lens of Pitman–Yor processes, combinatorial growth arguments, and the temperature window. Chapter 17 introduces Szpankowski's analytic information theory — motif statistics, trie depth, Mellin transforms — as the classical toolkit for sequence statistics that the LLM community should be reaching for. Chapters 18 and 19 are the strategic core: ten precisely stated open problems at the intersection of analytic combinatorics and LLMs, followed by a research roadmap that ranks them by tractability and proposes a concrete three-phase programme for a student entering the field.

Chapter 16

Zipf's Law and Token Distributions

Zipf's law is one of the oldest and most striking empirical regularities in language: a few token types are extremely common, while most are rare. This chapter does not pretend the mystery is solved. Its purpose is narrower:

- review several major explanation families,
- keep careful track of what is empirical and what is proved,
- and explain what a future analytic-combinatorics program would actually need to model.

16.1 Zipf's Law

Definition 16.1. Take a corpus and rank token *types* by decreasing empirical frequency. Zipf's law is the empirical claim that the relative frequency of rank r is approximately

$$f(r) \approx \frac{C}{r^s},$$

with s often close to 1.

This is a statement about ranked *types*. It is not yet a theorem, and it does not by itself say why such a pattern should arise.

A useful distinction is the following.

- If the ranked frequencies satisfy $p_r \propto r^{-s}$, then the *number of types* whose frequency exceeds a threshold f is of order $f^{-1/s}$, obtained by solving $p_r \approx f$ for r.
- This is different from asking for the total probability mass carried by all types above that threshold. That is a token-weighted question rather than a type-counting one.

So one must be careful not to slide between type tails and token-mass tails.

16.1.1 Mandelbrot's Refinement

Mandelbrot proposed the shifted form

$$f(r) \approx \frac{C}{(r+b)^s},$$

which often fits the highest-frequency ranks better. The shift mainly affects the head of the distribution rather than the long tail.

16.2 A Useful Skeptic: Piantadosi

Piantadosi's review [45] is valuable because it resists overinterpretation. Many very different models can be tuned to reproduce an approximate rank–frequency power law. So matching Zipf's law alone is weak evidence for any one mechanism.

That is the right methodological attitude for this chapter as well. A successful explanation should ideally connect Zipf behavior to other observables too: length, predictability, syntactic role, or temperature dependence.

16.3 Pitman–Yor Processes

A mathematically clean probabilistic explanation of heavy-tailed type frequencies comes from the Pitman–Yor process.

16.3.1 Chinese Restaurant Process View

With discount parameter $a \in (0, 1)$ and concentration parameter $\theta > -a$, the Pitman–Yor process may be described by a Chinese restaurant process. After $n - 1$ tokens have been generated, suppose type i currently has count c_i, and k distinct types have been seen so far. Then the next token is

- an existing type i with probability
$$\frac{c_i - a}{n - 1 + \theta},$$
- or a brand-new type with probability
$$\frac{\theta + ak}{n - 1 + \theta}.$$

The important qualitative facts are these.

- The process produces heavy-tailed ranked frequencies.
- The vocabulary size grows polynomially:
$$V(n) \asymp n^a$$
in the standard asymptotic sense.
- The discount parameter a controls how aggressively new types continue to appear.

For the present chapter, that last bullet is enough. We do not need a precise closed-form rank exponent formula to understand the role of the process. The main lesson is that a simple reinforcement-and-innovation mechanism can already produce long-tailed vocabularies.

16.3.2 Hierarchical / Two-Stage Versions

Hierarchical Pitman–Yor models, such as those studied by Goldwater, Griffiths, and Johnson [21], are more flexible. Linguistically, one may think of them as a cache or lexicon layer together with a token-generation layer. Those extra layers can fit natural-language data substantially better than a single flat process.

16.4 Berman's Combinatorial View

Berman's perspective is different in spirit. Rather than starting from a stochastic process prior, it starts from vocabulary growth. If the number of observed types after n tokens behaves like

$$V(n) \sim n^{\beta},$$

then one may ask what rank–frequency laws are compatible with that growth.

The chapter does not need the full derivation here. What matters is the conceptual shift: the exponent is constrained by a combinatorial growth law, not introduced as a free fitting parameter. This makes Berman's story complementary to the Pitman–Yor one rather than obviously contradictory to it.

It is also worth being precise about what is already standard: polynomial vocabulary growth of the form $V(n) \asymp n^{a}$ is well known in Pitman–Yor settings. The genuinely interesting question is not whether that happens, but whether different explanatory frameworks are telling the same deeper story in different languages.

16.5 Temperature Windows and Empirical Zipf Behavior

Mikhaylovskiy [39] studies how the empirical rank–frequency exponent changes with *local decoding temperature*. This chapter emphasizes the adjective "local": by Chapter 15's distinctions, this is not the same as the global Gibbs temperature.

The empirical claim is that there is often a window of local temperatures in which generated text looks approximately Zipfian, while lower temperatures make the distribution too concentrated and higher temperatures make it too flat. That is an interesting empirical phenomenon, but it should not be read as a theorem about the global partition function without further argument.

16.6 The Quantization Story

Michaud et al. [37] propose a neural-mechanistic story: token usage may reflect a distribution over latent "quanta" or reusable features, and those latent usage frequencies can themselves be heavy-tailed. This is an appealing analogy to hierarchical Pitman–Yor models, but the connection is still heuristic rather than a formal equivalence theorem.

16.7 A Simple Multinomial Proposition

To see what Zipf's law does *not* require, it is useful to record a very basic fact.

Proposition 16.2. *Assume a finite vocabulary*

$$\{w_1, \ldots, w_V\}$$

with strictly decreasing probabilities

$$p_1 > p_2 > \cdots > p_V > 0.$$

If tokens are drawn i.i.d. from this distribution, then the empirical frequency of the type with rank r converges almost surely to p_r.

Proof. By the strong law of large numbers, the empirical frequency of each type w_i converges almost surely to p_i. Because the probabilities are strictly ordered, the empirical rank order stabilizes almost surely for large enough sample size, so the empirical frequency at rank r converges to p_r. □

This proposition says that an i.i.d. model with Zipfian parameters produces Zipfian output, and one with non-Zipfian parameters does not. So the real explanatory question is not whether an i.i.d. model can display Zipf's law if hand-tuned that way; it is where the Zipf-like parameters come from in the first place.

16.8 A Worked Mini-Example

Consider a uniform unigram model on a vocabulary of size V. Then

$$p_r = \frac{1}{V} \qquad (1 \le r \le V),$$

so the limiting rank–frequency curve is flat, not Zipfian. This reminds us that long tails do not appear automatically from simple independence; some additional structure is needed.

16.9 A Speculative Bivariate Proposal

The natural analytic-combinatorics temptation is to package token length and rank together in a two-variable series, for example

$$F(z, u) = \sum_{w \in \mathcal{V}} z^{|w|} u^{r(w)},$$

where $r(w)$ is empirical rank.

This can be a useful *formal summary*, but it is not a standard symbolic-method bivariate generating function. The reason is that rank is not a local structural parameter like "number of leaves" or "number of occurrences of a letter." It is a global corpus statistic defined only after all types have been compared to each other.

So the honest status of such a two-variable series is:

- interesting as a proposal,
- potentially useful in asymptotic models,
- but not something to which the earlier symbolic marking theorems apply directly.

If a future asymptotic model turned rank into a genuine tractable parameter, then singularity drift in the extra variable might encode a length–rank tradeoff. At present, that is a research direction rather than a theorem.

Summary

This chapter's safest conclusions are:

1. Zipf's law is empirical and does not by itself identify the right mechanism.
2. Pitman–Yor, vocabulary-growth, and quantization stories are different explanation families, each illuminating a different aspect of the phenomenon.
3. The strongest analytic-combinatorics proposal here is still speculative, because rank is not a standard local marked parameter.

That is enough for the chapter's real role, which is to set up a sharper open problem rather than to pretend the theory is finished.

Further Reading

- **G. K. Zipf**, *Human Behavior and the Principle of Least Effort* (1949) [62] — the historical starting point.
- **S. T. Piantadosi**, "Zipf's word frequency law in natural language: a critical review and future directions" (2014) [45] — the best skeptical overview.
- **S. Goldwater, T. L. Griffiths, and M. Johnson**, "Producing power-law distributions and damping word frequencies with two-stage language models" (2011) [21] — the standard hierarchical Pitman–Yor reference.
- **V. Berman**, "Random text, Zipf's law, critical length, and implications for large language models" (2025) [5] — for the vocabulary-growth viewpoint.
- **N. Mikhaylovskiy**, "Zipf's law and the temperature window in transformer language models" (2025) [39] — for the empirical temperature-window story.
- **E. J. Michaud et al.**, "The quantization model of neural scaling laws" (2023) [37] — for the neural-mechanistic explanation family.

16.10 Exercises

Exercise 16.3. Give an example of a finite vocabulary distribution whose ranked probabilities are not close to a power law, and explain what its empirical rank–frequency plot converges to.

Exercise 16.4. In a finite vocabulary model with strictly decreasing probabilities $p_1 > \cdots > p_V$, explain why the distinctness assumption is needed in the multinomial proposition.

Chapter 17

Motif Statistics and Analytic Information Theory

This chapter points toward a mature classical toolkit rather than trying to redevelop it from first principles. Analytic information theory studies sequence statistics — motif counts, trie depth, match lengths, compression parameters — by encoding them into finite-state or transfer-matrix objects and then applying complex-analytic tools such as singularity analysis, Mellin transforms, and saddle-point methods.

The exact theorems belong to classical finite-state sources. The possible transfer to WFA approximations of LLMs is a research program, not an automatic corollary. The chapter will keep that distinction explicit.

17.1 A Classical Theme

For finite-state or Markov sources, many statistics of interest can be packaged into generating functions or matrix resolvents. Once that is done, asymptotic analysis can often be read off from poles, perturbations of dominant eigenvalues, Mellin poles, or saddle points. That is why analytic information theory fits so naturally into the rest of this book: it is the information-theoretic analogue of analytic combinatorics.

17.2 Motif Statistics: What Is the Generating Function?

Fix a source on strings over an alphabet Σ, and fix a motif M (a word or a regular-language pattern). There are two related but different generating functions one can study.

- A **counting** generating function that counts how many strings of length n have exactly k occurrences of the motif.
- A **weighted / probabilistic** generating function in which the coefficient is the probability that a random length-n output has exactly k occurrences.

For analytic information theory, the second is often the more natural object. We therefore define

$$F(z,u) = \sum_{n\geq 0}\sum_{k\geq 0} p_{n,k} z^n u^k,$$

where

$$p_{n,k} = \Pr(\text{a random length-}n\text{ string has exactly } k \text{ occurrences of } M).$$

If one wants the unweighted counting version instead, one simply replaces probabilities by counts. The same automaton technology underlies both.

17.3 Autocorrelation and Overlap

The structure of motif occurrences is governed by overlap.

Definition 17.1 (Autocorrelation polynomial)**.** If

$$M = m_1 m_2 \cdots m_\ell,$$

then its autocorrelation polynomial is

$$C_M(z) = \sum_{k=0}^{\ell-1} c_k z^k,$$

where $c_k = 1$ if the suffix of M of length k equals the prefix of M of length k, and $c_k = 0$ otherwise.

This polynomial records how one occurrence of the motif can overlap with the next. It matters because overlaps create dependence between nearby occurrence indicators.

Example 17.2 (The motif "aba")**.** For

$$M = \texttt{aba},$$

there is always the trivial overlap $k = 0$, and there is also a 1-letter overlap because the suffix 'a' equals the prefix 'a'. But there is no 2-letter overlap because 'ba' differs from 'ab'. Thus

$$C_M(z) = 1 + z.$$

a pattern with overlap can occur in strings like 'ababa', where two occurrences share one letter.

17.4 The Automaton Recipe

The usual workflow is mechanical.

1. Build a finite automaton whose states record how much of the motif has currently been matched (an overlap automaton, essentially the finite-state suffix-matching machine behind pattern search).
2. Add a mark u whenever a transition completes a new occurrence.
3. Weight each transition by z for length, and by the source probability if a probabilistic source is being used.
4. Form the transfer matrix $T(z, u)$ of this weighted automaton.

Then the bivariate generating function has the rational form

$$F(z, u) = \mathbf{v}^T (I - T(z, u))^{-1} \mathbf{e}$$

for appropriate initial and terminal vectors.

That is the finite-state miracle: once the pattern-tracking automaton is built, the rest is linear algebra.

Theorem 17.3 (Gaussian limit law for motif counts, black box)**.** *For a fixed motif tracked by a finite automaton under an ergodic Markov source, the number of motif occurrences in a random length-n string has mean*

$$\mu n + O(1),$$

variance

$$\sigma^2 n + O(1),$$

and a central-limit theorem after centering and scaling.

The mechanism is singularity perturbation in the auxiliary variable u together with a quasi-power theorem. For this chapter, the safe moral is enough: finite-state motif tracking leads naturally to linear growth of mean and variance and often to a Gaussian limit law.

17.5 A Simple "aba" Mean Calculation

For the uniform i.i.d. binary source and the motif 'aba', the leading mean can be obtained without any heavy machinery. Let

$$X_n = \#\text{occurrences of "aba" in a random binary string of length } n.$$

There are $n-2$ possible starting positions, and at each one the probability of seeing 'aba' is

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

So by linearity of expectation,

$$\mathbb{E}[X_n] = \frac{n-2}{8}.$$

Overlap matters for the variance and the full distribution, but not for this first-moment calculation.

17.6 Trie Depth and Entropy

Tries give another classical example. Build a trie from independent strings drawn from a finite-state source. The depth needed to distinguish one string from the others is closely tied to the source entropy rate.

Theorem 17.4 (Trie depth under a Markov source, black box)**.** *If strings are generated by an ergodic Markov source with Shannon entropy rate h, then the expected depth of a random key in a trie built from n independent samples satisfies*

$$\mathbb{E}[D_n] \sim \frac{1}{h} \log n.$$

A finer expansion often includes a bounded periodic fluctuation in $\log n$.

The important point is conceptual: the constant is the reciprocal of the *Shannon entropy rate of the source itself*, not a naive path-counting growth exponent. Chapter 9 already warned that support growth and Shannon entropy are different objects, and that warning remains in force here.

For the uniform binary source,

$$h = \log 2,$$

so

$$\mathbb{E}[D_n] \sim \log_2 n.$$

That is exactly what one should expect: among n roughly equally likely binary strings, one needs about $\log_2 n$ bits to isolate a typical one.

17.7 Two Classical Analytic Tools

Two techniques appear repeatedly in this literature.

Mellin transforms. Mellin methods are especially good for digital trees and periodic fluctuations. Non-real poles in Mellin space translate into oscillatory terms in $\log n$.

Saddle-point methods. When the relevant generating function is entire rather than singular at a finite point, one often evaluates coefficients by locating the saddle point of the Cauchy integral. This is an advanced tool and we use it here only as signposting.

A student should not feel compelled to master both techniques from this chapter alone. The important lesson is structural: different statistics call for different analytic machines, but the machines are reusable once the right finite-state encoding has been built.

17.8 What Transfers to WFA Approximations?

This is the chapter's main cautionary question.

If a WFA surrogate for an LLM preserves the statistic of interest to sufficient accuracy, then the exact finite-state theorems above become plausible approximations for the surrogate. But that "if" hides the hard part.

For example:

- a surrogate that matches next-token conditionals on short contexts may still get long-range motif counts wrong;
- a surrogate that matches the length PGF may still get phrase statistics wrong;
- a surrogate with excellent finite-length agreement may still have the wrong dominant poles asymptotically.

So the safe formulation is this:

> Analytic information theory supplies an exact toolkit for exact finite-state models and a research program for sufficiently faithful finite-state approximations.

It does not supply a theorem that every WFA approximation of an LLM automatically preserves every asymptotic statistic we care about.

Summary

This chapter's clean message is:

1. finite-state encodings of sequence statistics lead to rational or transfer-matrix generating functions;
2. once those encodings are in hand, classical analytic tools yield sharp asymptotics;
3. transferring those results to LLM approximations is promising, but conditional on the quality and target of the approximation.

That is enough to motivate the analytic information theory program without overselling what has already been proved for LLMs.

Further Reading

- **W. Szpankowski**, *Average Case Analysis of Algorithms on Sequences* (2001) [57] — the central monograph on analytic information theory.
- **L. J. Guibas and A. M. Odlyzko**, "String overlaps, pattern matching, and nontransitive games" (1981) [22] — for autocorrelation and overlap structure.
- **P. Nicodème, B. Salvy, and P. Flajolet**, "Motif statistics" (2002) [41] — for transfer-matrix motif counting and Gaussian laws.
- **P. Jacquet and W. Szpankowski**, "Analysis of digital tries with Markovian dependency" (1991) [27] — for trie depth and Mellin techniques.
- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (2009) [19] — for quasi-powers, singularity analysis, and saddle-point methods.

Exercises

Exercise 17.5. Compute the autocorrelation polynomial of the motif 'abab'. Which nontrivial overlaps occur?

Exercise 17.6. Explain why the strict distinction between counting and weighted motif generating functions matters by giving one statistic that changes when source probabilities are introduced and one that does not.

Chapter 18

Ten Open Problems at the Frontier

This chapter collects ten precisely stated open problems at the intersection of analytic combinatorics and large language models. Each problem is accompanied by a statement of what is known, why existing work stops short, and what a solution would look like. Some of the gaps depend only on classical mathematics; others depend on bridges to LLMs that remain conjectural in the present state of the subject. The numbering is for reference only and does not imply an ordering by importance; Chapter 19 addresses tractability.

18.1 Gap 1a: A Generating Function for LLM Length Distributions

Let μ be the distribution over Σ^* induced by an autoregressive language model (Chapter 10). The *length probability generating function* (length PGF) is

$$L(z) = \sum_{n=0}^{\infty} \Pr[|W| = n]\, z^n = \sum_{n=0}^{\infty} \Bigg(\sum_{\substack{w \in \Sigma^* \\ |w| = n}} \mu(w) \Bigg) z^n.$$

This is a well-defined formal power series with non-negative coefficients summing to 1 (by tightness, Chapter 8). Its radius of convergence R_L satisfies $R_L \geq 1$.

What is known. Du *et al.* 2023 [16] give a stopping-hazard criterion for tightness and prove that broad classes of softmax Transformers are tight. They do not construct $L(z)$ explicitly or characterize its singularity structure.

Why it stops short. Tightness is the statement $L(1) = 1$. The analytic-combinatorics programme needs far more: the location of the dominant singularity of $L(z)$, its type (polar, algebraic, logarithmic), and the resulting coefficient asymptotics $\Pr[|W| = n] \sim ?$ These would tell us, for example, whether LLM output lengths follow a geometric tail (pole), a subexponential power-law tail ($n^{-3/2}$ from a square-root branch point), or something else entirely.

What a solution looks like. An explicit formula for $L(z)$ in terms of the model's parameters — or at least a proof that $L(z)$ belongs to a specific function class (rational, algebraic, D-finite) — together with a singularity analysis yielding the asymptotic form of $\Pr[|W| = n]$. A weaker but still valuable route would be a tractable surrogate for $L(z)$ with certified approximation error.

18.2 Gap 1b: The Entropy-Rate Identity for LLM Counting Objects

Chapter 9 showed that an entropy-rate identity of the form $h = \log(1/R)$ can hold for a carefully chosen *entropy-typical counting object*. The delicate point is that this is not the same as naive

support counting, and it is not the same as the length PGF of Gap 1a.

What is known. In classical finite-state and context-free settings, several closely related growth quantities are available: measure-theoretic entropy, support-growth (topological) entropy, and typical-set growth. Earlier chapters emphasized that these must be separated carefully. Shannon's original work, Kontoyiannis 1998, and Takahashi and Tanaka-Ishii 2018 [58] estimate entropy-like quantities for natural language, while Scheibner, Smith, and Bialek 2025 [49] revisit the question with modern LLM-based predictors.

Why it stops short. The counting object for an LLM is not canonical. One could count support strings, entropy-typical strings, thresholded high-probability strings, or some other proxy. These lead to different radii and different interpretations. No one has carefully stated, let alone proved, the correct LLM analogue of $h = \log(1/R)$ with all these object-level distinctions made explicit.

What a solution looks like. A rigorous statement and proof of $h = \log(1/R_C)$ for a clearly specified entropy-related counting object $C(z)$ attached to a class of autoregressive models, with explicit hypotheses on the model class and an explicit distinction from both support growth and length PGFs.

18.3 Gap 2: Rigorous Boltzmann–Temperature Correspondence

Chapter 13 introduced the substitution $x = e^{-\beta}$ that turns a combinatorial generating function $A(x)$ into a partition function $Z(\beta) = A(e^{-\beta})$. Chapter 14 defined the LLM Gibbs measure $\mu_\beta(w) \propto \mu(w)^\beta$. The bridge between them is the *global* partition function

$$Z(\beta) = \sum_{w \in \Sigma^*} \mu(w)^\beta.$$

What is known. Blondel *et al.* 2025 [8] (arXiv:2512.15605) establish a bijection between autoregressive models and energy-based models via the soft Bellman equation. Kempton and Burrell 2025 [28] quantify the gap between local temperature sampling and the global Gibbs measure. Neither constructs $Z(\beta)$ as an analytic function of β or identifies its singularities. The global Gibbs object is mathematically clean; the empirically implemented object is usually local tokenwise temperature decoding. Reconciling those two is part of the gap, not a solved preliminary.

What a solution looks like. For a concrete non-trivial model class (e.g. a softmax Transformer trained on a PCFG corpus), construct $Z(\beta)$ explicitly, determine its domain of analyticity in β, and locate any phase-transition points β_c where Z or its derivatives become singular.

18.4 Gap 3: ACSV for Tokenized Models

When the alphabet $|\Sigma| > 2$ and we want joint statistics — for example, the joint distribution of the counts of two specific tokens in a sample of length n — the natural tool is multivariate generating functions and the Pemantle–Wilson framework of analytic combinatorics in several variables (ACSV) [43].

What is known. ACSV provides a complete asymptotic theory for coefficients of rational and algebraic multivariate generating functions, governed by the geometry of the singular variety. The framework is mature for combinatorial applications (lattice paths, queuing models).

Why it stops short. No one has applied ACSV to LLM output distributions. The multivariate GF of token-count statistics under an autoregressive model has not been constructed, and the singular variety has not been analyzed.

What a solution looks like. A multivariate GF $F(z_1, \ldots, z_k)$ tracking token counts under a precisely specified surrogate object — for example, a length-conditioned token-count law or a

weighted string-counting series for a WFA-approximated LLM — with ACSV applied to extract joint asymptotics.

18.5 Gap 4: A Chomsky–Schützenberger Test for LLM Support

Chapter 5 argued that generic nondegenerate square-root algebraic systems — including important unambiguous context-free examples under the usual irreducibility and aperiodicity hypotheses — have coefficients of the form $[z^n]C(z) \sim c \cdot n^{-3/2}\rho^{-n}$. This is a *testable signature*, but not a blanket property of all context-free systems.

What is known. Banderier and Drmota 2015 [3] classify possible asymptotic forms of coefficients of algebraic generating functions. The $n^{-3/2}$ exponent is the generic nondegenerate square-root signature, not the only algebraic possibility.

Why it stops short. No one has run this test on LLM outputs. It is not known whether the length distribution of samples from GPT-4, Llama, or any open-weight model is consistent with the algebraic asymptotic class.

What a solution looks like. An empirical study fitting $\Pr[|W| = n]$ or a support-proxy surrogate to the form $c \cdot n^{\alpha}\rho^{-n}$ and testing whether $\alpha = -3/2$ is consistent with the data. A positive result would be suggestive evidence for a square-root algebraic regime, not by itself a proof that the support language is approximately context-free.

18.6 Gap 5: Gaussian Limit Laws via Hwang's Quasi-Power Theorem

Hwang 1998 [25] proved a meta-theorem: if a bivariate generating function $F(z, u)$ tracking a statistic has the "quasi-power" form

$$[z^n]F(z, u) = A(u) \cdot B(u)^n \cdot \big(1 + O(1/n)\big)$$

uniformly near $u = 1$, then the statistic is asymptotically Gaussian with mean $\mu_n \sim n \cdot B'(1)/B(1)$ and variance $\sigma_n^2 \sim n \cdot V$ for an explicit V.

What is known. The quasi-power theorem is a workhorse of analytic combinatorics, applied to hundreds of combinatorial statistics (inversions, descents, pattern occurrences). Chapter 17's motif-count statistics are a direct instance.

Why it stops short. Verifying the quasi-power hypothesis for a statistic measured on *LLM output* — e.g. the count of a motif in a sample of length n — requires knowing the bivariate GF $F(z, u)$ of the LLM, which is Gap 1a generalized. For WFA-approximated LLMs, $F(z, u)$ is rational in (z, u) and the hypothesis is often checkable.

What a solution looks like. For a WFA-approximated LLM, verify the quasi-power form for a concrete statistic (e.g. occurrences of a specific bigram), state the Gaussian limit, and compare with empirical histograms.

18.7 Gap 6: Thermodynamic Formalism and Generating Functions

The thermodynamic formalism of Ruelle [48], Bowen [10], and Sinai studies shift-invariant measures on symbolic sequences via transfer operators and pressure functions. Chapter 13's Gibbs/GF correspondence is a special case. The full formalism provides a theory of equilibrium states, variational principles, and phase transitions that subsumes both the combinatorial Boltzmann picture and the information-theoretic entropy picture.

What is known. The thermodynamic formalism is fully developed for subshifts of finite type (which are the symbolic-dynamics analogues of regular languages / finite-state sources).

The pressure function $P(\phi)$ for a Hölder-continuous potential ϕ on a mixing subshift is real-analytic, and equilibrium states are unique.

Why it stops short. LLMs are not subshifts of finite type. Their potential functions (the log-probabilities) are computed by deep networks, not by finite-state rules. Extending the formalism to "soft" potentials arising from neural networks requires either (a) approximation by finite-state potentials (back to Chapter 12) or (b) new theorems about transfer operators with neural-network potentials.

What a solution looks like. A rigorous pressure function for a class of autoregressive models, with analyticity properties and a variational characterization, expressed in terms of a precisely chosen object — for example, the global partition sum $Z(\beta)$ or an explicitly defined weighted transfer operator — rather than the generic phrase "the model's generating function."

18.8 Gap 7: Classification of LLM Singularity Types

Analytic combinatorics has a well-developed zoology of singular behaviors: polar (rational GFs, geometric coefficients), algebraic-logarithmic (algebraic GFs, $n^{-3/2}$ and its cousins $n^{-(2k+1)/2}$), and essential (entire functions with super-exponential growth, requiring saddle-point methods). Which of these appear in LLM counting or length generating functions? Is there a new type?

What is known. WFA approximations give rational GFs (polar singularities). PCFG approximations give algebraic GFs (branch-point singularities). Real LLMs presumably have singularity structure that is neither purely rational nor purely algebraic.

What a solution looks like. A classification theorem: "for autoregressive models in class $\mathcal{C}$, the length PGF has singularities of type X, and the coefficient asymptotics are of form Y." Even a conjectural classification backed by numerical evidence would be a contribution.

18.9 Gap 8: Phase-Transition Classification via Composition Schemes

Banderier, Kuba and Wallner 2021 [4] classified the phase transitions arising from composition-scheme generating functions $A(B(z))$ where the inner and outer singularities interact. Chapter 15 argued that empirical LLM phase transitions (Nakaishi *et al.* [40], Arnold *et al.* [1]) may be instances of this classification.

What is known. The composition-scheme classification produces Airy-type universal behavior at critical points. Mikhaylovskiy 2025 [38] argues that empirical LLM phase transitions are consistent with this universality class. This comparison is indirect: the experiments vary local decoding temperature, while the composition-scheme story concerns a global analytic object.

Why it stops short. The match is qualitative. No one has derived a composition-scheme structure from an LLM's architecture or from a WFA approximation of one, and then matched the predicted critical exponents to the empirical ones.

What a solution looks like. For a concrete LLM surrogate (e.g. a Transformer trained on a PCFG), derive the composition-scheme structure of $Z(\beta)$, predict the critical exponents, and verify them empirically.

18.10 Gap 9: A WFA Pipeline with Analytic Error Bounds

Chapter 12 surveyed four approaches to WFA extraction. None comes with end-to-end *analytic* error bounds relating the approximation quality (in KL divergence, total variation, or Hankel-operator norm) to the rank of the extracted WFA and the spectral properties of the original model.

What is known. AAK theory (Section 12.4) gives optimal Hankel-operator-norm bounds for the one-letter case. Spectral learning gives Frobenius-norm bounds on the Hankel matrix. L^* has polynomial query complexity in the number of states of the target DFA. These bounds exist in isolation.

What a solution looks like. A combined pipeline: extract a WFA of rank k from an LLM, with a theorem stating that the KL divergence between the WFA's distribution and the LLM's distribution is at most $\varepsilon(k, \sigma_{k+1})$ where σ_{k+1} is the $(k+1)$st singular value of the empirical Hankel matrix. This is the most tractable of the ten gaps.

18.11 Gap 10: Phase Transitions as Confluent Singularities

A confluent singularity is a point in parameter space where two or more singular surfaces of a generating function merge. In the composition-scheme setting of Gap 8, these produce Airy-function asymptotics with universal critical exponents. The conjecture, motivated by Mikhaylovskiy 2025 [38] and the empirical evidence of Chapter 15, is that the observed phase transitions in LLM output are confluent singularities of $Z(\beta)$.

What is known. Confluent singularities are well understood for algebraic generating functions arising from context-free grammars (Flajolet and Sedgewick [19], Chapter VII). The Airy function $\mathrm{Ai}(\cdot)$ appears universally.

What a solution looks like. For a concrete model or surrogate, prove that a clearly specified global partition object has a confluent singularity at β_c, identify the Airy scaling, and match it to empirical critical exponents. Here again, the first step is to pin down the analytic object before trying to classify its singularity.

18.12 Summary of Gaps

Gap 1a Construct the length PGF $L(z)$ for an LLM and analyze its singularities.

Gap 1b Prove a correct $h = \log(1/R)$-type identity for an explicit entropy-related counting object attached to an LLM.

Gap 2 Construct $Z(\beta) = \sum_w \mu(w)^\beta$ as an analytic function and locate phase transitions.

Gap 3 Apply ACSV to multivariate token-count GFs from LLMs.

Gap 4 Test whether LLM length distributions exhibit the $n^{-3/2}$ algebraic signature.

Gap 5 Verify the quasi-power hypothesis and derive Gaussian limit laws for LLM statistics.

Gap 6 Extend thermodynamic formalism to neural-network potentials.

Gap 7 Classify the singularity types of LLM generating functions.

Gap 8 Match empirical LLM phase transitions to composition-scheme universality classes.

Gap 9 Build a WFA extraction pipeline with end-to-end analytic error bounds.

Gap 10 Prove that LLM phase transitions are confluent singularities with Airy scaling.

18.13 Research Suggestions

Gap 9 is the most tractable starting point: it synthesizes existing tools (AAK bounds, spectral learning, L^*) and does not require new theorems about LLMs themselves. Gap 5 is an attractive second target because Hwang's quasi-power theorem is a ready-made meta-theorem with

checkable hypotheses. Gap 10 is the prize: a proof would unify the empirical phase-transition observations with the analytic-combinatorics classification programme. Chapter 19 develops a concrete research roadmap from these priorities.

18.14 Notes and Further Reading

The ten gaps are distilled from the broader research programme described in the source literature. Flajolet and Sedgewick [19] remains the central reference for the analytic side; their Chapters VI–IX cover the singularity analysis, algebraic functions, and limit-law machinery that most of these gaps require. Pemantle and Wilson [43] is the reference for Gap 3. The empirical phase-transition literature (Nakaishi *et al.* [40], Arnold *et al.* [1], Mikhaylovskiy [38]) motivates Gaps 8 and 10 directly.

Recommended Reading

- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19] — Chapters VI–IX supply the singularity analysis, algebraic-function theory, transfer-matrix machinery, and limit-law framework that underpin Gaps 1a, 1b, 4, 5, 7, and 10. Required reading before attempting any of the analytic gaps.
- **R. Pemantle and M. C. Wilson**, *Analytic Combinatorics in Several Variables* (Cambridge University Press, 2013) [43] — The reference for analytic combinatorics in several variables (ACSV). Develops the critical-point theory and stratified-Morse-theory tools needed for Gap 3 (multivariate token-count generating functions).
- **H.-K. Hwang**, "On convergence rates in the central limit theorems for combinatorial structures," *European J. Combin.* (1998) [25] — Proves the quasi-power meta-theorem central to Gap 5. Reading this paper makes clear exactly what hypotheses need to be verified for a WFA-approximated LLM, and how Gaussian limit laws follow once they are.
- **D. Ruelle**, *Thermodynamic Formalism* (Addison-Wesley, 1978) [48] — The foundational reference for the thermodynamic formalism of equilibrium states, pressure functions, and transfer operators on subshifts of finite type. Essential background for Gap 6 (extending the formalism to neural-network potentials).
- **R. Bowen**, *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms* (Springer, 1975) [10] — Develops the Ruelle–Bowen theory of Gibbs states and the variational principle for topological pressure on Axiom A systems. Read alongside Ruelle for Gap 6; together they give the ergodic-theory substrate that Gap 6 seeks to extend.
- **N. Mikhaylovskiy**, "Zipf's law and the temperature window in transformer language models," *Findings of EMNLP* (2025) [39] — Empirically identifies temperature-driven phase transitions in transformer outputs and connects them to Zipf-exponent shifts. The primary empirical motivation for Gaps 8 and 10, and a concrete target for what a positive resolution would need to explain.

Chapter 19

A Research Roadmap

19.1 Where We Stand

Part I of this book developed the toolkit of analytic combinatorics: generating functions, singularity analysis, transfer matrices, and Boltzmann samplers [19]. Part II established the linguistic foundations: weighted finite automata, the Hankel characterisation, PCFGs, entropy rates, and the Chomsky–Schützenberger theorem. Part III introduced the Transformer architecture and surveyed the theoretical results bearing on its expressive power and limitations [60, 36, 47]. Part IV introduced the global partition function $Z(\beta)$ as a mathematically clean reference object, distinguished it from local decoding temperature, and surveyed a conjectural bridge from combinatorial singularity theory to empirical phase-transition evidence [4]. Part V surveyed Zipf distributions, motif statistics, and their analytic explanations [57], before Chapter 18 catalogued ten open problems — Gaps 1a, 1b, 2, 3, 4, 5, 6, 7, 8, 9, 10 — at the frontier. This final chapter answers a practical question: given that background, what should a researcher actually work on first?

19.2 Ranking the Gaps by Tractability

Not all ten gaps are equally open. Some are computation problems waiting for careful accounting; others require new mathematics that is not yet in sight. The following ranking is a judgement call, not a theorem.

Most tractable: Gap 9 (WFA-approximation pipeline with analytic error bounds). The four approximation strategies surveyed in Chapter 12 — spectral learning from moment matrices, AAK-based Hankel truncation, L^*-style automaton learning, and tensor-network contraction — are each well understood in isolation. What is missing is a unified error budget: given that each step introduces its own approximation error, what does the compound bound look like? The ingredients are already on the shelf. The AAK singular-value bounds were made explicit for weighted automata by Lacroce, Panangaden, and Rabusseau [31]; the L^* convergence guarantees are classical. The missing piece is to compose these bounds and state a single theorem of the form: "for a rank-r WFA approximation to a clearly specified target — for example, the full string distribution over sequences of length at most n, the total-variation error is at most $f(r, n)$." This is, in essence, a careful accounting exercise. It does not require proving anything new about LLMs themselves.

Second: Gap 5 (Gaussian limit laws via the Hwang quasi-power theorem). Hwang's 1998 meta-theorem [25] gives Gaussian limits once the relevant probability generating function satisfies a quasi-power hypothesis. Verifying that hypothesis for a concrete LLM surrogate — say, a small PCFG with a softmax output head — is a calculation, not an open problem in the usual sense. The difficulty is making the surrogate tractable enough that the PGF can be computed or tightly bounded. A PCFG with k nonterminals has an algebraic GF

of degree at most k; the softmax perturbs it in a controlled way. Working through this for the simplest non-trivial case would constitute a publishable result and, more importantly, would establish whether the quasi-power hypothesis is robust to the softmax perturbation.

Third: Gap 1b (the entropy-rate identity $h = \log(1/R)$ for LLM counting GFs). Once Gap 9 is resolved and a WFA approximation exists for a clearly specified counting target, Gap 1b becomes a surrogate-verification problem rather than an immediate corollary. For the surrogate itself, Perron–Frobenius gives exact exponential growth rates for rational counting series. The real question is whether the counting object one has chosen is the right analogue of Chapter 9's entropy-typical construction, and how close its radius-based quantity is to an independently estimated entropy quantity for the original model.

One must also be precise about the distinction between the length probability generating function (Gap 1a, concerning $\sum_n p_n x^n$ where $p_n = \Pr[\text{length} = n]$) and a counting GF (Gap 1b, concerning some sequence $\sum_n c_n x^n$ built from a counting or entropy-typical object). These are not the same object and the singularities can differ. So the right Phase 2 task is to decide which counting object is being tested, compute its radius on a surrogate, and compare the resulting $\log(1/R)$ with the relevant entropy-like measurement rather than assume the identity automatically.

Big prize: Gap 10 (phase transitions as confluent singularities). The empirical observations of Nakaishi et al. [40], Arnold et al. [1], and Mikhaylovskiy [38] all point to sharp transitions in LLM-generated text statistics at critical temperatures. The Banderier–Kuba–Wallner composition-scheme framework [4] gives a rigorous mechanism for such transitions: they arise when two dominant singularities of the partition function $Z(\beta, x)$ coalesce as β varies. A positive result connecting these two bodies of work — proving that the observed transitions correspond to singularity confluence in $Z(\beta, x)$ for an explicit surrogate model — would be a significant contribution. This is the right long-term target but not the right starting point. This phase also inherits the Chapter 14–15 distinction between local decoding temperature and the global Gibbs parameter of $Z(\beta)$; any successful bridge must make that relationship explicit rather than silently identifying the two.

Long shot in the sense of ease of entry, not in probability of success: Gap 4 (Chomsky–Schützenberger test). This is an empirical project. The idea is simple: sample strings from a medium-sized open-weight LLM at fixed temperature, fit a counting GF to the empirical length distribution, and ask whether the dominant singularity is algebraic of the type $\sim C \cdot n^{-3/2} r^{-n}$ (the generic square-root signature of a non-degenerate context-free-like system). Anyone with a GPU cluster and access to an open-weight model can start this tomorrow. The theoretical interpretation of a positive or negative result is less straightforward, but the empirical finding would itself be informative. This gap is listed last in tractability not because it is hard to attempt but because the result is harder to interpret without the theoretical scaffolding of Phases 1–2. An empirical $n^{-3/2}$ fit here would be suggestive, not a direct proof that the support language is approximately context-free.

19.3 A Three-Phase Programme

The following programme is designed so that each phase produces publishable intermediate results while also building the substrate for the next phase.

Phase 1: Implement and benchmark a WFA-extraction pipeline. Choose a small LLM surrogate: a 2-layer Transformer trained on a corpus generated by a known PCFG. Using the PCFG, the ground-truth distribution is computable, so errors can be measured exactly. Fix the primary approximation target up front: the most natural first choice is the truncated full string distribution on strings of length at most L. Extract the Hankel matrix H of the surrogate's string probabilities up to length L, compute its singular-value decomposition, and

truncate to rank r. Record the singular-value decay as a function of r (this is a diagnostic of how well a WFA can approximate the model). Construct the rank-r WFA by the spectral-learning procedure of Balle–Mohri or equivalently via Lacroce–Panangaden–Rabusseau [31]. Measure the empirical KL divergence between the WFA distribution and the surrogate distribution as a function of r and L. Compare to the theoretical AAK bound.

Deliverables: a codebase, a table of singular-value plots, and a graph of empirical KL divergence versus rank. The gap between the empirical error and the AAK bound, if large, is itself an interesting finding.

Phase 2: Verify the entropy-rate identity. Using the rank-r WFA from Phase 1, compute the transition matrix M and extract $\rho(M)$. Compute the exact rational objects associated with the surrogate's chosen counting target, including the relevant radius R, and compare $\log(1/R)$ to an independently measured entropy-like quantity on the surrogate (for example, a cross-entropy estimate on held-out strings). Do this for several values of r and record how quickly the discrepancy decays.

This does not automatically close Gap 1b for the original LLM. What it does provide is a controlled testbed: which counting object supports a log-radius identity on the surrogate, and how sensitive that identity is to approximation error.

Phase 3: Scan temperature and look for singularity confluence. Fix the surrogate from Phases 1–2. Define the temperature-parameterised finite-size proxy

$$Z_L(\beta) = \sum_{|\mathbf{x}| \le L} p(\mathbf{x})^\beta,$$

where the sum runs only over strings up to length L. For each β in a grid, compute the dominant singularity of the associated surrogate WFA (the reciprocal spectral radius $R(\beta)$) and track how it moves with β. This is a finite-size probe, not yet a literal nonanalyticity of the infinite-volume partition function. A sharp kink or crossover in $R(\beta)$ is evidence for a singularity crossing, which in the Banderier–Kuba–Wallner framework [4] corresponds to a phase transition. If a transition is found, attempt to classify the type of confluence (smooth, cusp, or higher order) by examining the local expansion of the surrogate partition object near the critical point. In parallel, one may scan the empirically implemented local decoding temperature on the teacher model and compare observables, but the two temperature operations should not be conflated.

A positive finding here constitutes empirical evidence for Gap 10. A negative finding is also informative: it would suggest that LLM phase transitions, when they occur, are not of the composition-scheme type and would require a different theoretical explanation.

19.4 Parallel Tracks and What to Avoid First

Gap 4 (the Chomsky–Schützenberger numerical test) is an orthogonal project that does not depend on the WFA pipeline. It can be pursued in parallel by a researcher with access to a real open-weight LLM [8, 16] and a good sampling implementation. The main risk is that the length distribution of LLM output is heavily confounded by prompt length and stopping heuristics; controlling for these requires care. Another confounder is the local-vs-global temperature distinction from Chapter 14: fitting a global analytic model to tails produced under local decoding temperature can be conceptually ambiguous.

Gap 5 (Hwang quasi-power limits) can be pursued as a theoretical side-project by a combinatorics-oriented researcher who does not have access to GPU resources. The calculation for the PCFG-plus-softmax surrogate can be carried out on paper or with a computer-algebra system [25].

What not to work on first. Three gaps should be deferred.

Gap 6 (thermodynamic formalism) is where the cleanest theory lives: the Ruelle–Bowen–Sinai transfer operator gives a complete account of the free energy and its singularities. But this background is a semester course on its own, with prerequisites in ergodic theory that most combinatorics or machine-learning students will not have. Return to it after Phases 1–3 have built intuition.

Gap 7 (classification of all singularity types arising in LLM partition functions) is probably the ultimate theoretical destination of this research programme, but it presupposes Gap 10 being resolved and a rich collection of examples to classify. It is not a starting point. It also presupposes deciding what analytic class the partition object $Z(\beta)$ actually belongs to.

Gap 3 (ACSV for tokenized models) requires the Pemantle–Wilson multivariate machinery [43] in its full generality. This is technically demanding — the critical-point analysis for multivariate GFs involves stratified Morse theory — and the payoff depends on first establishing that LLM generating functions have the right structure to make the analysis applicable. Defer until the univariate picture (Phases 1–2) is settled.

19.5 For a Student Entering the Field

The following reading order is recommended for a graduate student who has not yet read this book and wants to enter this research area efficiently.

1. **Flajolet–Sedgewick [19], Chapters I–IV.** This covers symbolic combinatorics, ordinary and exponential GFs, and the singularity-analysis framework. These chapters correspond to Chapters 1–5 of this book. This material is non-negotiable; everything else builds on it.

2. **This book, Part II** (WFAs, PCFGs, entropy rates). The primary mathematical objects are the Hankel matrix and the Perron–Frobenius theorem applied to weighted transition matrices.

3. **Szpankowski [57].** For researchers whose interest is in word statistics, motif occurrences, and large-deviation results. Chapter 17 of this book is an appetiser; Szpankowski's monograph is the full treatment.

4. **Rizvi et al. [47] and Merrill–Sabharwal–Smith [36].** For the LLM side: what Transformers can and cannot express as formal languages, and what circuit-complexity arguments say about their computational depth. Read alongside Strobl et al. [54] for the broader survey of formal language results.

5. **Duchon–Flajolet–Louchard–Schaeffer [17].** The foundational reference for Boltzmann sampling. Read this before attempting Phase 3; the connection between the Boltzmann parameter and the thermodynamic β is made explicit there.

This reading list is not comprehensive, but it covers the minimum required to engage with the three-phase programme above without repeatedly needing to pause and build background. Chapters 1–8 and the exact finite-state parts of Chapters 12 and 17 are best read as classical or exact toolkit. Chapters 9, 14, and 15 are more bridge-like and conjectural in places; they are important, but they should be read with that status in mind.

19.6 Closing

The purpose of this book has been to place in one document the pieces of two literatures — analytic combinatorics and large language model theory — that have developed largely independently. The mathematical community has Flajolet–Sedgewick, Szpankowski, and Pemantle–Wilson; the machine-learning community has Transformer theory, scaling laws, and empirical

phase-transition observations. These are not separate subjects. The formal-language perspective forces precision about what a language model is computing; the singularity-analysis perspective gives a vocabulary for what happens at critical temperatures. If the book has done its job, a reader should be able to move between a chapter of Flajolet–Sedgewick and a survey such as Strobl et al. [54] without feeling that they have changed disciplines. The ten gaps of Chapter 18 and the programme sketched here are an invitation to close that distance further. They are not a manifesto; they are a list of things that seem doable.

Further Reading

The reading list in Section 19.5 covers the minimum required for the three-phase programme. The items below are the core references specifically for the roadmap itself.

- **P. Flajolet and R. Sedgewick**, *Analytic Combinatorics* (Cambridge University Press, 2009) [19] — The analytical backbone of Phases 1–3. Chapters VI–IX on singularity analysis, algebraic functions, and the saddle-point method are used directly; Chapter IX on words and tries is the template for the WFA transition-matrix computations in Phase 2.

- **W. Szpankowski**, *Average Case Analysis of Algorithms on Sequences* (Wiley, 2001) [57] — The bridge between analytic combinatorics and information theory. For researchers entering from the combinatorics side, this monograph makes entropy rates, trie depths, and suffix-tree statistics into singularity-analysis problems, exactly the translation needed for Phase 2.

- **R. Pemantle and M. C. Wilson**, *Analytic Combinatorics in Several Variables* (Cambridge University Press, 2013) [43] — Required for Gap 3 and for Phase 3 once the analysis extends beyond univariate generating functions. The critical-point and stratified-Morse-theory toolkit for multivariate coefficient asymptotics.

- **H.-K. Hwang**, "On convergence rates in the central limit theorems for combinatorial structures," *European J. Combin.* (1998) [25] — The quasi-power meta-theorem that converts a bivariate generating function in quasi-power form into a Gaussian limit law. A short paper with large returns: reading it clarifies exactly what Phase 1's WFA extraction must deliver for Gap 5 to follow automatically.

- **D. Ruelle**, *Thermodynamic Formalism* (Addison-Wesley, 1978) [48] — Background for Gap 6 and for understanding why Phase 3's $Z(\beta)$ behaves like a thermodynamic pressure function. The Ruelle transfer operator is the infinite-alphabet generalisation of the WFA transition matrix.

- **R. Bowen**, *Equilibrium States and the Ergodic Theory of Anosov Diffeomorphisms* (Springer, 1975) [10] — Companion to Ruelle for the variational principle and Gibbs-state characterisation. Together these two short books give the ergodic-theoretic language that the roadmap's Phase 3 is, in spirit, trying to instantiate for neural-network potentials.

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